Testing Instrument Validity with Covariates *

Thomas Carr[†], Toru Kitagawa[‡]

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Abstract

Instrumental variable methods are widely used in causal inference, but the validity of the identifying assumptions is often disputed. Tests of these identifying assumptions exist, but are rarely used in practice as they struggle to accommodate conditioning covariates, which researchers often use to justify instrument validity. We develop a novel test of the instrumental variable identifying assumptions for heterogeneous treatment effect models that can easily accommodate conditioning covariates without compromising power. We assume semiparametric dependence between potential outcomes and conditioning covariates. This allows us to replace testable implications for the outcome variable which hold conditional on covariates with unconditional testable implications for estimable partial residuals, effectively performing dimensionality reduction. We obtain testable equality and inequality restrictions among the subdensities of partial residuals, and discuss how they can be employed to detect invalid instruments. To improve power, we propose *distillation*, where a trimmed sample is used to test the inequality restrictions. We show asymptotic validity of our test, and in Monte Carlo exercises we find gains in finite sample power from testing restrictions jointly and distillation. We apply our test procedure to three instruments and reject the null for one.

Keywords: Program evaluation, local average treatment effect, marginal treatment effect.

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[†]Department of Economics, Bristol University

[‡]Department of Economics, Brown University. Email: toru_kitagawa@brown.edu

1 Introduction

Instrumental variable (IV) methods are widely used in empirical economics. The credibility of IV estimates rests on the validity of the identifying assumptions, which is often debated. In IV estimation, conditioning covariates are frequently used to enhance the credibility of the identifying assumptions or improve the precision of estimates. For example, when proximity to college is used as an instrument to estimate the return to schooling, researchers control for location characteristics as they may otherwise induce dependence between the instrument and the outcome. Huber and Mellace (2015), Kitagawa (2015), and Mourifié and Wan (2017) propose statistical tests for instrument validity in the heterogeneous treatment effect model of Imbens and Angrist (1994). However, the state of the art literature jointly tests a set of implications which grows in size with the dimension of conditioning covariates, leading to a lack of computationally feasible tests for more than a small number of covariates. A test procedure which avoids these issues has the potential to be widely adopted in empirical studies, but the development of such a procedure has been an open problem.

The first contribution of this paper is to propose an easy-to-implement test procedure for instrumental validity in the heterogeneous treatment effect model that can more easily accommodate conditioning covariates without compromising power. Building on the semiparametric approach of Carneiro et al. (2011), we specify that conditioning covariates are independent of unobserved heterogeneity and affect potential outcomes linearly. Under these restrictions, the conditional expectation of the outcome given conditioning covaraites and the instrument follows a partially linear model that depends nonparametrically on the probability of receiving treatment (propensity score). This allows the conditioning covariates to be partialed out and partial residuals obtained. We obtain a set of testable implications for the subdensities of these partial residuals. These testable implications are unconditional: they hold without reference to covariate values. By basing our test on these implications, we avoid the computational hurdles existing test procedures face. In our approach, increasing the number of conditioning covariates increases only the complexity of estimating the propensity score and partially linear model, not the number of implications to be tested.

Two testable implications are available: *index sufficiency* and *nesting inequalities*. Index sufficiency states that the partial residual subdensities depend on the instrument through the propensity score only. The nesting inequalities are monotonicity inequalities for the joint distributions of the partial residuals and treatment status. They state that, for two distinct values of the propensity score, the distribution of partial residuals for treated units with the higher propensity score nests that of units with the lower propensity score, and the converse is true for untreated units. These

implications can be generalised to the case of multiple and continuous instruments, and a version holds under the weaker partial monotonicity assumption considered in Mogstad et al. (2020). The two testable implications complement each other: index sufficiency holds the propensity score fixed and compares subdensities for different values of the instrument, while the nesting inequalities compare subdensities for different propensity scores. To exploit this complementarity, we propose jointly testing the two.

As the subdensity nesting inequalities are ordered with respect to the propensity score, a natural approach to testing them is to coarsen the propensity score and test the monotonicity of the subdensities conditional on this coarsened propensity score. However, this approach can suffer from low power. When the propensity score is correlated with the conditioning covariates, classifying observations by propensity score homogenises the distribution of instrument values across propensity score bins, which can mask violations of instrument validity.

Our second contribution is to propose a novel approach to testing the nesting inequalities. We suggest testing monotonicity of the conditional partial residual subdensities given the instrument. This approach ensures that observations with different instrument values are kept separate. We show that, when the identifying assumptions are satisfied, the nesting inequalities hold over instrument values if the conditional distribution of propensity scores is monotonic in the instrument in the sense of first-order stochastic dominance. If first-order stochastic dominance does not hold, we propose trimming the sample and testing the nesting inequalities for a sub-sample where first-order stochastic dominance is satisfied. We refer to this process as *distillation* as it distills the sample down to the observations best able to detect violations.

We show the benefits of implementing distillation analytically and numerically. Specifically, we present a class of data generating processes for which distillation dominates testing using a coarsened propensity score in terms of its ability to detect violations of instrument validity. For this class, any violation of the nesting inequalities conditional on the coarsened propensity score implies a larger violation of the nesting inequalities with a distilled sample, but not vice versa. That is, a violation of the nesting inequalities with a distilled sample does not imply the existence of a violation of the nesting inequalities conditional on the coarsened propensity score. Consequently, there exist data generating processes where instrument validity does not hold, but the violation can only be detected by the nesting inequalities when performing distillation.

We show asymptotic validity of our proposed test procedure under the null where index sufficiency and the nesting inequalities jointly hold and the model is correctly specified. This is not trivial due to the presence of multiple generated variables and a non-standard trimming procedure. We build on and extend the results of Linton et al. (2005) and Linton et al. (2010) to accommodate the estimated propensity score and distillation.

We perform Monte Carlo exercises to evaluate the size and power of the proposed test procedure and compare it to two alternatives: the Kitagawa (2015) test that does not control for covariates, and the test of nesting inequalities using a coarsened propensity score. We consider processes where the instrument is independent of the covariates, and processes where it is not. In the first case, an instrument that is valid conditional on the covariates will also be valid without conditioning on covariates, so the size of the Kitagawa (2015) test will not be distorted. In the second, size will only be controlled if the conditioning covariates are appropriately controlled for. We find that the size of the proposed test is controlled in all cases. For processes where the Kitagawa (2015) test has the correct size, we find our test procedure has comparable power. For all processes, it has higher power than the test of nesting inequalities using a coarsened propensity score. We also investigate the contributions of the index sufficiency and nesting inequality tests to the power of the joint test. These results suggest that index sufficiency has most power when the distributions of propensity scores for different instrument values are similar, so that there are observations with different instrument values and a similar propensity score. On the other hand, for the processes considered, rejection probabilities when testing the nesting inequalities do not depend strongly on the similarity of the propensity score distributions across instrument values. This is consistent with the two testable implications being complementary.

To demonstrate the feasibility of the test procedure and illustrate how it can be used to resolve questions of instrument credibility, we present three applications. The first is Card (1993). This instrument has been tested in Huber and Mellace (2015), Kitagawa (2015), Mourifié and Wan (2017) and Sun (2023) but, for computational reasons, in each case only a subset of the conditioning covariates of the original specification is used. The proposed test procedure allows all conditioning covariates to be included and remains computationally feasibile. Second, we turn to the research design of Angrist and Evans (1998), which features both the same-sex instrument and the multiple birth instrument. The validity of the same-sex instrument is questioned in Rosenzweig and Wolpin (2000) and its validity has been formally tested by Huber (2015) and Mourifié and Wan (2017). As with Card (1993), these studies use only a subset of the conditioning covariates, while our test procedure allows us to include all conditioning covariates. We perform a joint test of the validity of both instruments under the partial monotonicity assumption of Mogstad et al. (2019). Finally, we consider the research design of Bleakley and Chin (2010). This study estimates the effect of English speaking ability on a variety of outcomes using a sample of childhood immigrants. This design features both a multivalued treatment and a multivalued instrument, and hence provides an example of how our procedure can be adapted to this context.

The remainder of the paper proceeds as follows. The related literature is reviewed in Section 1.1 below. Section 2 introduces the semiparametric testable implications. Section 3 describes the

test procedure. Section 4 presents Monte Carlo exercises. Section 5 discusses our three empirical applications. Section 6 concludes.

1.1 Related Literature

This paper contributes to the active literature on the identification and estimation of the local average treatment effect (LATE) model of Angrist et al. (1996) and Imbens and Angrist (1994) and the marginal treatment effect model (MTE) of Heckman and Vytlacil (1999, 2001a,b, 2005). This literature is reviewed in Mogstad and Torgovitsky (2018) and Mogstad and Torgovitsky (2024). A testable implication of the LATE identifying assumptions was derived in Balke and Pearl (1997), Imbens and Rubin (1997) and Heckman and Vytlacil (2005). Kitagawa (2015) shows that this testable implication is sharp, and proposes a test procedure using a Kolmogorov-Smirnov statistic. Mourifié and Wan (2017) reformulates this testable implication as a conditional moment inequality, and proposes a test based on the intersection bounds framework of Chernozhukov et al. (2013). It is straightforward to generalize this testable implication to incorporate conditioning covariates, but assessing it nonparametrically in a finite sample is limited by the capacity of nonparametric methods to handle multiple conditioning covariates.

To facilitate the estimation of marginal treatment effects while controlling for multiple conditioning covariates, Carneiro et al. (2011) further imposes that potential outcomes are linear in the conditioning covariates, unobserved heterogeneities enter additively, and these unobservables are statistically independent of the covariates and instrument. The test procedure presented in this paper corresponds to a test for the validity of the Carneiro et al. (2011) identifying assumptions. Maestas et al. (2013), Eisenhauer et al. (2015), Kline and Walters (2016), Cornelissen et al. (2018), Felfe and Lalive (2018), Bhuller et al. (2020), and Coury et al. (2022) among others apply the approach of Carneiro et al. (2011) to recover marginal treatment effects. Our test procedure can be applied in any of these contexts to assess the validity of the identifying assumptions. Furthermore, even if the validity of the instrument is undisputed, the test procedure can still be used to detect errors in the functional form specification for the relationship between potential outcomes and conditioning covariates. Brinch et al. (2017) further develops the approach of Carneiro et al. (2011), and show how marginal treatment effects can be identified in a variety of settings. Mogstad et al. (2018) discusses how to extrapolate marginal treatment effects to set identify policy relevant treatment effects.

Several alternative tests of the LATE identifying assumptions have been proposed. Huber and Mellace (2015) considers a weaker set of identifying restrictions than Kitagawa (2015), derives a testable implication, and proposes several test procedures. Laffers and Mellace (2017) shows that the testable implication of Huber and Mellace (2015) is sharp for these identifying assumptions. Sun

(2023) derives testable implications when the treatment is an ordered or unordered multi-valued variable, and proposes a power improving refinement of the Kitagawa (2015) test procedure. Arai et al. (2022) extends the Kitagawa (2015) test to fuzzy regression discontinuity designs. Farbmacher et al. (2022) proposes a test procedure that uses random forests and classification and regression trees to find violations of the monotonicity and exclusion assumptions. Pearl (1995), Kitagawa (2021), and Kédagni and Mourifié (2020) derive joint testable implications of the instrument exclusion and independence assumption without imposing monotonicity. These testable implications take the form of a set of inequalities. Kédagni and Mourifié (2020) reformulates these inequalities as a set of conditional expectations. and develops a test procedure using the framework of Chernozhukov et al. (2013). Machado et al. (2019) discusses tests for instrument validity in the case of a binary outcome which responds monotonically with respect to the treatment. Frandsen et al. (2023) develops a test of the identifying assumptions of research designs which exploit random assignment of judges. The closest paper to ours is Mao and Sant'Anna (2020), which also uses the MTE framework of Carneiro et al. (2011), and derives index sufficiency and the nesting inequalities. Mao and Sant'Anna (2020) focuses on testing the nesting inequalities, and does not jointly test index sufficiency. The nesting inequalities are tested by considering many possible discretisations of the propensity score. We differ in that we jointly test index sufficiency and the nesting inequalities. In addition, for the nesting inequalities, rather than examining subdensities indexed by the propensity score, we propose constructing a distilled sample so that the inequalities hold for subdensities indexed by the instrument, and show the advantages of this approach.

We contribute to several empirical literatures by testing the validity of commonly used instruments and identification strategies. The college proximity instrument of Card (1993) has also been used in Cameron and Taber (2004) to estimate the effects of borrowing constraints on education, Carneiro et al. (2011) to estimate the marginal return to education, and Heckman et al. (2018) to estimate the effects of education on labour market and health outcomes. The same-sex instrument of Angrist and Evans (1998) is widely used to identify the effects of family size. Black et al. (2010),Conley and Glauber (2006), Angrist et al. (2010) and Becker et al. (2010) use this instrument to empirically investigate the existence of a quantity-quality trade-off for children, while Cruces and Galiani (2007) uses it to investigate the relationship between family size and maternal labour supply.

2 Model and Testable Implications

2.1 Setting

The data is a random sample (Y, D, X, Z), where Y is an observed outcome continuously distributed on $\mathbb{R}$, $D \in \{1, 0\}$ is an observed binary treatment status indicating treated ($D = 1$) or untreated ($D = 0$), $X \in \mathcal{X} \subset \mathbb{R}^{k_X}$ is a vector of individual pretreatment observable covariates, and $Z \in \mathcal{Z} \subset \mathbb{R}^{k_Z}$ is a vector of instrumental variables. For the moment, we impose no restrictions on the instrument. Let $\{Y_{dz}\}_{d\in\{1,0\},z\in\mathcal{Z}}$ be the set of potential outcomes indexed by treatment status $d \in \{1, 0\}$ and instrument value $z \in \mathcal{Z}$. The observed outcome Y can be written as

$$Y = Y_{1Z}D + Y_{0Z}(1 - D).$$

We write a selection equation of unconstrained form as

$$D = 1\{\mu(X, Z, V) \geq 0\}.$$

We interpret this selection equation as follows. With V fixed at v, $D_{xz} = 1\{\mu(x, z, v) \geq 0\}$ generates the counterfactual selection response of an individual whose pre-treatment characteristics X and instrument Z are exogenously set to x and z.

The following three assumptions identify the marginal treatment effect (MTE):

MTE Identification Assumptions (Heckman and Vytlacil (2005)):

(A1) *Instrument exclusion restriction*: Potential outcomes do not depend on the instrument in the sense $Y_{1z} = Y_1$ and $Y_{0z} = Y_0$ for all $z \in \mathcal{Z}$ with probability one.

(A2) *Random assingnment*: $(\{Y_{1z}\}_{z\in\mathcal{Z}}, \{Y_{0z}\}_{z\in\mathcal{Z}}, V)$ are statistically independent of Z conditional on X, and V is statistically independent of X.

(A3) *Instrument monotonicity*: The latent utility in the selection equation satisfies $\mu(x, z, v) = \tilde{\mu}(x, z) - v$ with $v \in \mathbb{R}$ having a continuous distribution given $X = x$ for any x.

The instrument exclusion restriction (A1) precludes the instrument from having a direct causal impact on the outcome for any member of the population. With this restriction imposed, an individual's potential outcomes are reduced to a pair indexed only by their treatment status, i.e, Y_1 denotes their potential outcome with treatment and Y_0 their potential outcome without treatment. The observed outcome Y can then be written as $Y = Y_1D + Y_0(1 - D)$.

Assumption (A2) states that, conditional on X, the instrument Z is assigned without reference to underlying potential outcomes and unobserved heterogeneity in selection response. This corresponds to the conventional assumption of instrument exogeneity. In the heterogeneous treatment effect model, however, it should be noted that instrument exogeneity takes the form of joint statistical independence of potential outcomes and selection heterogeneities, rather than the zero-correlation exogeneity restriction. Assumption (A3) states that a hypothetical change in the instrument from z' to z can induce some individuals to opt in to treatment or some individuals to opt out of treatment, but never both simultaneously. Equivalently, in the terminology of Imbens and Angrist (1994), there is a sub-population of compliers associated with this hypothetical change, but no sub-population of defiers. Instrument monotonicity of the form (A3) allows for multiple instrumental variables. Mogstad et al. (2020) argue that, in the presence of multiple instruments, (A3) implies severe restrictions on individual selection responses. Our test procedure allows for multiple instruments and can be used to assess a necessary testable implication of (A3) jointly with the exclusion and random assignment restrictions. Section 3.2.2 discusses in detail the implications of multiple instruments for instrument validity testing.

Following Heckman and Vytlacil (2001a, 2005), assumptions (A1) - (A3) imply a heterogeneous treatment effect model of the following form:

$$\begin{cases} Y_1 = m_1(X) + \tilde{U}_1; \\ Y_0 = m_0(X) + \tilde{U}_0; \\ Y = DY_1 + (1-D)Y_0; \\ D = \mathbb{1}\{p(X,Z) - U_D \geq 0\}, \end{cases} \tag{1}$$

with

$$(Y_1, Y_0, U_D) \perp Z|X \text{ and } U_D \perp X. \tag{2}$$

Here, $E[Y_1|X = x] = m_1(x)$ and $E[Y_0|X = x] = m_0(x)$ are regression equations of potential outcomes (Y_1, Y_0) on the conditioning covariates, $\tilde{U}_1$ and $\tilde{U}_0$ are unobserved heterogeneities in the potential outcomes (Y_1, Y_0), $p(x,z) = \Pr(D = 1|X = x, Z = z)$ is the propensity score, and $U_D = F_V(V)$ is uniformly distributed on $[0,1]$,independent of (X,Z), and by the independence between V and (X,Z) assumed in (A2), the value of U_D is comparable across different covariate values.

2.2 Semiparametric Testable Implications

The MTE identifying assumptions introduced above involve counterfactual variables that are never jointly observed for the same individual, so cannot be directly verified using the distribution of observables. However, *necessary* testable implications have been established. These conditions

relate to the distribution of observables, so may be examined empirically. They consist of *single index sufficiency* and *nesting inequalities*. We present a proof in Appendix A for completeness.

Proposition 1 *Assume (A1) - (A3) and that, conditional on $X = x$, the propensity score $p(X, Z)$ has a nondegenerate distribution. Then, at any such x, the following two conditions hold:*
(i) ***Index sufficiency****: The conditional distribution of (Y, D) given (X, Z) depends on Z only through the propensity score $p(X, Z)$, i.e., for any measurable set $A \subset \mathcal{Y}$ and $d \in \{0, 1\}$,*

$$\Pr(Y \in A, D = d | X = x, Z = z) = \Pr(Y \in A, D = d | p(X, Z) = p, X = x) \tag{3}$$

holds for any (x, z) with $p = p(x, z)$.
(ii) ***Nesting inequalities****: The following inequalities hold for every $p \geq p'$ in the support of $p(X, Z)$ and every measurable subset $A \subset \mathcal{Y}$:*

$$\begin{aligned} &\Pr(Y \in A, D = 1 | p(X, Z) = p, X = x) \geq \Pr(Y \in A, D = 1 | p(X, Z) = p', X = x), \\ &\Pr(Y \in A, D = 0 | p(X, Z) = p, X = x) \leq \Pr(Y \in A, D = 0 | p(X, Z) = p', X = x). \end{aligned} \tag{4}$$

Condition (i), index sufficiency, restricts the conditional distribution of (Y, D) given (X, Z) to depend on Z through the propensity score $p(X, Z)$ only. That is, if there are multiple values of Z, say $z \neq z' \in \mathcal{Z}$, that yield the same value of the propensity score given $X = x$, the conditional distributions of (Y, D) given $(X = x, Z = z)$ and $(X = x, Z = z')$ must be identical. Condition (ii) provides distributional monotonicity inequalities among the conditional distributions of (Y, D) given (p, x) with respect to $p = p(X, Z)$. This corresponds to the testable monotonicity shown by Heckman and Vytlacil (2005). The difference between the two sides of (4) can be expressed as

$$\begin{aligned} & \Pr(Y_1 \in A, p' < U_D \leq p | X = x) \\ = \; & \Pr(Y_1 \in A | p' < U_D \leq p, X = x) \times \Pr\left(p' < U_D \leq p | X = x\right). \end{aligned}$$

Individuals whose selection heterogeneity U_D falls in $(p', p]$ can be viewed as compliers conditional on $X = x$. The nesting inequality (4) imposes non-negativity of the treated outcome probability density function for these conditional compliers. Detecting any violation of the conditions of Proposition 1 allows the joint restrictions (A1) - (A3) to be refuted. Neither of these testable implications restrict the support of the instrument Z, i.e., Z can be discrete, continuous, or multi-dimensional.

The two testable implications shown in Proposition 1 are distinct and assess different aspects of the distribution of observables. Holding X fixed, index sufficiency compares conditional distributions of (Y, D) given $(p(X, Z), X)$ at a *fixed* value of the propensity score, while the nesting inequalities compare conditional distributions across *different* values of the propensity score. Consequently, there are scenarios in which one testable implication is useful for assessing instrument

validity but the other is not. For instance, if Z is a scalar and $p(x,z)$ is strictly monotonic in z at every x, instrument validity cannot be assessed through index sufficiency since their is no variation in the instrument conditional on $(p(X,Z),X)$, whereas it can be assessed using the nesting inequalities. Conversely, when the propensity score does not vary with the instrument conditional on X (i.e. the instrument is irrelevant), index sufficiency does have content for assessing instrument validity while the nesting inequalities do not. As such, the two testable implications are complementary.

When the dimension of X is not small, nonparametrically testing the implications listed in Proposition 1 is challenging. Kitagawa (2015) proposes testing the inequalities of Proposition 1 (ii) with a Kolmogorov-Smirnov type test statistic where a supremum is taken over a large class of instrument functions defined on the product space of Y and X. Similar to optimization in the empirical risk minimizing classification problem, the computational complexity of calculating the supremum increases with the dimension of X. In a sample of N observations with k_X conditioning covariates, it can be shown that finding the supremum requires up to $N^{2(k_X+1)}$ function evaluations. Due to this computational barrier, the finite sample power of this test has yet to be characterised, and it has rarely been implemented in practice.

We develop a test for instrument validity that circumvents these implementation issues. This test *jointly* assesses index sufficiency and the nesting inequalities. In addition, the test procedure can accommodate continuous instruments.

To achieve this, we adopt the semiparametric restrictions of Carneiro et al. (2011), which have been used in many empirical studies of marginal treatment effects.

(A4) *Linear Functional Form*: The regression equations for potential outcomes have the parametric form

$$\begin{cases} Y_1 = \alpha_1 + X'\theta_1 + \tilde{U}_1; \\ Y_0 = \alpha_0 + X'\theta_0 + \tilde{U}_0. \end{cases} \tag{5}$$

In addition, strengthen the exogeneity restriction (A2) to:

(A5) *Strong Exogeneity*: Let $(\tilde{U}_1, \tilde{U}_0)$ be the residuals of the potential outcome regression equations defined in (5). $(\tilde{U}_1, \tilde{U}_0, U_D)$ is independent of (X,Z).

(A4) restricts the conditional mean potential outcomes to depend linearly on the covariates. (A5) implies that the mean zero unobserved terms $\left(\tilde{U}_1, \tilde{U}_0\right)$ are independent of the covariates. (A4) and (A5) jointly imply that covariates affect the potential outcomes only through their conditional means. Note that this assumption does not imply that the treatment effect is homogeneous, since the individual causal effect $Y_1 - Y_0$ remains random conditional on X. Note also that, while only

conditional mean treatment effects may depend on X, the distribution of treatment effects remains otherwise unconstrained.

Under (A1) - (A3), the MTE at $X = x$ and $U_D = p \in \mathcal{P}$ is nonparametrically identified by

$$\frac{\partial}{\partial p} E\left[Y | p\left(X, Z\right) = p, X = x\right].$$

Carneiro et al. (2011) shows that imposing the parametric functional forms of (A4) together with strengthening the exogeneity assumption of (A2) to (A5), yields the following partially linear regression equation for the observed outcome Y:

$$E\left[Y | p\left(X, Z\right) = p, X = x\right] = x'\theta_0 + px'\left(\theta_1 - \theta_0\right) + \phi\left(p\right), \tag{6}$$

where $\phi\left(\cdot\right)$ is a unknown function of the propensity score, which absorbs the intercept term $p\alpha_1 + \left(1 - p\right)\alpha_0$. As a result, the MTE at $X = x$ and $U_D = p$ is identified by

$$MTE(x, p) \equiv E[Y_1 - Y_0 | X = x, U_D = p] = x'\left(\theta_1 - \theta_0\right) + \phi'\left(p\right). \tag{7}$$

Since the only nonparametric component in this regression equation is $\phi\left(\cdot\right)$, which is a function of the scalar-valued propensity score, the computational complexity of estimation does not explode as the dimension of X increases. Carneiro et al. (2011) argues that this is a practical advantage of the functional form specification (A4) and the stronger instrument exogeneity assumption (A5).

Since the partially linear regression (6) identifies (θ_1, θ_0), it is possible to compute

$$U_1 \equiv \alpha_1 + \tilde{U}_1 = Y_1 - X'\theta_1$$

for every treated observation, and

$$U_0 \equiv \alpha_0 + \tilde{U}_0 = Y_0 - X'\theta_0$$

for every untreated observation.

The joint restrictions (A1), (A3), (A4) and (A5) simplify not only the estimation of MTE, but also the testable implications. To see how, consider the distribution of U_1 conditional on $(D = 1, p\left(X, Z\right) = p, X = x)$. Under (A1), (A3) and (A4),

$$\begin{aligned} U_1 | \left(D = 1, p\left(X, Z\right) = p, X = x\right) &\sim U_1 | \left(U_D \leq p, p\left(X, Z\right) = p, X = x\right) \\ &\sim U_1 | \left(U_D \leq p\right), \end{aligned} \tag{8}$$

and similarly,

$$\begin{aligned} U_0 | \left(D = 0, p\left(X, Z\right) = p, X = x\right) &\sim U_0 | \left(U_D > p, p\left(X, Z\right) = p, X = x\right) \\ &\sim U_0 | \left(U_D > p\right). \end{aligned} \tag{9}$$

That is, for $d = 0, 1$, the distribution of the residual, $U_d = Y_d - X'\theta_d$, conditional on $(D = d, p(X, Z), X)$ depends only on $p(X, Z)$. Exploiting this single index sufficiency for the distributions of U_1 and U_0, we obtain the following necessary testable implications for the joint restrictions (A1), (A3), (A4) and (A5).

Proposition 2 *Let $U = DU_1 + (1 - D)U_0$. Assume the propensity score $p(X, Z)$ has a nondegenerate distribution. If (A1), (A3), (A4) and (A5) hold, then the following two conditions hold:*
(i) ***Index sufficiency****: The conditional distribution of (U, D) given (X, Z) depends only on the propensity score $p(X, Z)$, i.e., for any measurable set $A \subset \mathbb{R}$ and $d \in \{0, 1\}$*

$$\Pr(U \in A, D = d | X = x, Z = z) = \Pr(U \in A, D = d | p(X, Z) = p) \tag{10}$$

holds for any (x, z) where $p = p(x, z)$.
(ii) ***Nesting inequalities****: For every $p \geq p'$ in the support of the propensity score distribution and every measurable subset $A \subset \mathcal{Y}$:*

$$\begin{aligned} &\Pr(U \in A, D = 1 | p(X, Z) = p) \geq \Pr(U \in A, D = 1 | p(X, Z) = p'), \\ &\Pr(U \in A, D = 0 | p(X, Z) = p) \leq \Pr(U \in A, D = 0 | p(X, Z) = p'). \end{aligned} \tag{11}$$

We refer to the joint restrictions of (A1), (A3), (A4), and (A5) as *instrument validity* and base our test on the testable implications of Proposition 2. Compared to Proposition 1, Proposition 2 replaces the outcome variable Y with the partial residual U, and removes the requirement to condition on X. Index sufficiency is strengthened such that the distribution of (U, D) depends on (X, Z) only through the propensity score $p(X, Z)$. The nesting inequalities are strengthened to imply distributional monotonicity with respect to changes in $p(X, Z)$ regardless of the underlying variation in (X, Z). Theorem 2.2 of Mao and Sant'Anna (2020) also derives these testable implications and shows that they are sharp.

As the testable implications of Proposition 2 follow from assumptions (A1), (A3), (A4) and (A5), any test based on them is a joint test of these assumptions. That is, in addition to the conventional exclusion, random assignment and monotonicity, we test the functional form assumption (A4) and strong exogenienty assumption (A5). The testable implications of Proposition 2 will be violated if the functional form (5) is misspecified or if the restrictions on MTE heterogeneity with respect to observable characteristics do not hold, i.e., MTEs are not additive separable as in (7). In addition, when the propensity score is estimated, misspecification of the propensity score function may also lead to a rejection of the null.

As index sufficiency holds for all (X, Z) it leads to a large number of inequalities and testing all of them may be impractical. In this paper, we assume that the strong exogeneity of X holds

and focus on testing the validity of Z. As such, we consider testing that the invariance of the joint distribution of (U, D) given $P(X, Z)$ for different values of Z.

2.3 Potential Difficulties Detecting Violation of Instrument Validity

In this section, we discuss how the conditional distribution of the propensity score given Z affects the usefulness of the testable implications characterized in Proposition 2. We use the case of a single binary instrument to illustrate.

Index sufficiency (10) can be tested by examining whether the difference

$$\Pr(U \in A, D = d | p(X, Z) = p, Z = 1) - \Pr(U \in A, D = d | p(X, Z) = p, Z = 0) \tag{12}$$

is zero.

For a data generating process to have empirical content to assess index sufficiency, there must exist some p where observations with both $Z = 0$ and $Z = 1$ are available. This occurs if we can find covariate values $x \neq x'$ such that $p(x, 1) = p(x', 0)$. If no such p exists, then for all p either $\Pr(p(X, Z) = p, Z = 0) = 0$ or $\Pr(p(X, Z) = p, Z = 1) = 0$ and (12) cannot be calculated. Hence, the practical relevance of (10) is determined by the degree of overlap between the propensity score distributions conditional on $Z = 0$ and $Z = 1$.

Now consider the nesting inequalities (11). These can be viewed as analogous to those of Kitagawa (2015) with the observed outcome Y replaced by the outcome residual U and the instrument replaced by the propensity score $p(X, Z)$. If the propensity score has continuous variation, a direct approach to testing the nesting inequalities would invoke nonparametric smoothing or coarsening techniques. If the propensity score were coarsened, e.g., by binning, the test procedure with no conditioning covariates and a discrete instrument of Kitagawa (2015) could be applied. Although this approach would attain an asymptotically valid test size,[1] there are circumstances where conditioning on the propensity score has a detrimental effect on power.[2]

Maintain the simplifying assumption of a single binary instrument. Assume that the conditioning covariates X are statistically independent of the unobservables $(\tilde{U}_1, \tilde{U}_0, U_D)$, as required by (A5). Furthermore, assume that they are also independent of Z. Consider the following simple violation of the exclusion restriction (A1) where the residuals of the potential outcome regression for $D = 1$ depend on Z

$$U_1 = \alpha_1 + \nu Z + \tilde{U}_1, \tag{13}$$

[1] Assuming the estimation errors in U and $p(X, Z)$ asymptotically vanish

[2] Instead of coarsening, one can consider smoothing the conditional distributions of (U, D) with respect to the conditioning propensity scores. In our exposition, we focus on coarsening as it is more straightforward to analyze. Smoothing can be viewed as a more flexible form of coarsening with a non-degenerate weighting function, and the power loss discussed in this section would also apply with smoothing.

with $\nu \neq 0$. Note that, since U_1 is a sum of quantities that are independent of X, it will be independent of X.

If $\Pr(Z = 1|p(X, Z) = p)$ is bounded away from 0 and 1, the additive separability of the selection equation and the independence of X permits the following decomposition:

$$\begin{aligned}
\Pr(U \in A, D = 1|p(X, Z) = p) &= \Pr(U \in A, D = 1|p(X, Z) = p, Z = 0) \Pr(Z = 0|p(X, Z) = p) \\
&\quad + \Pr(U \in A, D = 1|p(X, Z) = p, Z = 1) \Pr(Z = 1|p(X, Z) = p) \\
&= \Pr(U_1 \in A, U_D \leq p|p(X, Z) = p, Z = 0) \Pr(Z = 0|p(X, Z) = p) \\
&\quad + \Pr(U_1 \in A, U_D \leq p|p(X, Z) = p, Z = 1) \Pr(Z = 1|p(X, Z) = p) \\
&= \Pr(U_1 \in A, U_D \leq p|Z = 0) \Pr(Z = 0|p(X, Z) = p) \\
&\quad + \Pr(U_1 \in A, U_D \leq p|Z = 1) \Pr(Z = 1|p(X, Z) = p). \qquad (14)
\end{aligned}$$

The first equality states that, for a given p where $\Pr(Z = 1|p(X, Z) = p)$ is bounded away from 0 and 1, we have a mixture of observations with $Z = 0$ and $Z = 1$ with conditioning covariates implicitly adjusting to hold the propensity score constant. The second equality uses that the event $(U \in A, D = 1)$ is equivalent to $(U_1 \in A, U_D \leq p)$. To see the third, note that conditioning on $(p(X, Z) = p, Z = z)$ is equivalent to conditioning on $Z = z$ and $X \in \{x|p(X = x, Z = z) = p\}$. X is independent of U_D and U_1, so $X \in \{x|p(X = x, Z = z) = p\}$ can be dropped from the conditioning set.

Assume that the violation of the exclusion restriction assumption leads there to be some $A \subset \mathcal{Y}$ and $p' < p$ where the following inequality holds:

$$\Pr(U_1 \in A, U_D \leq p'|Z = 0) - \Pr(U_1 \in A, U_D \leq p|Z = 1) > 0. \qquad (15)$$

Notice that if the instrument were valid the reverse inequality would hold.

To see how (15) can drive a violation of the first nesting inequality (11), apply the decomposition (14) to the difference between the left and right side of (11)

$$\begin{aligned}
&\Pr(U \in A, D = 1|p') - \Pr(U \in A, D = 1|p) \qquad (16) \\
&= \left[(\Pr(Z = 1|p) - \Pr(Z = 1|p')\right] \left[\Pr(U_1 \in A, U_D \leq p'|Z = 0) - \Pr(U_1 \in A, U_D \leq p|Z = 1)\right] \\
&\quad - \Pr(Z = 0|p) \cdot \Pr(U_1 \in A, p' < U_D \leq p|Z = 0) - \Pr(Z = 1|p') \cdot \Pr(U_1 \in A, p' < U_D \leq p|Z = 1).
\end{aligned}$$

The violation (15) appears in the second line of (16) multiplied by $(\Pr(Z = 1|p) - \Pr(Z = 1|p')$. The remaining terms in the third line are all negative. In order for (15) to result in a violation of the nesting inequality (i.e. for (16) to be positive), it is desirable that $\Pr(Z = 1|p) - \Pr(Z = 1|p')$ is of large magnitude.

This magnitude depends on the strength of the relationship between the covariates X and the propensity score. If variation in the propensity score is driven mainly by X, $\Pr(Z = 1|p)$ will change little in p, and the magnitude of $(\Pr(Z = 1|p) - \Pr(Z = 1|p')$ will be small. This will result in limited detectability of violations of instrument validity through the nesting inequality (16).

In the Monte Carlo studies in Section 4, we find that tests of the nesting inequalities that directly examine the monotonicity of $\Pr(U \in A, D = 1|p)$ in p suffer from low power if variation in the propensity score is driven by the conditioning covariates X.

2.4 Distillation

To improve power when testing nesting inequalities, we consider conditioning on Z rather than conditioning on the propensity score. Consider the difference between the probabilities of $\{U \in A, D = 1\}$ conditional on Z,

$$\Pr(U \in A, D = 1|Z = 0) - \Pr(U \in A, D = 1|Z = 1) \tag{17}$$
$$= \int_0^1 \left[\Pr(U \in A, D = 1|p(X, Z) = p, Z = 0)f(p|Z = 0) - \Pr(U \in A, D = 1|p(X, Z) = p, Z = 1)f(p|Z = 1)\right] dp,$$

where $f(p|Z = z)$ denotes the conditional probability density (or probability mass function) of the propensity score $p(X, Z)$ given $Z = z$.

A positive sign for (17) can imply three possibilities:

Case 1: If the distribution of $p(X, Z)|Z = 0$ is first-order stochastically dominated by the distribution of $p(X, Z)|Z = 1$, a positive sign implies index sufficiency is violated, the nesting inequalities are violated, or both. If index sufficiency holds it implies violation of the nesting inequalities, i.e., monotonicity of $\Pr(U \in A, D = 1|p(X, Z) = p)$ in p fails for some value of $p \in (0, 1)$ with a positive measure.

Case 2: If $f(p|Z = 0) = f(p|Z = 1)$, a positive sign for (17) implies violation of index sufficiency at some value of $p \in (0, 1)$ with a positive measure.

Case 3: If the distribution of $p(X, Z)|Z = 0$ is not first-order stochastically dominated by the distribution of $p(X, Z)|Z = 1$, a positive sign does *not* imply violation of index sufficiency or the nesting inequalities. i.e., (17) can be positive even under the null.

Hence, (17) is informative for detecting violations of instrument validity only if first-order stochastic ordering holds between the distributions of $p(X, Z)|Z = 0$ and $p(X, Z)|Z = 1$. First-order stochastic dominance holds, for instance, if the propensity score is given by a probit function $\Phi(X'\delta+\gamma Z)$ with $\gamma > 0$, and X and Z are uncorrelated. However, first-order stochastic dominance is not implied

by (A1), (A3), (A4) and (A5). If $\gamma > 0$, $\delta > 0$ and X and Z are negatively correlated, first-order stochastic dominance can fail even if (A1) - (A5) hold. Therefore, in general, a positive value of (17) cannot be taken as evidence against instrument validity.

Case 3 renders positive values of (17) inconclusive about violations of instrument validity. To rule out Case 3, first-order stochastic dominance can be checked as part of the test procedure. In cases where it does not hold, we consider trimming the sample so that the conditional distribution of the propensity score $p(X, Z)$ given Z is stochastically monotonic in Z. We then assess the sign of an inequality analogous to (17) using the potentially trimmed sample. We refer to this process as *distillation*.

Given an instrument, $Z \in \mathcal{Z} \subset \mathbb{R}$, define a binary indicator $S_1 \in \{0, 1\}$ such that $S_1 = 1$ indicates that an observation is included in the sample used for testing the nesting inequalities, and $S_1 = 0$ indicates that the observation is discarded. The distribution of $S_1 \in \{0, 1\}$ can depend on (X, Z), but we require that S_1 is independent of (U_1, U_0, U_D).

Under the assumptions of Proposition 2, index sufficiency of the propensity score for the distribution of (U_d, U_D) implies

$$\begin{aligned}
&\Pr(U \in A, D = 1 | Z = z, S_1 = 1) \\
&= \int_0^1 \Pr(U_1 \in A, D = 1 | p(X, Z) = p, Z = z, S_1 = 1) dP_{p|ZS_1}(p|z, S_1 = 1) \\
&= \int_0^1 \Pr(U_1 \in A, U_D \leq p | p(X, Z) = p, Z = z, S = 1) dP_{p|ZS_1}(p|z, S_1 = 1) \\
&= \int_0^1 \Pr(U_1 \in A, U_D \leq p) dP_{p|ZS_1}(p|z, S_1 = 1), \qquad (18)
\end{aligned}$$

where $P_{p|ZS_1}$ is the conditional distribution of the propensity score $p(X, Z)$ given $Z = z$ and $S_1 = 1$. The third equality follows by Assumption (A5) and the independence of S_1 and (U_1, U_0, U_D). Since the integrand in (18) is monotonic in p by Proposition 2, $\Pr(U \in A, D = 1 | Z = z, S_1 = 1)$ is monotonically increasing in z if $P_{p|ZS_1}(\cdot|z, S_1 = 1)$ is monotonic in z in terms of first-order stochastic dominance. Similarly, the stochastic monotonicity of $P_{p|ZS_1}(\cdot|z, S_1 = 1)$ in z implies

$$\Pr(U \in A, D = 0 | Z = z, S_1 = 1) = \int_0^1 \Pr(U_0 \in A, U_D > p) dP_{p|ZS_1}(p|z, S_1 = 1)$$

is monotonically decreasing in z.

We now formally define distillation

Definition 1 (Distillation) *Let $S_1 \in \{0, 1\}$ be an indicator for inclusion in the sample. (Z, S_1) is a distillation rule if $(U_1, U_0, U_D) \perp S_1$ and the conditional distribution of the propensity score given $\{Z, S_1\}$, $P_{p|ZS_1}(\cdot|z, S_1 = 1)$, is stochastically increasing in $z \in \mathcal{Z}$ in terms of the first-order stochastic dominance.*

As discussed above, the nesting inequalities shown in Proposition 2 remain valid even when we replace the propensity score $p(X, Z)$ with the instrument Z and restrict the sample to a subsample with $S_1 = 1$. We hence obtain the next proposition.

Proposition 3 *Suppose the assumptions of Proposition 2 holds. Let $(Z, S_1) \in \mathcal{Z} \times \{0, 1\}$ be a distilled sample as defined in Definition 1. For any $z \geq z' \in \mathcal{Z}$ and every measurable set $A \subset \mathcal{Y}$,*

$$\begin{aligned} &\Pr(U \in A, D = 1 | Z = z, S_1 = 1) \geq \Pr(U \in A, D = 1 | Z = z', S_1 = 1), \\ &\Pr(U \in A, D = 0 | Z = z, S_1 = 1) \leq \Pr(U \in A, D = 0 | Z = z', S_1 = 1). \end{aligned} \tag{19}$$

Proposition 3 shows that a testable implication exists in terms of the subdensities conditional on Z provided that the sample is appropriately trimmed. However, it does not show any benefits of this approach relative to directly testing the nesting inequalities in Proposition 2 with a coarsened propensity score. In the proposition below, we characterise a class of processes and a restriction on trimming such that distillation will outperform a test of the nesting inequalities for observations binned by propensity score values in terms of detecting violations of instrument validity.

Proposition 4 *Let P^- and P^+ be disjoint intervals with the lower bound of P^+ weakly greater than the upper bound of P^- such that $\Pr(p(X, Z) \in P^-, Z = z)$ and $\Pr(p(X, Z) \in P^+, Z = z)$ are positive for $z = 0, 1$. Consider the differences*

$$\Pr(U \in A, D = 1 | p \in P^-) - \Pr(U \in A, D = 1 | p \in P^+), \tag{20}$$

$$\Pr(U \in A, D = 0 | p \in P^+) - \Pr(U \in A, D = 0 | p \in P^-), \tag{21}$$

and

$$\Pr(U \in A, D = 1 | Z = 0, S_1 = 1, p \in P^- \cup P^+) - \Pr(U \in A, D = 1 | Z = 1, S_1 = 1, p \in P^- \cup P^+)), \tag{22}$$

$$\Pr(U \in A, D = 0 | Z = 1, S_1 = 1, p \in P^- \cup P^+) - \Pr(U \in A, D = 0 | Z = 0, S_1 = 1, p \in P^- \cup P^+). \tag{23}$$

Assume that the selection indicator S_1 is chosen so that the following hold
(i) ***Valid Distillation****: S_1 satisfies the conditions of Definition 1*
(ii) ***Trimming Rule****: $S_1 = 1$ for all observations with $Z = 0$ and $p \in P^-$, and all observations with $Z = 1$ and $p \in P^+$.*

Let $\mathbb{P}$ be the class of data generating processes satisfying
(iii) ***Labeling****: $Z = 0$ and $Z = 1$ are chosen such that*

$$\Pr(p \in P^+ | Z = 0, p \in P^- \cup P^+) \leq \Pr(p \in P^+ | Z = 1, p \in P^- \cup P^+) \tag{24}$$

(iv) ***Conditional Nesting****: For every $p \geq p'$ in the support of the propensity score distribution, every measurable subset $A \subset \mathcal{Y}$, and $z \in \{0,1\}$:*

$$\Pr(U \in A, D = 1 | p(X,Z) = p, Z = z) \geq \Pr(U \in A, D = 1 | p(X,Z) = p', Z = z), \quad (25)$$

$$\Pr(U \in A, D = 0 | p(X,Z) = p, Z = z) \leq \Pr(U \in A, D = 0 | p(X,Z) = p', Z = z). \quad (26)$$

The following statements hold:

1. *For all data generating processes in $\mathbb{P}$, for any measurable subset $A \subset \mathbb{R}$ and P^-, P^+ where (20) is positive*

$$\Pr(U \in A, D = 1 | Z = 0, S_1 = 1, p \in P^- \cup P^+) -$$
$$\Pr(U \in A, D = 1 | Z = 1, S_1 = 1, p \in P^- \cup P^+) \geq$$
$$\Pr(U \in A, D = 1 | p \in P^-) - \Pr(U \in A, D = 1 | p \in P^+).$$

   Similarly, for any measurable subset $A \subset \mathbb{R}$ where (21) is positive

$$\Pr(U \in A, D = 0 | Z = 1, S_1 = 1, p \in P^- \cup P^+) -$$
$$\Pr(U \in A, D = 0 | Z = 0, S_1 = 1, p \in P^- \cup P^+) \geq$$
$$\Pr(U \in A, D = 0 | p \in P^+) - \Pr(U \in A, D = 0 | p \in P^-).$$

2. *There exist data generating processes in $\mathbb{P}$ where, given P^- and P^+, a positive value of (22) for some $A \subset \mathbb{R}$ does not imply a positive value of (20) for the same $A \subset \mathbb{R}$. Similarly, there exist processes in $\mathbb{P}$ where, given P^- and P^+, a positive value of* (23) *for some $A \subset \mathbb{R}$ does not imply a positive value of* (21)*.*

3. *Given P^- and P^+, there exist data generating processes in $\mathbb{P}$ such that distillation can detect violation of instrument validity while the original nesting inequalities cannot, i.e., (22) or (23) is positive at some $A \subset \mathbb{R}$, while (20) and (21) are less than or equal to zero for any $A \subset \mathbb{R}$.*

A proof is presented in Appendix B.

Regarding the construction of S_1, condition (ii) requires that all observations with $Z = 0$ and $p \in P^-$ are retained, and all observations with $Z = 1$ and $p \in P^+$ are retained. On the other hand, no restriction is placed on the trimming of observations with $p \in P^+$ and $Z = 0$ or observations with $p \in P^-$ and $Z = 1$. In other words, first order stochastic dominance should be attained solely through trimming observations with $Z = 0$ and a high propensity score, and observations

with $Z = 1$ and a low propensity score. In Section 2.6 we present an algorithm that satisfies these conditions.

The inequality (24) can always be satisfied by setting $Z = 1$ for the instrument value with a higher proportion of observations in P^+. Condition (iv) imposes restrictions on the dependence between the partial residuals and Z. For example, condition (iv) follows if Z affects the density of partial residuals, and this relationship is independent of X i.e. the partial residuals admit the representation

$$U_d = \nu Z + \tilde{U}_d \quad \tilde{U}_d \perp (X, Z).$$

The first statement of Proposition 4 says that, for any A where a nesting inequality calculated using propensity score bins is positive, the nesting inequality calculated using a distilled sample will also be positive and at least as large in magnitude. That is, any violation of instrument validity that can be detected by a comparison of the data across the propensity score bins can also be detected by a comparison of the data across different instrument values, and the magnitude of the violation can be larger when comparing across instrument values. The second statement says that, for some A where the nesting inequality calculated using a distilled sample is positive, the nesting inequality calculated with propensity score bins is not positive. In other words, for such A the violation becomes undetectable if we compare across propensity score bins rather than instrument values. The second statement compares the nesting inequalities and distilled inequalities for a common A, while the third statement compares their maxima over A. The third statement says that there exist data generating processes where instrument validity does not hold, but this violation can only be detected by comparing across instrument values. This proposition characterises a class of data generating processes where distillation is guaranteed to outperform the coarsened propensity score approach in terms of detecting invalid instruments. Thus it provides theoretical support for implementing distillation.

Proposition 4 does not contradict the sharpness of the testable implications of Proposition 2. In Proposition 4, we compare distillation with coarsening, and coarsening discards some of the nesting inequalities to be assessed. That is, we do not check nesting inequalities for any interval of p within the coarsened intervals P^+ and P^-. This means that we do not screen out some violations of instrument validity. In terms of identification, coarsening would never be necessary, but with a finite sample, some degree of coarsening will always be required. This opens the door to the benefits of distillation seen in Proposition 4.

2.5 Example Process

Proposition 4 implies that we can more easily detect violations of instrument validity by testing the nesting inequalities with a distilled sample. To verify the practical relevance of this gain, we

perform a numerical exercise using the following process

$$
\begin{aligned}
&Y = \nu Z - \nu(1-Z) + D(X'\theta_1 + \tilde{U}_1) + (1-D)(X'\theta_0 + \tilde{U}_0),\\
&D = \mathbb{1}\{Z(X'\delta_1 + \alpha) + (1-Z)(X'\delta_0 - \alpha) + U_D \geq 0\},\\
&\Pr(Z=1) = 1/2,\\
&X \sim N(0, I),\\
&(U_D, \tilde{U}_1, \tilde{U}_0) \sim N(\mu, \Sigma),\\
&\mu = \begin{pmatrix} 0 \\ \mu_1 \\ \mu_0 \end{pmatrix} \quad \Sigma = \begin{pmatrix} 1 & \sigma_1\rho_{D,1} & \sigma_0\rho_{D,0} \\ \sigma_1\rho_{D,1} & \sigma_1^2 & \sigma_1\sigma_0\rho_{1,0} \\ \sigma_0\rho_{D,0} & \sigma_1\sigma_0\rho_{1,0} & \sigma_0^2 \end{pmatrix}.
\end{aligned}
$$

X is assumed to consist of $k_X \geq 3$ covariates. We assume $\nu \neq 0$, so the instrument has a direct causal effect on the outcome. In addition, we impose $\alpha > 0$ and $\delta_0'\delta_0 = \delta_1'\delta_1 = \delta'\delta > 0$.

We consider the asymptotic setting, with both propensity scores and the parameters θ_1 and θ_0 estimated. We assume that propensity scores are estimated consistently. Given the above specification, the distribution of propensity scores conditional on $Z = 1$ will first-order stochastically dominate the distribution conditional on $Z = 0$. Conditions (i) and (ii) of Proposition 4 can then be satisfied by including the entire sample, and condition (iii) is immediate.

Let $\hat{\theta}_1^*$ and $\hat{\theta}_0^*$ be the probability limits of the estimates of θ_1 and θ_0. We assume these estimates are obtained semiparametrically.[3] The partial residuals are

$$U = Y - DX'\hat{\theta}_1^* - (1-D)X'\hat{\theta}_0^*.$$

In Appendix C we show that condition (iv) of Proposition 4 follows immediately if these estimates are consistent i.e. $\hat{\theta}_1^* = \theta_1$ and $\hat{\theta}_0^* = \theta_0$. However, as the exclusion restriction is violated, in general $\hat{\theta}_1^* \neq \theta_1$ and $\hat{\theta}_0^* \neq \theta_0$. Proposition 7 in Appendix C shows that the effect of the asymptotic bias on the partial residual subdensities can be summarised by two constants which depend on the parameters ν, α and $\delta'\delta$ introduced above, and ρ_δ which is defined below

$$\rho_\delta \equiv \frac{\delta_0'\delta_1}{\delta'\delta}.$$

Furthermore, ν enters only as a scalar multiplier. Hence, the class of processes in this numerical exercise can be indexed by values of $\delta'\delta$ and ρ_δ. Proposition 7 further shows that the effect of the bias is the same for the density of $(U, D = 1)$ conditional on $Z = 1$ and the density of $(U, D = 0)$ conditional on $Z = 0$, and the density of $(U, D = 0)$ conditional on $Z = 1$ and the density of $(U, D = 1)$ conditional on $Z = 0$. As the conditional densities for $(U, D = 0)$ thus mirror those for $(U, D = 1)$, we present results for the $(U, D = 1)$ densities only.

[3] Specifically, we assume these estimates are obtained as described in Section 3.1.2.

The numerical exercise is as follows. μ_1 is set to 0.3, μ_0 to 0, and the elements of Σ are drawn randomly. We set $\alpha = 0.3$, $\nu = 0.2$, and loop over values of ρ_δ and $\delta'\delta$. We define P^- to include propensity scores below the median and P^+ to include propensity scores above the median. At each set of values, we calculate the violation of the nesting inequality with observations split into bins by the median propensity score as

$$\max_{A \in \mathcal{C}(\mathcal{U})} \left\{ \Pr(U \in A, D = 1 | p \in P^-) - \Pr(U \in A, D = 1 | p \in P^+) \right\}. \tag{27}$$

and the violation of the nesting inequality with a distilled sample as

$$\max_{A \in \mathcal{C}(\mathcal{U})} \left\{ \Pr(U \in A, D = 1 | Z = 0, S_1 = 1, p \in P^- \cup P^+) - \Pr(U \in A, D = 1 | Z = 1, S_1 = 1, p \in P^- \cup P^+) \right\}. \tag{28}$$

where $\mathcal{C}(\mathcal{U})$ is the set of connected intervals. We then compare these two violations. Figure 1 plots the difference between the violation of the nesting inequality with a distilled sample and the nesting inequality with propensity score bins. For most parameter values, the violation with a distilled sample is larger, with the gain largest for high values of $\delta'\delta$ and ρ_δ close to zero. Figure 2 plots the region of the parameter space where the violation with a distilled sample is larger than the violation with propensity score bins, and the region where Statement 3 of Proposition 4 holds. That is, the violation can only be detected by using a distilled sample. Testing with a distilled sample outperforms coarsening the propensity score for the majority of parameter values, and the region where Statement 3 holds is large.

[Figure 1 about here.]

[Figure 2 about here.]

2.6 A Distillation Procedure

Definition 1 provides conditions which a distilled sample must satisfy. Here we introduce an algorithm to construct a sample which satisfies these conditions. As in Proposition 5, consider data such that $p \in P^- \cup P^+$. Let i index observations. Sort observations $i = 1, ..., N$ into ascending order by propensity score, so that observation $i = 1$ has the lowest propensity score and observation $i = N$ the highest. Split ties by ordering observations with $Z = 0$ before observations with $Z = 1$. Let $S_{1,i}$ be the sample inclusion indicator for observation i. First-order stochastic dominance holds when

$$\frac{\sum_{i=1}^{j} S_{1,i} Z_i}{\sum_{i=1}^{N} S_{1,i} Z_i} \leq \frac{\sum_{i=1}^{j} S_{1,i}(1 - Z_i)}{\sum_{i=1}^{N} S_{1,i}(1 - Z_i)} \quad \text{for } j = 1, ..., N. \tag{29}$$

These constraints restrict the conditional CDF of propensity scores for $Z = 1$ to never exceed the conditional CDF for $Z = 0$.

It is immediate from (29) that

1. $S_{1,i} = 0$ for all i with $Z_i = 1$ and $p(X_i, Z_i) < \min_j(p(X_j, Z_j) : Z_j = 0)$,
2. $S_{1,i} = 0$ for all i with $Z_i = 0$ and $p(X_i, Z_i) > \max_j(p(X_j, Z_j) : Z_j = 1)$.

That is, in the trimmed sample the observation with the lowest propensity score must have $Z = 0$ and the observation with the highest propensity score must have $Z = 1$. Assume that we begin with a sample that has been trimmed in this manner, so that $Z_i = 0$ for $i = 1$ and $Z_i = 1$ for $i = N$. Let n_0 be the number of observations with $Z = 0$ and n_1 be the number with $Z = 1$. Define Δ_j to be the difference between the conditional propensity score distributions for $Z = 1$ and $Z = 0$ at point j

$$\Delta_j \equiv \frac{\sum_{i=1}^{j} Z_i}{n_1} - \frac{\sum_{i=1}^{j}(1 - Z_i)}{n_0}.$$

The algorithm is as follows

1. Set $S_{1i} = 1 \; \forall \; i$
2. Calculate Δ_j for $j = 1, ..., N$. If $\max(\Delta_j) \leq 0$, stop. If $\max(\Delta_j) > 0$, continue to steps 3 and 4.
3. Let $J^- = \{j : p_j \in P^-\}$
   (a) Find

$$j^- = \underset{j \in J^-}{\operatorname{argmax}} \, d_{1j},$$
$$d_1 = \max_{j \in J^-} d_{1j},$$

   where

$$d_{1j} = \text{ceil}\left(\frac{n_0}{n_0 - \sum_{i=1}^{j}(1 - Z_i)} n_1 \Delta_j \right),$$

   (b) Loop over $j = 1, ..., j^-$. If $Z_j = 1$
      i. Calculate

$$\delta_{1j} = \frac{\sum_{i=1}^{j} S_{1,i} Z_i}{n_1 - d_1} - \frac{\sum_{i=1}^{j}(1 - Z_i)}{n_0}.$$

      ii. If $\delta_{1j} > 0$, set $S_{1j} = 0$
4. Let $J^+ = \{j : p_j \in P^+\}$

(a) Find

$$j^+ = \underset{j \in J^+}{\text{argmax}}\, d_{0j},$$

$$d_0 = \max_{j \in J^+} d_{0j},$$

where

$$d_{0j} = \text{ceil}\left(n_0 \frac{n_1 - d_1}{\sum_{i=1}^{j} Z_i - d_1} \left(\Delta_j - \frac{d_1}{n_1 - d_1} \frac{n_1 - \sum_{i=1}^{j} Z_i}{n_1} \right) \right).$$

(b) Loop over $j = N, ..., j^+ + 1$. If $Z_j = 0$

i. Calculate

$$\delta_{0j} = \frac{\sum_{i=j}^{N} S_{1,i}(1 - Z_i)}{n_0 - d_0} - \frac{\sum_{i=j}^{N} Z_i}{n_1 - d_1}.$$

ii. If $\delta_{0j} > 0$, set $S_{1j} = 0$

The algorithm begins by including all possible observations. In step 2, it checks if first-order stochastic dominance holds with all possible observations included. If stochastic dominance would not hold, we proceed to steps 3 and 4. Step 3 considers observations with $p \in P^-$. In step 3a, the algorithm calculates for each index j the number of observations with $i \leq j$ and $Z_i = 1$ that need to be trimmed for there to be no violation of stochastic dominance at $p(X_j, Z_j)$. Taking the maximum gives the overall number of observations that must be trimmed, d_1, and the index by which this trimming must take place, j^-. Step 3b then selects the observations to be trimmed. Looping over j from 1 to j^-, an observation is trimmed whenever its inclusion would lead the trimmed propensity score distribution for $Z = 1$ to cross the propensity score distribution for $Z = 0$. Step 4 is analogous to step 3, but instead considers observations with $p \in P^+$. Step 4a calculates the number of observations with $i \geq j$ and $Z_i = 1$ that need to be trimmed so that there is no violation of stochastic dominance at $p(X_j, Z_j)$. Taking the maximum of this gives a number of observations to trim, d_0, and an index above which this trimming must take place j^+. In step 4b, an observation is trimmed whenever its inclusion would lead the trimmed complementary propensity score distribution for $Z = 0$ to cross the trimmed complementary distribution for $Z = 0$.

Figure 3 presents an example with artificial data. Panel a plots the empirical distributions of the propensity scores conditional on Z. First-order stochastic dominance is violated as the distribution for $Z = 1$ lies above the distribution for $Z = 0$ for some p. Panel b plots d_{1j} for each $j \in J^-$. In step 3a, d_1 and j^- are found by taking the maximum. In this case, $d_1 = 6$. Panel c shows step 3b. An observation is trimmed whenever the trimmed propensity score distribution for $Z = 1$ would rise above the distribution for $Z = 0$. Panel d plots d_{0j} for each $j \in J^+$. In step 4a, d_0 and j^+ are found by taking the maximum. In this case $d_0 = 11$. Panel e shows step 4b. Here an observation is

trimmed whenever the trimmed complementary propensity score distribution for $Z = 0$ would rise above the trimmed complementary distribution for $Z = 1$. Panel f shows the resulting trimmed distributions, where first-order stochastic dominance holds as required by distillation.

[Figure 3 about here.]

This algorithm satisfies Definition 1 as first order stochastic dominance is guaranteed to hold for the trimmed sample, and trimming is performed based on propensity score values only. In addition, as all observations with $Z_i = 0$ and $p \in P^-$ or $Z_i = 1$ and $p \in P^+$ are retained

$$\Pr(P \leq p | Z = 0, S_1 = 1, p \in P^-) = \Pr(P \leq p | Z = 0, p \in P^-)$$

$$\Pr(P \leq p | Z = 0, p \in P^+) = \Pr(P \leq p | Z = 0, S_1 = 1, p \in P^+)$$

which is sufficient to satisfy condition (ii) of Proposition 4.

In general there will be many trimmed samples that satisfy Definition 1 and condition (ii) of Proposition 4. To select a particular sample, in this algorithm we choose the observations that are trimmed in order to make the propensity score distributions for $Z = 0$ and $Z = 1$ as similar as possible. This can be motivated heuristically as follows: variation in subdensities driven by propensity score can mask variation in the subdensities driven by Z. Hence, we wish to compare subdensities for similar propensity score distributions.

As the second step of the algorithm does not take into account any subsequent trimming of observations with $Z_i = 0$, it may trim more observations with $Z_i = 1$ than is necessary to attain stochastic dominance. Appendix D details how the algorithm may be modified to avoid unnecessary trimming. The additional step essentially determines d_0 and d_1 by iterating using the algorithm above, so satisfies the conditions of Definition 1 and condition (ii) of Proposition 4 in the same manner.

3 Test Statistics and Procedure

We set the null hypothesis for our test to be the joint restrictions of index sufficiency (10) and the nesting inequalities with a distilled sample (43) shown in Proposition 3. This section presents the construction of our test statistics and an implementation procedure with a bootstrap algorithm for computing p-values. Our exposition of the test here is restricted to a binary instrument $Z \in \{1, 0\}$, while the support of covariates is unconstrained. Appendix F presents a test procedure for a single multi-valued instrument.

3.1 Test statistics and bootstrap algorithm

Let $\lambda = \Pr(Z = 1) \in (0, 1)$ and $(Z, S_1) \in \{0, 1\} \times \{0, 1\}$ be a binary instrument and a sample inclusion indicator as defined in Definition 1. With binary Z, we can rewrite the nesting inequalities with a distilled sample (43) as the following moment inequalities: for every measurable subset $A \subset \mathcal{U}$,

$$E[1\{U_i \in A\}D_i|Z = 0, S_1 = 1] - E[1\{U_i \in A\}D_i|Z = 1, S_1 = 1] \leq 0, \tag{30}$$

$$E[1\{U_i \in A\}(1 - D_i)|Z = 1, S_1 = 1] - E[1\{U_i \in A\}(1 - D_i)|Z = 0, S_1 = 1] \leq 0. \tag{31}$$

The inequalities (30) and (31) express the nesting inequalities of Proposition 3 as conditional expectations given $(Z, S_1 = 1)$.

Let $q(p) = \Pr(Z = 1|p(X, Z) = p)$. Define the sample inclusion indicator S_2 to be

$$S_2 = \mathbb{1}\{\iota \leq q(p) \leq 1 - \iota\}. \tag{32}$$

where $\iota \in (0, 0.5)$ is a trimming constant. The index sufficiency equalities of Proposition 2 may be written as: for every measurable subset $A \subset \mathcal{U}$

$$E\left[\frac{1\{U_i \in A\}D_i(1 - \lambda)}{1 - q(p)}|Z = 0, S_2 = 1\right] - E\left[\frac{1\{U_i \in A\}D_i\lambda}{q(p)}|Z = 1, S_2 = 1\right] = 0, \tag{33}$$

$$E\left[\frac{1\{U_i \in A\}(1 - D_i)(1 - \lambda)}{1 - q(p)}|Z = 0, S_2 = 1\right] - E\left[\frac{1\{U_i \in A\}(1 - D_i)\lambda}{q(p)}|Z = 1, S_2 = 1\right] = 0. \tag{34}$$

Using Bayes rule, the moment equality hypotheses (33) and (34) express the index sufficiency restrictions of Proposition 2 in terms of the conditional expectation given $(Z, S_2 = 1)$. Trimming the sample through S_2 avoids dividing by near zero probabilities. As S_2 depends only on the value of the propensity score, trimming in this manner does not affect the validity of the index sufficiency restrictions.

Denote the conditional distributions of (U, D, X) given $Z = 1$ and $Z = 0$ by P and Q, respectively, and the corresponding empirical distributions by P_{n_1} and Q_{n_0}, where n_1 and n_0 are the number of observations with $Z = 1$ and $Z = 0$, respectively. Following the notational conventions of empirical process theory (e.g., Van Der Vaart and Wellner (1996)), we denote the expectation of a function of (U, D, X, S_1), f, with respect to a generic distribution, P, by Pf and the sample analogue (sample average) by $P_{n_1}f$. Similarly, for a function g of (U, D, X, S_2), the expectation with respect to a generic distribution P and its sample analogue are denoted by Pg and $P_{n_1}g$, respectively.

Let $\mathcal{C}(\mathcal{U})$ be the class of closed and connected intervals in $\mathcal{U}$. Let $\hat{U}_i = Y_i - D_iX_i'\theta_1 - (1 - D_i)X_i'\theta_0$ be an estimated partial residual, $\hat{p}(X_i, Z_i)$ be an estimated propensity score, and $\hat{q}(p)$ be

an estimated instrument propensity score. Define

$$f_i^z(A,d) = \frac{1\{\hat{U}_i \in A, D_i = d\} S_{1i}}{\Pr(S_1 = 1 | Z = z)}, \tag{35}$$

$$g_i^z(A,d) = \frac{1\{\hat{U}_i \in A, D_i = d\} \hat{\lambda}^z (1-\hat{\lambda})^{1-z} S_{2i}}{\hat{q}(\hat{p})^z (1-\hat{q}(\hat{p}))^{1-z} \Pr(S_2 = 1)}. \tag{36}$$

Consider the following test statistic,

$$T \equiv \max \left\{ \sup_{A \in \mathcal{C}(\mathcal{U}), d \in \{0,1\}} T_1(A,d), \sup_{A \in \mathcal{C}(\mathcal{U}), d \in \{0,1\}} T_2(A,d), \sup_{A \in \mathcal{C}(\mathcal{U}), d \in \{0,1\}} -T_2(A,d) \right\}, \tag{37}$$

$$T_1(A,d) = \sqrt{\frac{n_1 n_0}{n_1 + n_0}} \left[\left(Q_{n_0} f_i^0(A,1) - P_{n_1} f_i^1(A,1) \right) d + \left(P_{n_1} f_i^1(A,0) - Q_{n_0} f_i^0(A,0) \right)(1-d) \right],$$

$$T_2(A,d) = \sqrt{\frac{n_1 n_0}{n_1 + n_0}} \left(Q_{n_0} g_i^0(A,d) - P_{n_1} g_i^1(A,d) \right).$$

Statistic T jointly tests (30) - (34) by searching for a maximal violation of their sample analogues. As in the test of Kitagawa (2015), searching over the closed and connected intervals $\mathcal{C}(\mathcal{U})$ is sufficient to detect violations in any measurable set in $\mathcal{U}$.

3.1.1 Bootstrap

We want to resample $T_1(\cdot,\cdot)$ and $T_2(\cdot,\cdot)$ from a distribution such that the null hypothesis holds and their variance-covariance structure is preserved. Specifically, we impose the null by recentering. Consider drawing a bootstrap sample of N observations from the original sample. Let $\tilde{n}_0$ and $\tilde{n}_1$ be, respectively, the number of observations in the bootstrap sample with $Z = 0$ and $Z = 1$. A bootstrap analogue of T under a least favorable null can be constructed as

$$\tilde{T} \equiv \max \left\{ \sup_{A \in \mathcal{C}(\mathcal{U}), d \in \{0,1\}} \tilde{T}_1(A,d), \sup_{A \in \mathcal{C}(\mathcal{U}), d \in \{0,1\}} \tilde{T}_2(A,d), \sup_{A \in \mathcal{C}(\mathcal{U}), d \in \{0,1\}} -\tilde{T}_2(A,d) \right\}, \text{ where}$$

$$\begin{aligned}
\tilde{T}_1(A,d) = &\sqrt{\frac{n_1 n_0}{n_1 + n_0}} \left(\tilde{Q}_{\tilde{n}_0} \tilde{f}_i^0(A,1) - Q_{n_0} f_i^0(A,1) - \left(\tilde{P}_{\tilde{n}_1} \tilde{f}_i^1(A,1) - P_{n_1} f_i^1(A,1) \right) \right) d \\
&+ \sqrt{\frac{n_1 n_0}{n_1 + n_0}} \left(\tilde{P}_{\tilde{n}_1} \tilde{f}_i^1(A,0) - P_{n_1} f_i^1(A,0) - \left(\tilde{Q}_{n_0} \tilde{f}_i^0(A,0) - Q_{n_0} f_i^0(A,0) \right) \right) (1-d) \\
\tilde{T}_2(A,d) = &\sqrt{\frac{n_1 n_0}{n_1 + n_0}} \left[\tilde{Q}_{\tilde{n}_0} \tilde{g}_i^0(A,d) - Q_{n_0} g_i^0(A,d) - \left(\tilde{P}_{\tilde{n}_1} \tilde{g}_i^1(A,d) - P_{n_1} g_i^1(A,d) \right) \right],
\end{aligned}$$

where $\tilde{P}$ and $\tilde{Q}$ are the bootstrap sample empirical conditional distributions of $(\tilde{U}, D, X)$ given $Z = 1$ and $Z = 0$, and $\tilde{f}_i^z(A,D)$ and $\tilde{g}_i^z(A,d)$ are as defined in (35) and (36) but calculated with the bootstrap sample. Note that any quantity which is estimated in the main sample will be re-estimated with the bootstrap sample.

Our test obtains critical values from a least favorable null distribution. An alternative is to introduce a moment selection step, as suggested in Andrews and Soares (2010) and Chernozhukov et al. (2019) among others. This can potentially improve power, though at the cost of introducing additional tuning parameters.

3.1.2 Test Procedure

An implementation of the test is as follows.

1. Estimate the propensity scores $\Pr(D = 1|Z, X)$, and obtain the fitted values $\hat{p}(x_i, z_i)$.

2. Estimate the partially linear regression model (6) and obtain estimates of θ_1 and θ_0. Specifically, let $\hat{\mu}_{x,i} = \hat{E}(x_i|\hat{p}(x_i, z_i))$ be nonparametric regression estimates (e.g. local linear predicted values) of the covariate vector x_i onto the estimated propensity scores, $\hat{p}_i \equiv \hat{p}(x_i, z_i)$, and let $\hat{\mu}_{y,i} = \hat{E}(y_i|\hat{p}(x_i, z_i))$ be a nonparametric regression estimate of the outcome onto the estimated propensity scores. (θ_0, θ_1) can be estimated by running OLS on

$$y_i - \hat{\mu}_{y,i} = \hat{p}_i (x_i - \hat{\mu}_{x,i}) \theta_1 + (1 - \hat{p}_i)(x_i - \hat{\mu}_{x,i}) \theta_0 + \epsilon_i.$$

3. Construct partial residuals as $\hat{u}_i = D_i \left(y_i - x_i'\hat{\theta}_1\right) + (1 - D_i)\left(y_i - x_i'\hat{\theta}_0\right)$.

4. Construct the inclusion indicators S_1. This can be done using the algorithm of Section 2.5.

5. Estimate $q(p(X, Z))$ and generate the inclusion indicators S_2

6. Using $(\hat{u}_i, D_i, z, \hat{q}(\hat{p}_i), S_{1,i}, S_{2,i})$ as data, calculate the test statistic of Section 3.1

The test procedure described above involves semiparametric estimation of a partially linear model where each of the k_X conditioning covariates and the outcome are non-parametrically regressed on the propensity score. Given that the computational complexity of a workhorse nonparametric regression for sample size N is of the order N^2, the overall computational complexity of Steps 1 and 2 is $(1 + k_X)O(N^2)$. Holding N fixed, as the dimension of X grows, computational complexity increases by $O(N^2)$, which is constant given N. Hence, the computational complexity of this step is linear in the dimension of X. Calculating the test statistic involves at worst $(N + 1)N/2$ calculations, independent of the dimension of X. In contrast, the complexity of the Kitagawa (2015) procedure with covariates can be shown to be $N^{2(k_X+1)}$.

3.1.3 Asymptotic Theory

Define $q(p)$ to be the instrument propensity score

$$q(p) \coloneqq \Pr(Z = 1 | p(X, Z) = p)$$

We impose the following function form restriction on the propensity score and the instrument propensity score.

Assumption 1

(i) $\beta \in \mathbf{B}$ *where* $\beta \coloneqq (\beta_0, \beta_1)$ *and* $\mathbf{B} \coloneqq \mathbf{B}_0 \times \mathbf{B}_1$ *with* $\mathbf{B}_0 \subset R^{k_X}$ *and* $\mathbf{B}_1 \subset R^{k_X}$

(ii) The functions $p(X, Z)$ *and* $q(p)$ *have the form*

$$p(X, Z) = p(X, Z; \delta_p) = \varphi_p(\kappa_p(X, Z)^T \delta_p)$$
$$q(p) = q(X, Z; \delta_p, \delta_q) = \varphi_q(\kappa_q(p(X, Z; \delta_p))^T \delta_q)$$

where, for $j = q, p$, $\varphi_j(\cdot) : \mathbb{R} \to [0, 1]$ *is a known monotonically increasing function,* $\kappa_j(\cdot)$ *is a vector of known functions, and* $\delta_j \in \mathbf{D}_j \subset \mathbb{R}^{k_j}$ *is a vector of parameters. In addition, there exists a constant* C_p *such that for any* $(x, z) \in \mathcal{X} \times \{0, 1\}$ *and any* $\delta_p, \delta_p' \in \mathbf{D}_p \times \mathbf{D}_p$

$$|\varphi_p(\kappa_p(x, z)^T \delta_p) - \varphi_p(\kappa_p(x, z)^T \delta_p')| \leq C_p ||\delta_p - \delta_p'||$$

and a constant C_q *such that for any* $p \in [0, 1]$ *and* $p' \in [0, 1]$*, and any* $(\delta_q, \delta_q') \in \mathbf{D}_q \times \mathbf{D}_q$

$$|\varphi_q(\kappa_q(p)^T \delta_q) - \varphi_q(\kappa_q(p)^T \delta_q')| \leq C_q ||\delta_q - \delta_q'||$$
$$|\varphi_q(\kappa_q(p)^T \delta_q) - \varphi_q(\kappa_q(p')^T \delta_q)| \leq C_q ||p - p'||$$

The conditions of Assumption 1 are satisfied by commonly used parametric estimators of limited dependent variables, such as probit and logit. Further Assumptions are stated in Appendix E. These assumptions are technical in nature, and do not impose restrictions on functional form.

Theorems 4 and 5 in Appendix E establish the asymptotic distributions of the index sufficiency and distilled test statistics. In the case of the distilled test, we consider the case without trimming. [4] The following theorem combines these results to give the asymptotic distribution of T

Theorem 2 *Suppose that Assumption 1 and the additional assumptions stated in Appendix E hold. Then, under the worst case null where the distilled nesting inequalities hold with equality for all* $A \in \mathcal{C}(\mathcal{U})$*, as* $N \to \infty$

$$T \implies \max \left\{ \sup_{A \in \mathcal{C}(\mathcal{U}), d \in \{0,1\}} \varepsilon_1(A, d), \sup_{A \in \mathcal{C}(\mathcal{U}), d \in \{0,1\}} \varepsilon_2(A, d), \sup_{A \in \mathcal{C}(\mathcal{U}), d \in \{0,1\}} \varepsilon_3(A, d) \right\},$$

[4] While the nesting test statistic is not of primary interest, Theorem 3 shows its asymptotic distribution.

where each $\varepsilon_j(A,d)$ *is a Gaussian process on* $\mathcal{C}(\mathcal{U})\times\{0,1\}$ *with covariance kernel defined in Appendix E.*

Size control of the test procedure in Section 3.1.2 can be shown in a similar manner to Linton et al. (2005, 2010). This requires further assuming that the bootstrap can mimic the distributions of the estimated parameters, and appealing to Theorem 3.7.1 of Vaart and Wellner (2023).

3.2 Practical Considerations

3.2.1 Testing Causal Intepretability of 2SLS

The test proposed here does not directly test instrument validity in the context of linear 2SLS. This is because the set of identifying assumptions considered in Proposition 2 is distinct from the set of conditions that allows us to interpret the linear two-stage least square (2SLS) estimand as a weighted average of LATEs/MTEs with positive weights. Specifically, causal interpretability of linear 2SLS requires only the weaker exogeneity condition (A2), not the strong exogeneity of (A5), and does not require the functional form specification for the potential outcome equations of (A4). On the other hand, as shown in Abadie (2003), Kolesár (2013), Słoczyński (2021), and Blandhol et al. (2022), causal interpretability of 2SLS relies crucially on the linearity of $E(Z|X)$ in the covariate vector X, whereas the identifying assumptions here considered place no restrictions on $E(Z|X)$.

However, we believe our test is useful as a specification check when one reports linear 2SLS estimates. First, non-rejection in our test means the data do not reject strong exogeneity (A5), implying that the data also do not contradict weak exogeneity (A2). With credible evidence for the linearity of $E(Z|X)$, non-rejection of our test can be used to support causal interpretability of linear 2SLS. Second, with specifications of the potential outcome equations and the propensity score, our test is useful for checking which variables should be included as controlling covariates, since rejection of our test can signal a violation of instrument exclusion conditional on the chosen set of covariates. Hence, p-values of our test can also suggest which set of covariates should be included in the linear 2SLS estimation.

It is a common practice to run overidentification test for instrument validity in the context of linear IV regression. However, rejection of the null in an overidentification test can be driven by heterogeneity of treatment effects rather than violations of IV validity. Our test, in contrast, allows for unrestricted heterogeneity of the treatment effects in unobservables. Hence, if the researcher believes treatment effects are heterogeneous, we would recommend performing our test.

3.2.2 Multiple Instruments

The discussion to this point has considered only a scalar Z, here we consider the case where Z is a vector of multiple instruments. Propositions 1 and 2 hold irrespective of the number of instruments. Hence, both testable implications remain available. However, a test of the nesting inequalities using a coarsened propensity score would be subject to the same concerns regarding power as the single instrument case.

Consider the case of two binary instruments $Z = (Z_1, Z_2)$, $Z_1 \in \{0, 1\}$, $Z_2 \in \{0, 1\}$. When testing nesting inequalities and index sufficiency, Z can be treated as a single instrument taking four values. That is, the estimation step is performed taking Z to be vector, but when testing we map from $(0, 0), (0, 1), (1, 0), (1, 1)$ to four values of a single categorical variable and implement the test for a multi-valued instrument described in Appendix F. In cases where this is not viable, one approach is to aggregate the multiple instruments into a single index, and then treat this index as single multivalued instrument. Suppose the propensity score has a generalised linear form with a probit link

$$p(X, Z) = \Phi(\rho X + \varphi(Z)), \tag{38}$$

where $\varphi(\cdot)$ reflects the researchers assumptions about the relationship between the instruments and Z and ρ is a vector of coefficients. In this case, we view $\varphi(Z)$ as an aggregated index of instruments, and perform the test as if it was a single instrument.

Jiang and Sun (2023) extends the test of Kitagawa (2015) to the setting of Mogstad et al. (2019, 2020), where the montonicity assumption Imbens and Angrist (1994) is replaced by partial monotonicity. Our test procedure can be extended to this setting in a similar manner. Mogstad et al. (2019, 2020) show that inference using a single instrument remains valid as long as the remaining instruments are controlled for. In the context of our test procedure, the estimation of the propensity score and residuals would be performed using the entire set of instruments, but the final step would be performed for each instrument. That is, violation of index sufficiency and the nesting inequalities are evaluated by the test statistic (2) for every instrument, with a distilled sample used in the test of nesting inequalities. The joint test statistic would then be obtained as the maximum of test statistics for individual instruments, with bootstrap test statistics calculated in the same manner.

3.2.3 Ordered Multivalued Treatments

In the case of an ordered multivalued treatment variable, a straightforward approach to testing the validity of an instrumental variable is to binarise the treatment variable and apply the test procedure described above. However, as described in Angrist and Imbens (1995) and Marshall (2016), a valid

instrumental variable that affects choices at multiple intensities of a multivalued treatment variable will not satisfy the exclusion restriction for that treatment's binarised counterpart. When the treatment is a binarised variable, our test procedure becomes a test of the validity of binarisation as well as the validity of the instrumental variable. If the researcher does not intend to binarise the treatment variable in their analysis, then this procedure tests an irrelevant assumption.

As an alternative, we develop testable implications for an ordered multivalued treatment. To do so, we follow Heckman et al. (2006) and consider a heterogenous treatment effect model with a multivalued treatment.

Let $D \in \mathcal{D} = \{0, ..., D^{\max}\}$ with $D^{\max} > 1$. Assumptions for an ordered mutlivalued treatment are presented in Appendix G. They are minor modifications of the assumptions of Section 2. These assumptions imply the following heteogeneous treatment effect model

$$\begin{cases} Y_d = \alpha_d + X'\theta_d + \tilde{U}_d \text{ for all } d \in \mathcal{D}; \\ Y = Y_0 + \sum_{d \geq 1} \mathbb{1}\{D \geq d\}(Y_d - Y_{d-1}); \\ D = \sum_{d \geq 1} \mathbb{1}\{\nu(X, Z) \geq V_{d-1}\}. \end{cases} \tag{39}$$

Here $\nu(X, Z)$ is an index function which replaces the propensity score in the binary treatment case. Following Vytlacil (2006), we can impose the normalisation $\nu(X, Z) = E[D|X, Z]$. V_{d-1} is a threshold such that $v(X, Z) \geq V_{d-1}$ implies $D \geq d - 1$. The thresholds $\{V_d\}$ are heterogeneous in the sense that they vary over units. Define partial residuals to be

$$\begin{aligned} U_d &= \alpha_d + \bar{U}_d, \\ U &= U_0 + \sum_{d \geq 1} \mathbb{1}\{D \geq d\}(U_d - U_{d-1}). \end{aligned} \tag{40}$$

The following proposition extends Proposition 2 to ordered mulitvalued treatments.

Proposition 5 *Assume the index function $v(X, Z)$ has a nondegenerate distribution. If (A1 MV), (A3 MV), (A4 MV) and (A5 MV) hold, then the following two conditions hold:*
(i) ***Index sufficiency****: The conditional distribution of (U, D) given (X, Z) depends only on the expectation $v(X, Z)$, i.e., for any measurable set $A \subset \mathbb{R}$ and $d \in \mathcal{D}$*

$$\Pr(U \in A, D = d|X = x, Z = z) = \Pr(U \in A, D = d|v(X, Z) = v) \tag{41}$$

holds for any (x, z).
(ii) ***Nesting inequalities****: For every $v \geq v'$ in the support of the distribution of $v(X, Z)$ and every measurable subset $A \subset \mathcal{Y}$:*

$$\begin{aligned} &\Pr(U \in A, D = D^{\max}|v(X, Z) = v) \geq \Pr(U \in A, D = D^{\max}|v(X, Z) = v'), \\ &\Pr(U \in A, D = 0|v(X, Z) = p) \leq \Pr(U \in A, D = 0|v(X, Z) = v'). \end{aligned} \tag{42}$$

Appendix G presents a heuristic derivation of these testable implications. Index sufficiency carries over to the multivalued treatment setting, with $v(X,Z)$ in place of the propensity score $p(X,Z)$. The nesting inequalities (42) are analagous to those derived in Sun (2023). They are available only for the highest and lowest treatment intensities. For all intermediate intensities variation in the value of the index function can induce both flows in and flows out, which means that no nesting relationship exists. [5]

Proposition 3 can be extended to the multivalued treatment variable case as follows

Proposition 6 *Suppose the assumptions of Proposition 5 holds. Let $S_1 \in \{0,1\}$ be an inclusion indicator that satisfies*

i $(\{U_d\}_{d\in\mathcal{D}}, \{V_d\}_{d\in\mathcal{D}}) \perp S_1$

ii The conditional distribution of the index function $v(X,Z)$ given $\{Z, S_1\}$, $P_{v|ZS_1}(\cdot|z, S_1 = 1)$, is stochastically increasing in $z \in \mathcal{Z}$ in terms of the first-order stochastic dominance.

Then for any $z \geq z' \in \mathcal{Z}$ and every measurable set $A \subset \mathcal{Y}$,

$$\begin{aligned} &\Pr(U \in A, D = D^{\max}|Z = z, S_1 = 1) \geq \Pr(U \in A, D = D^{\max}|Z = z', S_1 = 1), \\ &\Pr(U \in A, D = 0|Z = z, S_1 = 1) \leq \Pr(U \in A, D = 0|Z = z', S_1 = 1). \end{aligned} \tag{43}$$

The definition of distillation here is unchanged from the binary treatment case, except that $v(X,Z)$ replaces the propensity score. Hence, the algorithm of Section 2.6 remains valid.

The index sufficiency testable implication of Proposition 5 and the distilled nesting inequalities of Proposition 6 can be tested jointly using a modified version of the test procedure of Section 3.1.2. The modification relates to the first step of the procedure. Assumptions 1 (MV) - 5 (MV) imply a partially linear model of the form

$$\begin{aligned} Y &= X'\theta_0 + \sum_{d\geq 1} \Pr(D \geq d|X,Z) X'\theta_d + \phi(v(X,Z)) + \varepsilon \\ v(X,Z) &= \sum_{d\geq 1} \Pr(D \geq d|X,Z) \end{aligned} \tag{44}$$

This differs from the binary treatment case as multiple propensity scores enter the outcome equation. To reflect this, the first step of the test procedure in Section 3.1.2 is replaced with

[5] Kwon and Roth (2024) derive a further testable implication that holds for all treatment intensities. Their testable implication cannot be easily accommodated by our proposed test procedure, but could be tested separately.

1. For $d \geq 1$ estimate $\Pr(D \geq d|Z, X)$ and obtain the fitted values $\hat{\Pr}(D \geq d|Z, X)$. Construct fitted values for the index function $\hat{v}(X, Z)$ as

$$\hat{v}(X, Z) = \sum_{d \geq 1} \hat{\Pr}(D \geq d|Z, X)$$

Under the null hypothesis, the propensity scores satisfy the monotonicity condition $\Pr(D \geq d - 1|X, Z) \geq \Pr(D \geq d|X, Z)$ for all $d \geq 1$. They can be estimated separately without imposing these restrictions by a standard method such as probit. An alternative is to estimate a model that explicitly imposes these constraints, such as the hierarchical ordered probit model described in Greene (2024).

4 Monte Carlo

We perform two sets of Monte Carlo exercises. The first evaluates the size and power of the joint test of nesting inequalities with a distilled sample and index sufficiency, and compares it to the Kitagawa (2015) test which does not control for covariates and the test of the nesting inequalities using a coarsened propensity score. The second looks separately at the power of the tests of index sufficiency and nesting inequalities with a distilled sample.

All exercises use

$$X \sim \mathrm{N}_3(0_3, I_3) \tag{45}$$

$$Z = \mathbb{1}\{X'\gamma + U_Z \geq 0\} \tag{46}$$

$$D = \mathbb{1}\{\alpha_0(1 - Z) + \alpha_1 Z + X'\delta + U_D \geq 0\} \tag{47}$$

$$Y = DY_1 + (1 - D)Y_0 \tag{48}$$

$$Y_0 = X'\theta + \tilde{U}_0 \tag{49}$$

$$U_Z \sim N(0, 1) \tag{50}$$

$$(\tilde{U}_0, U_D) \sim N(0, \Sigma), \tag{51}$$

with diagonal entries of Σ set to 1 and off-diagonal entries to 0.3. Elements of θ are drawn from the uniform distribution with bounds $(-1, 1)$. The distributions for γ and δ are described below. α_0, α_1, and the specification for Y_1 differ for the size and power processes. When checking size we present results for a valid but irrelevant instrument[6]

$$\alpha_0 = \alpha_1 = 0, \tag{52}$$

$$Y_1 = X'\theta + 1 + \tilde{U}_1, \text{ with } \tilde{U}_1 = \tilde{U}_0. \tag{53}$$

[6] Addition results for a relevant instrument are presented in Appendix H. Rejection rates with a relevant instrument are very similar to those with an irrelevant instrument.

Here, conditional on the propensity score, the joint distribution of (U, D) conditional on Z is identical for $Z = 0$ and $Z = 1$. Four specifications are used to check power. They share

$$\alpha_0 = \Phi^{-1}(0.45), \tag{54}$$

$$\alpha_1 = \Phi^{-1}(0.55), \tag{55}$$

$$Y_1 = X'\theta + \tilde{U}_1, \tag{56}$$

where $\Phi^{-1}()$ is the standard normal quantile function. The specifications for $\tilde{U}_1$ are

$$\begin{aligned}
&\text{DGP1:} \quad \tilde{U}_1 = Z \times \tilde{U}_0 + (1 - Z) \times (-0.7 + \tilde{U}_0), \\
&\text{DGP2:} \quad \tilde{U}_1 = Z \times \tilde{U}_0 + (1 - Z) \times 1.675 \times \tilde{U}_0, \\
&\text{DGP3:} \quad \tilde{U}_1 = Z \times \tilde{U}_0 + (1 - Z) \times 0.515 \times \tilde{U}_0, \\
&\text{DGP4:} \quad \tilde{U}_1 = Z \times \tilde{U}_0 + (1 - Z) \times (\mu_l + 0.125 \times \tilde{U}_0),
\end{aligned}$$

where μ_l takes values $(-1, -0.5, 0, 0.5, 1)$ with probabilities $(0.15, 0.2, 0.3, 0.2, 0.15)$. Figure 4 plots the densities of $(\tilde{U}_1, D = 1|Z)$ for each of the four processes for one draw of parameters. In the case of DGP1 the distribution for $Z = 0$ is shifted, with DGP2 it has thicker tails, with DGP3 it is more concentrated, and with DGP4 there are violations at each value of μ_l. In all cases, for a given propensity score, the distribution of $(\tilde{U}_1, D = 1)$ conditional on Z does not coincide for $Z = 0$ and $Z = 1$.

We consider three values of the sample size N: 200, 500, and 1000. For each process and sample size, we generate 1,000 samples. When comparing test procedures, the same simulated samples are used for each test procedure. The proposed test procedure requires estimating propensity scores, a partially linear model for outcomes, and $\Pr(Z = 1|p(X, Z))$. Propensity scores are estimated by probit with the correct specification supplied. For the class of process we consider, if the instrument is valid then the partially linear model reduces to linear regression with a generated variable. For these functional forms, a closed-form expression for $\Pr(Z = 1|p(X, Z))$ is available as a function of α_0, α_1, δ and γ. We obtain estimates $\hat{\Pr}(Z = 1|p(X, Z))$ by plugging-in estimated parameter values to this expression. In the case of DGP1-DGP4, the specification for the partially linear model is incorrect, so estimates will be biased.[7] We set the sample inclusion indicator for the index sufficiency test to $S_2 = \mathbb{1}\{\Pr(Z = 1|p(X, Z)) \in [0.05, 0.95]\}$. P-values are calculated using 500 bootstrap samples.

[7]For the size process, this specification is correct. However, as the instrument is irrelevant, the model is not identified. In particular, it can be shown that asymptotically there is perfect multicollinearity among the columns of $X - E[X|p]$. In the finite samples used in these Monte Carlo exercises it is still possible to calculate an estimate of θ. This estimate reflects finite sample noise, so the associated partial residuals can be used to check size.

4.1 Size and Power

Here we evaluate the size and power of the proposed test procedure, and compare it to the Kitagawa (2015) test that does not control for covariates and the test of nesting inequalities with a binarised propensity score.

For this exercise, the elements of δ are drawn from the uniform distribution with bounds $(-1, 1)$. For γ, we consider two cases. The first sets all elements of γ to 0. X and Z are then independent, so an instrument that is valid conditional on the covariates will also be valid without conditioning on covariates and failure to control appropriately for X should not affect size. In the second case each element of γ is drawn from the uniform distribution with bounds $(-1, 1)$. Here X and Z will not, in general, be independent, and the instrument can only be valid conditional on X.

Tables 1 and 2 present rejection rates. Turning first to the proposed test procedure, rejection rates for the size process are below nominal size irrespective of whether X and Z are independent. For the four power DGPs, with the exception of the smaller sample sizes for DGP4, all rejection rates are well above the nominal size. Similar rejection rates are obtained irrespective of whether X and Z are independent.

Next consider the test with a binarised propensity score. This test performs well in terms of size: rejection rates for the size processes are below nominal size. However, consistent with the results of Proposition 4, there is a considerable loss in power compared to the proposed test procedure. Rejection rates are below nominal size for all power DGPs and all sample sizes.

Figures 5 and 6 illustrate why the test with a binarized propensity score lacks power. To construct these figures, for each of the four power DGPs, we generate a single draw for θ and δ. To simplify, γ is set to the zero vector. We approximate the asymptotic estimates of θ, $\hat{\theta}^\star$, by generating a sample of a million observations and regressing $Y - E[Y|p]$ on $X - E[X|p]$. Partial residuals are then calculated as $U = Y - X'\hat{\theta}^\star$. Figure 5 plots the densities of $(U, D = 1)$ conditional on $Z = 0$ and $Z = 1$, whereas Figure 6 plots the subdensities conditional on the binarised propensity score Z^*. In Figure 5, when we condition on Z, violations of the null are apparent in all cases, but in Figure 6 violations are either small or have vanished entirely. Each value of Z^* mixes observations with $Z = 0$ and $Z = 1$. The subdensities in Figure 6 are thus a mixture of those in Figure 5, which masks differences in shape. Furthermore, by construction, all observations with $Z^* = 0$ have a lower propensity score than observations with $Z^* = 1$, so the $D = 1$ subdensity for $Z^* = 0$ is guaranteed to have less total mass than the subdensity for $Z^* = 1$.

The Kitagawa (2015) test, which does not control for covariates, performs reasonably well when the instrument is independent of the covariates. The size of the test is controlled, but power is lower than the proposed test procedure. This is for two reasons. First, the Kitagawa (2015) test tests nesting inequalities only, and some violations are more easily detected by index sufficiency.

Second, failure to control for covariates adds noise to the outcome and which makes violations of the nesting inequalities more difficult to detect. For a single draw of θ and δ, and with γ set to zero, Figure 7 plots the densities of $(Y, D = 1)$ conditional on $Z = 0$ and $Z = 1$ for each DGP. These differ from the densities of Figure 4 as the effect of the covariates on the outcome has not been partialed out. Compared to the densities in Figure 4, violations are either reduced or have vanished entirely. As expected, when Z and X are not independent, this test procedure has rejection rates above nominal size regardless of whether the instrument is valid conditional on covariates.

[Table 1 about here.]

[Table 2 about here.]

4.2 Comparison of Index Sufficiency and Nesting Inequalities

Our second exercise compares the nesting inequality and index sufficiency tests. Section 2 argued that the relative power of testing nesting inequalities and index sufficiency will depend on the similarity between propensity score distributions conditional on Z. To examine this more formally, we compare rejection rates for the nesting inequalities and index sufficiency in two cases. Both set all elements of γ to 0, so that the instrument and covariates are independent. In the first, the elements of δ are drawn from the uniform distribution with bounds $(-1, 1)$, as in section 4.1. In the second, the elements of δ are drawn from the uniform distribution with bounds $(-0.1, 0.1)$. This second case limits the influence of covariates on the propensity score, which in turn limits the overlap of the conditional propensity score distributions for $Z = 0$ and $Z = 1$. Figure 8 plots the density of propensity scores conditional on $Z = 0$ and $Z = 1$ for different values of δ. As the elements of δ increase in magnitude, the densities overlap more. Figure 9 plots the $\Pr(Z = 1|p(X, Z))$ function associated with each value of δ. For the smallest δ this function is close to a step function but, as the elements of δ grow in magnitude, it becomes flatter.

Table 3 presents rejection rates. When the elements of δ are drawn from the uniform distribution with bounds $(-1, 1)$, index sufficiency has higher rejection rates. When the elements of δ are drawn from the uniform distribution with bounds $(-0.1, 0.1)$, index sufficiency loses power in all cases. Nesting inequalities now have considerably higher power for DGP1 and DGP3, and slightly higher power for DGP4, while index sufficiency still exhibits more power for DGP2.

[Table 3 about here.]

[Figure 4 about here.]

[Figure 5 about here.]

[Figure 6 about here.]

[Figure 7 about here.]

[Figure 8 about here.]

[Figure 9 about here.]

5 Applications

In this section we apply the proposed test procedure to the instrumental variables of Card (1993), Angrist and Evans (1998), and Bleakley and Chin (2010). In all cases we maintain the outcome, treatment, instrument and conditioning covariates of the original design, and estimate a potential outcome model of the form described in Section 2.

5.1 Card (1993)

Card (1993) estimates returns to education. The outcome Y is the logarithm of weekly earnings, D is years of education, and Z is growing up in a local labour market with an accredited four year college. The headline specification uses four ordered discrete, three categorical and six binary conditioning covariates. These include potential experience, parental education, and residence in a standard metropolitan statistical area (SMSA). The sample consists of 3,010 observations.

The validity of this instrument has previously been tested by Huber and Mellace (2015), Kitagawa (2015), Mourifié and Wan (2017) and Sun (2023). However, for computational reasons, at most only a subset of the conditioning covariates have been used. Sun (2023) does not control for covariates, Huber and Mellace (2015) conditions on two covariates, Mourifié and Wan (2017) three, and Kitagawa (2015) five. In addition, all of the conditioning covariates that have been used are binary. The proposed test procedure allows us to include all conditioning covariates.

The treatment variable is an ordered discrete variable, we can apply the procedure described in Section 3.2.3. However, the number of observations at the highest and lowest treatment intensities is extremely limited. Instead, we binarize the treatment, with the binarized variable defined to be 1 if years of education is at least 16. As discussed in Section 3.2.3, binarisation can introduce coarsening bias, so this is not equivalent to testing validity with the underlying multivalued treatment.

When estimating the partially linear model, we specify the propensity score as a probit with the instrument, conditioning covariates and interactions between the two as controls. For the outcome, we allow coefficients on the covariates to depend on the treatment. The estimation procedure is as described in step 2 of Section 3.1.2, with local linear regressions used to estimate $E[X|p]$ and

$E[Y|p]$. $\Pr(Z = 1|p)$ is estimated by local constant regressions. Bandwidths are selected by least squares cross validation.

Panel a of Figure 10 plots joint (Y, D) subdensities for $Z = 0$ and $Z = 1$. Kitagawa (2015) finds that, without conditioning on covariates, the null of instrument validity can be rejected. This violation can be seen in the $Y(0)$ subdensities. Figure 11 plots the conditional distributions and smoothed conditional densities for the estimated propensity score. The distribution for $Z = 1$ lies below the distribution for $Z = 0$, which indicates that first-order stochastic dominance holds, and there is considerable overlap between the densities. Table 5 shows the number of observations in the sample and the number used for each test. No observations are removed for either the test of nesting inequalities with a distilled sample or the test of index sufficiency. Panel b of Figure 10 plots the subdensities of estimated partial residuals used in the test of nesting inequalities. After controlling for covariates, the violation in the $D = 0$ subdensities is no longer evident. Panel c plots the reweighted densities used to test index sufficiency. These are similar to those in panel b. p-values are presented in Table 4. A p-value below 0.05 corresponds to a rejection of the null hypothesis at the 5% level of significance. p-values are reported for a range of values of the trimming parameter ξ. p-values are lower for the index sufficiency test than the test of nesting inequalities with a distilled sample, but the null is not rejected at any conventional level of significance.

[Figure 10 about here.]

[Figure 11 about here.]

[Table 4 about here.]

5.2 Angrist and Evans (1998)

Angrist and Evans (1998) estimate the effect of family size on parental labour supply. In their research design, D is a dummy that takes a value of 1 when a mother has more than two children. They consider two binary instruments: the multiple birth instrument (Z_M) and the same-sex instrument (Z_S). Z_M is set to one when the second child is part of a multiple birth, and Z_S is set to one for mothers whose eldest two children are of the same sex.

Angrist and Evans (1998) argue that Z_S is a valid instrumental variable as a child's sex is randomly determined and there is a parental preference for mixed sex children, so parents whose first two children are the same sex are more likely to have a third. However, Rosenzweig and Wolpin (2000) challenge its validity, and show in a simple model of labour supply that additional restrictions on preferences are needed for this instrument to satisfy the exclusion restriction. Its validity is formally tested in Huber (2015) and Mourifié and Wan (2017). In both cases, the

Angrist and Evans (1998) sample is split into subgroups by covariates, and validity is tested for each subgroup. Neither Huber (2015) or Mourifié and Wan (2017) uses the full set of covariates from Angrist and Evans (1998), and some variables are coarsened.

We follow the replication files of Angrist and Evans (1998) to construct the sample and variables. We focus on the 1980 Census sample of married mothers, which consists of married mothers aged between 21 and 35 who have at least two children and were at least 15 years old at the time of their first birth. This sample contains 254,652 observations. For computational reasons, we do not use the full sample. The sample contains 2,121 observations with $Z_M = 1$ and 252,531 with $Z_M = 0$. We retain all observations with $Z_M = 1$, and randomly choose 10% of those with $Z_M = 0$. Angrist and Evans (1998) presents results for several measures of labour supply. We set Y to be the logarithm of annual family labour income. The headline specification controls for demographic characteristics of the mother, the sex of the first child, and the sex of the second child.

The definitions of D and Z_M imply that $D = 1$ whenever $Z_M = 1$, so any observation with $Z_M = 1$ has a propensity score of 1. Propensity scores for observations with $Z_M = 0$ are estimated using a probit specification. When estimating the partially linear model, we use the entire sample. As both instruments and all covariates are discrete, we estimate a parametric model with a polynomial in the propensity score. However, index sufficiency is testable for observations with $Z_M = 1$. In this subsample, we set $\Pr(Z_S = 1|p(X, Z_M, Z_S))$ equal to its sample analogue.

We perform a joint test of the following testable implications

(i) In the subsample with $Z_M = 1$, index sufficiency holds for Z_S.

(ii) In the subsample with $Z_M = 0$, the nesting inequalities hold for Z_S.

(iii) In the subsample with $Z_S = 1$, the $D = 1$ subdensity nesting inequality holds for Z_M.

(iv) In the subsample with $Z_S = 0$, the $D = 1$ subdensity nesting inequality holds for Z_M.

(v) The $D = 1$ subdensity for $Z_M = 1$ nests the subdensity for $Z_M = 0, Z_S = 1$

Implications (i) through (iv) hold under partial monotonicity, as discussed in Section 3.2.2. That is, they are a test of the validity of one instrument with the other held constant. The implication (v) holds only if monotonicity extends to the entire instrument vector. Notice that the nesting inequality for the $D = 0$ subdensity is guaranteed to hold when testing the validity of Z_M, so we omit it.

Table 6 presents results and Table 7 presents the number of observations used in each test. The first row of Table 6 contains P-values for the joint test of all testable implications. P-values are all above 0.1. The remaining rows of Table 6 contain P-values for the individual tests of

nesting inequalities and index sufficiency. For some values of ξ, the null is rejected for testable implication (v). Figure 12b plots the subdensities used in the test of index sufficiency with Z_M held constant at 1. The subdensity conditional on $Z_S = 1$ appears more concentrated than the subdensity conditional on $Z_S = 0$, although the null would not be rejected at any conventional level of significance. Subdensities used in the tests of nesting inequalities are shown in Figures 13b, 14b and 15b. No violation of the nesting inequalities is apparent for any of these subdensities.

[Table 5 about here.]

[Figure 12 about here.]

[Figure 13 about here.]

[Figure 14 about here.]

[Figure 15 about here.]

5.3 Bleakley and Chin (2010)

Bleakley and Chin (2010) investigates the relationship between English speaking ability and social integration among immigrants to the US. Their sample is drawn from the 2000 US Census. It consists of immigrants who arrived in the US aged 14 or younger and were between 25 and 55 years of age at the time of the 2000 census. The treatment variable is a self-reported measure of English speaking ability with four ordered levels.[8] The instrumental variable is

$$Z = \mathbb{1}\{\text{non-English speaking birth country}\} \times \max(0, \text{age at time of arrival} - 9). \tag{57}$$

This variable takes value 0 for immigrants from English speaking countries and immigrants who arrived in the US at 9 years of age or younger. Its value increases with age at arrival, reaching a maximum value of 5 for immigrants who arrived in the US when they were 14 years old. Bleakley and Chin (2010) argues that this variable will be negatively related to English speaking ability as language acquistion is more difficult for immigrants who arrive in the US at a greater age.

Bleakley and Chin (2010) considers a variety of outcomes. For this exercise, we use 'spouse log wage' as the outcome. The main specification of the paper includes a sex dummy, a 'hispanic' dummy, and fixed effects for race, age, age at the time of arrival in the US and country of birth. [9]

[8] The four levels are speaks English 'not at all', 'not well', 'well', and 'very well'.

[9] While Bleakley and Chin (2010) do not explicitly adopt a functional form assumption, their instrument is a deterministic function of country and birth and age at the time of arrival. Hence, it does not nonparametrically identify an average of treatment effects.

Both the treatment and instrument of Bleakley and Chin (2010) are ordered discrete multivalued variables. To implement the test, we combine the procedure for multivalued treatments described in Section 3.2.3 with the test procedure for multivalued instruments described in Section F. Of the testable implications introduced in Section 3.2.3, we use only the nesting inequality for the highest level of the treatment. The main specification of Bleakley and Chin (2010) includes only fixed effects and dummy variables, so index sufficiency has no testable content. The sample of Bleakley and Chin (2010) consists of 73,680 observations. Only 662 of these have an observed treatment intensity of the lowest level, while 62,080 have the highest treatment intensity. Hence, a test of the nesting inequality for the lowest treatment intensity would have little power. We test the nesting inequality for the highest treatment intensity for each neighbouring pair of instrument values, as described in Appendix F. As the treatment has four intensities, we have to estimate three propensity scores. We estimate these separately by probit. When estimating the partially linear model, we assume a cubic polynomial for the nonparametric component.

Figure 16 plots pairs of joint (Y, D = speaks English very well) subdensities for neighbouring values of Z. Figure 17 plots conditional distributions of the estimated index function, and Figure 18 plots conditional smoothed densities. For each pair of neighbouring instrument values, first-order stochastic dominance of the distribution of $v(X, Z)$ generally holds, although there are some violations at higher values of $v(X, Z)$. Figure 19 plots pairs of joint (U, D = speaks English very well) subdensities. Violations of the nesting inequalities can be seen in the tails of the densities for some pairs of Z values. Table 8 presents P-values from the test. The first five columns show P-values when the test is conducted using a single pair of Z values only, the final column is an overall test where an overall test statistic is formed as the maximum of the test statistics for individual pairs. We reject the null at the 5% level of significance for three of the pairs of Z values and overall.

[Figure 16 about here.]

[Figure 17 about here.]

[Figure 18 about here.]

[Figure 19 about here.]

[Table 6 about here.]

6 Conclusion

This paper develops a novel test for instrument validity in the LATE/MTE framework that can accommodate conditioning covariates. We follow Carneiro et al. (2011) in assuming a linear relationship between potential outcomes and conditioning covariates and that unobserved heterogeneities

are independent of the instrument and covariates. Expected outcomes then follow a partially linear model, depending linearly on conditioning covariates and non-parametrically on the propensity score. With this functional form, partial residuals can be obtained by partialing out the conditioning covariates. We derive testable implication for the subdensities of these partial residuals: index sufficiency and the nesting inequalities. We propose a test procedure consisting of an initial step where the partially linear model is estimated and partial residuals obtained, followed by a joint test of index sufficiency and the nesting inequalities. Crucially, as conditioning covariates are partialed out, they increase only the complexity of estimating the propensity score and the partially linear model. In this way, the test procedure overcomes the computational hurdles of implementing tests such as Kitagawa (2015), Huber and Mellace (2015) and Mourifié and Wan (2017) in the presence of conditioning covariates.

The testable implications are complementary. Index sufficiency holds the propensity score fixed and compares subdensities for different values of the instrument, whereas the nesting inequalities are distributional monotonicity relations ordered with respect to the propensity score. A straightforward approach to testing the nesting inequalities is to coarsen the propensity score and check the inequalities for subdensities indexed by this coarsened variable. However, in cases where the propensity score is strongly correlated with the conditioning covaraites, the distribution of instrument values can become homogenised across propensity score bins, leading to a lack of power. We showed that the nesting inequalities hold with the original instrument in place of the propensity score if the conditional distribution of the propensity score given the instrument is stochastically monotonic in the sense of first-order stochastic dominance. Since first-order stochastic dominance is not implied by the identifying assumptions, we proposed testing the nesting inequalities using a trimmed sample where first-order stochastic dominance does hold. We refer to this process as distillation, and present an algorithm for constructing distilled samples. We show analytically a class of data generating processeses where distillation is guaranteed to outperform a test of the nesting inequalities over propensity score bins in terms of detecting violations of instrument validity. Monte Carlo exercises confirm this finding, with consistent gains in power when using the proposed test procedure in place of testing the nesting inequalities with a coarsened propensity score.

We believe that there are two areas which future research could address. First, formal characterisation of the size and power of this test, incorporating uncertainty over estimation of the partial residuals and the distillation process, is left for future research. Second, as noted in Section 2, in general there will be many possible distilled samples that satisfy the first-order stochastic dominance condition. In Proposition 4, we showed a restriction on trimming that is guaranteed to improve detection of violations of instrument validity for a class of data generating processes, and we provided an algorithm that satisfies this restriction. However, other possibilities exist, and the

optimal choice of distillation process is an open question.

Appendix

A Proofs for Propositions 1 and 2

Proof of Proposition 1. Fix the value of the conditioning variable X at x, and consider an arbitrary Borel subset $A \subset \mathcal{Y}$. Then, the additive error representation of the selection equation in (173) implies that the event $\{Y \in A, D = 1 | X = x, Z = z\}$ is equivalent to $\{Y_1 \in A, U_D \leq p | p(X, Z) = p, X = x, Z = z\}$. We therefore obtain

$$\begin{aligned} \Pr(Y \in A, D = 1 | X = x, Z = z) &= \Pr(Y_1 \in A, U_D \leq p | p(X, Z) = p, X = x, Z = z) \\ &= \Pr(Y_1 \in A, U_D \leq p | X = x), \end{aligned}$$

where the second equality follows from the random assignment condition (A2). Similarly, for $D = 0$ case, we have

$$\Pr(Y \in A, D = 0 | X = x, Z = z) = \Pr(Y_0 \in A, U_D > p | X = x).$$

These prove index sufficiency, i.e., conditional on X, the distribution of (Y, D) depends on Z through the propensity score $p(X, Z)$ only.

To obtain the nesting inequalities, we note

$$\begin{aligned} & \Pr(Y \in A, D = 1 | p(Z, X) = p, X = x) - \Pr(Y \in A, D = 1 | p(Z, X) = p', X = x) \\ = \; & \Pr(Y_1 \in A, U_D \leq p | X = x) - \Pr(Y_1 \in A, U_D \leq p' | X = x) \\ = \; & \Pr(Y_1 \in A, p' < U_D \leq p | X = x) \geq 0, \end{aligned} \tag{58}$$

follows. The inequality for $D = 0$ case is proven in a similar manner. ■

Proof of Proposition 2. Noting that the event $\{U \in A, D = 1 | X = x, Z = z\}$ is equivalent to $\{U \in A, U_D \leq p | p(X, Z) = p, X = x, Z = z\}$, we have

$$\begin{aligned} \Pr(U \in A, D = 1 | X = x, Z = z) &= \Pr(U_1 \in A, U_D \leq p | p(X, Z) = p, X = x, Z = z) \\ &= \Pr(U_1 \in A, U_D \leq p), \end{aligned}$$

where the second equality follows from the strong exogeneity condition of (A4). A similar argument gives

$$\Pr(U \in A, D = 0 | X = x, Z = z) = \Pr(U_0 \in A, U_D > p)$$

Hence, we obtain index sufficiency: the distribution of (U, D) depends on (X, Z) through the propensity score $p(X, Z)$ only.

To obtain the nesting inequalities, by (8) and recalling that U_D is uniformly distributed, we have

$$\begin{aligned}
&\Pr(U \in A, D = 1 | p(X, Z) = p) - \Pr(U \in A, D = 1 | p(X, Z) = p') \\
=&\Pr(U_1 \in A, U_D \leq p) - \Pr(U_1 \in A, U_D \leq p') \\
=&\Pr(U_1 \in A, p' < U_D \leq p) \\
\geq& 0. \qquad (59)
\end{aligned}$$

The other inequality of the proposition can be shown similarly. ■

B Proof of Proposition 4

This section presents a proof for Proposition 4. We first introduce notation, then show several lemmas, and then use these lemmas to prove each of the three statements of Proposition 4. We prove the statements of Proposition 4. for $D = 1$. The proof for $D = 0$ is identical.

B.1 Notation

Let P be an inerval of propensity scores. Define

$$\begin{aligned}
\mathcal{F}(P, z) &\equiv \int_P \Pr(U \in A, D = 1 | p, Z = z) f(p | p \in P, Z = z) dp, \\
w(z|P) &\equiv \Pr(Z = z | p \in P).
\end{aligned}$$

The joint probability of $(U \in A, D = 1)$ conditional on $p \in P$ may be written as

$$\Pr(U \in A, D = 1 | p \in P) = (1 - w(1|P))\mathcal{F}(P, 0) + w(1|P)\mathcal{F}(P, 1). \qquad (60)$$

Thus the nesting inequality with a coarsened propensity score, (20), subtracts a weighted sum of $\mathcal{F}(P^+, 0)$ and $\mathcal{F}(P^+, 1)$ from a weighted sum of $\mathcal{F}(P^-, 0)$ and $\mathcal{F}(P^-, 1)$

$$(1 - w(1|P^-))\mathcal{F}(P^-, 0) + w(1|P^-)\mathcal{F}(P^-, 1) - (1 - w(1|P^+))\mathcal{F}(P^+, 0) - w(1|P^+)\mathcal{F}(P^+, 1). \quad (61)$$

We refer to (61) as the coarsened propensity score nesting violation.

Define

$$\begin{aligned}
\mathcal{G}(P, z) &\equiv \int_P \Pr(U \in A, D = 1 | p, S_1 = 1, Z = z) f(p | p \in P, S_1 = 1, Z = z) dp, \\
v(P|z) &\equiv \Pr(p \in P | Z = z, S_1 = 1, p \in P^- \cup P^+).
\end{aligned}$$

The joint probability of $(U \in A, D = 1)$ conditional on $Z = z$, $S_1 = 1$ and $p \in P^- \cup P^+$ can be written

$$\Pr(U \in A, D = 1 | Z = z, p \in P^- \cup P^+) = v(P^-|z)\mathcal{G}(P^-, z) + v(P^+|z)\mathcal{G}(P^+, z). \tag{62}$$

Thus the nesting inequality with a distilled sample, (22), subtracts a weighted sum of $\mathcal{G}(P^-, 1)$ and $\mathcal{G}(P^+, 1)$ from a weighted sum of $\mathcal{G}(P^-, 0)$ and $\mathcal{G}(P^+, 0)$,

$$v(P^-|0)\mathcal{G}(P^-, 0) + v(P^+|0)\mathcal{G}(P^+, 0) - v(P^-|1)\mathcal{G}(P^-, 1) - v(P^+|1)\mathcal{G}(P^+, 1). \tag{63}$$

We refer to (63) as the distilled nesting violation.

B.2 Lemmas

Lemma 1 *The labeling condition of Proposition 4 implies*

$$w(1|P^+) \geq w(1|P^-). \tag{64}$$

Proof of Lemma 1.

Write out the labeling condition (24) and rearrange

$$\begin{aligned}
&\frac{\Pr(Z = 1, p \in P^+)}{\Pr(Z = 1, p \in P^+) + \Pr(Z = 1, p \in P^-)} \geq \frac{\Pr(Z = 0, p \in P^+)}{\Pr(Z = 0, p \in P^-) + \Pr(Z = 0, p \in P^+)} \\
\implies &\Pr(Z = 1, p \in P^+) \Pr(Z = 0, p \in P^-) \geq \Pr(Z = 1, p \in P^-) \Pr(Z = 0, p \in P^+) \\
\implies &\Pr(Z = 1, p \in P^+) \Pr(p \in P^-) \geq \Pr(Z = 1, p \in P^-) \Pr(p \in P^+) \\
\implies &\Pr(Z = 1 | p \in P^+) \geq \Pr(Z = 1 | p \in P^-).
\end{aligned}$$

The third line adds $\Pr(Z = 1, p \in P^+) \Pr(Z = 1, p \in P^-)$ to both sides and simplifies. By definition, the left hand side of the last line $w(1|P^+)$, and the right hand side is $w(1|P^-)$. ■

Lemma 2 *The labeling condition and trimming rule jointly imply*

$$v(P^+|1) \geq v(P^+|0). \tag{65}$$

Proof of Lemma 2.

Consider first $v(P^+|1)$

$$\begin{aligned}
v(P^+|1) &\equiv \frac{\Pr(Z = 1, p \in P^+, S_1 = 1)}{\Pr(Z = 1, p \in P^+, S_1 = 1) + \Pr(Z = 1, p \in P^-, S_1 = 1)} \\
&= \frac{\Pr(Z = 1, p \in P^+)}{\Pr(Z = 1, p \in P^+) + \Pr(Z = 1, p \in P^-, S_1 = 1)} \\
&\geq \frac{\Pr(Z = 1, p \in P^+)}{\Pr(Z = 1, p \in P^+) + \Pr(Z = 1, p \in P^-)}
\end{aligned}$$

The second line uses that the trimming under imposes that all observations with $Z = 1$ and $p \in P^+$ are retained. The third line follows from $\Pr(Z = 1, p \in P^-) \geq \Pr(Z = 1, p \in P^-, S_1 = 1) \geq 0$. The expression on the last line is equivalent to $\Pr(p \in P^+|Z = 1)$, hence

$$v(P^+|1) \geq \Pr(p \in P^+|Z = 1) \tag{66}$$

Following the same steps with $v(P^+|0)$, and using that the trimming rule imposes that all observations $Z = 0$ and $p \in P^-$ are retained yields

$$\Pr(p \in P^+|Z = 0) \geq v(P^+|0) \tag{67}$$

Combine (66), (67) and the labeling condition (24)

$$v(P^+|1) \geq \Pr(p \in P^+|Z = 1) \geq \Pr(p \in P^+|Z = 0) \geq v(P^+|0)$$

■

Lemma 3 *The conditional nesting assumption implies*

$$\mathcal{F}(P^+, z) \geq \mathcal{F}(P^-, z), \tag{68}$$

for $z \in \{0, 1\}$.

Proof of Lemma 3.

Let $\hat{p}$ be the upper bound of P^-. Equation (25) implies

$$\begin{aligned}
\mathcal{F}(P^+, z) &\equiv \int_{P^+} \Pr(U \in A, D = 1|Z = z, p) f(p|p \in P^+, Z = z) dp \\
&\geq \Pr(U \in A, D = 1|Z = z, p = \hat{p}) \\
&\geq \int_{P^-} \Pr(U \in A, D = 1|Z = z, p) f(p|p \in P^-, Z = z) dp \\
&\equiv \mathcal{F}(P^-, z).
\end{aligned}$$

■

Lemma 4 *Under the trimming rule*

$$\mathcal{G}(P^-, 0) = \mathcal{F}(P^-, 0), \tag{69}$$

$$\mathcal{F}(P^+, 1) = \mathcal{G}(P^+, 1). \tag{70}$$

Proof of Lemma 4.

Equation (69) is shown as follows

$$\begin{aligned}\mathcal{G}(P^-,0) &\equiv \int_{P^-} \Pr(U \in A, D=1|p, S_1=1, Z=0) f(p|p \in P^- S_1=1, Z=0)dp, \\ &= \int_{P^-} \Pr(U \in A, D=1|p, Z=0) f(p|p \in P^-, Z=0)dp \\ &\equiv \mathcal{F}(P^-,0).\end{aligned}$$

The second line uses that, under the trimming rule, $S_1 = 1$ for all observations with $Z = 0$ and $p \in P^-$. By using that $S_1 = 1$ for all observations with $Z = 1$ and $p \in P^+$, (70) can be shown similarly. ■

Lemma 5 *The conditional nesting assumption and trimming rule jointly imply*

$$\mathcal{G}(P^+,1) \geq \mathcal{G}(P^-,1), \tag{71}$$

and either

$$\mathcal{G}(P^+,0) \geq \mathcal{G}(P^-,0), \tag{72}$$

$$v(P^+,0) > 0, \tag{73}$$

or

$$\mathcal{G}(P^+,0) = 0, \tag{74}$$

$$v(P^+,0) = 0. \tag{75}$$

Proof of Lemma 5.

To show (71), use Lemma 4 to impose $\mathcal{F}(P^+,1) = \mathcal{G}(P^+,1)$. The inequality can then be shown by following the proof for Lemma 3.

For (72) and (74), note that under the trimming rule it is possible that all observations with $Z = 0, p \in P^+$ are trimmed. If the trimmed sample includes observations with $Z = 0, p \in P^+$, then $\Pr(p \in P^+, Z = 1, S_1 = 1) > 0$, which implies (73). Equation (72) can ten be shown by following the proof for Lemma 3. If all observations with $Z = 0, p \in P^+$ are trimmed, then $\Pr(p \in P^+, Z = 1, S_1 = 1) = 0$, and (74) and (75) follow immediately. ■

B.3 Proof of Proposition 4

Proof of Statement 1.

Suppose that the coarsened propensity score nesting violation (61) is positive. For a weighted sum of $\mathcal{F}(P^-,0)$ and $\mathcal{F}(P^-,1)$ to exceed a weighted sum of $\mathcal{F}(P^+,0)$ and $\mathcal{F}(P^+,1)$, it must be that

$$\max(\mathcal{F}(P^-,0),\mathcal{F}(P^-,1)) > \min(\mathcal{F}(P^+,0),\mathcal{F}(P^+,1)). \tag{76}$$

Combining (76) and Lemma 3, there are two possible orderings of $\mathcal{F}(P^+,0)$, $\mathcal{F}(P^-,0)$, $\mathcal{F}(P^+,1)$, and $\mathcal{F}(P^-,1)$

$$\mathcal{F}(P^-,0) \leq \mathcal{F}(P^+,0) \leq \mathcal{F}(P^-,1) \leq \mathcal{F}(P^+,1), \tag{77}$$

$$\mathcal{F}(P^-,1) \leq \mathcal{F}(P^+,1) \leq \mathcal{F}(P^-,0) \leq \mathcal{F}(P^+,0). \tag{78}$$

Rewrite (61) as

$$w(1|P^-)(\mathcal{F}(P^-,1)-\mathcal{F}(P^+,1)) + (1-w(1|P^-))(\mathcal{F}(P^-,0)-\mathcal{F}(P^+,0)) \\ -(w(1|P^+)-w(1|P^-))(\mathcal{F}(P^+,1)-\mathcal{F}(P^+,0)). \tag{79}$$

Lemma 3 implies that the first two terms are (weakly) negative. Turning to the final term, by Lemma 1

$$w(1|P^+)-w(1|P^-) \geq 0,$$

If (77) holds, $\mathcal{F}(P^+,1) > \mathcal{F}(P^+,0)$, and this last term will also be negative, which implies that the nesting inequality for a coarsened propensity can never be positive. Hence, for this nesting inequality to be positive, (78) must hold.

Assuming (78) holds, we obtain the inequality

$$v(P^+|0)\mathcal{G}(P^+,0) + v(P^-|0)\mathcal{G}(P^-,0) \geq \mathcal{F}(P^-,0) \\ \geq (1-w(1|P^-))\mathcal{F}(P^-,0) + w(1|P^-)\mathcal{F}(P^-,1). \tag{80}$$

The first inequality holds by Lemma 4 and Lemma 5. The second inequality follows directly from (78). Maintaining (78), we can obtain the additional inequality

$$(1-w(1|P^+))\mathcal{F}(P^+,0) + w(1|P^+)\mathcal{F}(P^+,1) \geq \mathcal{F}(P^+,1) \\ \geq v(P^+|1)\mathcal{G}(P^+,1) + v(P^-|1)\mathcal{G}(P^-,1). \tag{81}$$

Here the first inequality follows from (78), while the second uses Lemma 4 and Lemma 5.

Summing (80) and (81)

$$v(P^+|0)\mathcal{G}(P^+,0) + v(P^-|0)\mathcal{G}(P^-,0) + (1-w(1|P^+))\mathcal{F}(P^+,0) + w(1|P^+)\mathcal{F}(P^+,1) \geq \\ (1-w(1|P^-))\mathcal{F}(P^-,0) + w(1|P^-)\mathcal{F}(P^-,1) + v(P^+|1)\mathcal{G}(P^+,1) + v(P^-|1)\mathcal{G}(P^-,1),$$

which can be rearranged to

$$v(P^+|0)\mathcal{G}(P^+,0) + v(P^-|0)\mathcal{G}(P^-,0) - v(P^+|1)\mathcal{G}(P^+,1) - v(P^-|1)\mathcal{G}(P^-,1) \geq$$
$$(1 - w(1|P^-))\mathcal{F}(P^-,0) + w(1|P^-)\mathcal{F}(P^-,1) - (1 - w(1|P^+))\mathcal{F}(P^+,0) - w(1|P^+)\mathcal{F}(P^+,1).$$

The left-hand side is the distilled nesting violation (63). The right-hand side is the coarsened propensity score nesting violation (61).

■

Proof of Statement 2.

Our strategy to show the second statement is as follows. First, we make assumptions such that, for a given A, the coarsened propensity score nesting violation (61) is not positive. We then show that this does not preclude the distilled nesting violation (63) from being positive.

The above argument established that (78) must hold for the nesting inequality conditional on bins of p to be positive. Therefore, to guarantee that the nesting inequality conditional on the coarsened propensity score is negative, we assume that (78) does *not* hold i.e.

$$\mathcal{F}(P^-,0) < \mathcal{F}(P^+,1). \tag{82}$$

Write the distilled nesting violation (63) as

$$v(P^-|1)(\mathcal{G}(P^-,0) - \mathcal{G}(P^-,1)) + (v(P^-|0) - v(P^-|1))(\mathcal{G}(P^-,0) - \mathcal{G}(P^+,1))$$
$$+ (1 - v(P^-|0))(\mathcal{G}(P^+,0) - \mathcal{G}(P^+,1)). \tag{83}$$

Expressed this way, (63) is a weighted sum of the differences $\mathcal{G}(P^-,0) - \mathcal{G}(P^-,1)$, $\mathcal{G}(P^-,0) - \mathcal{G}(P^+,1)$ and $\mathcal{G}(P^+,0) - \mathcal{G}(P^+,1)$. Lemma 2 guarantees that $v(P^-|0) - v(P^-|1) \geq 0$, so the weights are all (weakly) positive. By Lemma 4, (82) is equivalent to

$$\mathcal{G}(P^-,0) < \mathcal{G}(P^+,1). \tag{84}$$

Assuming (84), (83) can only be positive if at least one of $\mathcal{G}(P^-,0) - \mathcal{G}(P^-,1)$ and $\mathcal{G}(P^+,0) - \mathcal{G}(P^+,1)$ is positive.

Take the four $\mathcal{G}(P,z)$ as given and solve for $v(P^-|0)$ and $v(P^-|1)$ such that (83) is positive subject to the constraints

$$0 \leq v(P^-|0) \leq 1,$$
$$0 \leq v(P^-|1) \leq 1,$$
$$v(P^-|1) \leq v(P^-|0).$$

Doing so yields the following bounds for $v(P^-|0)$

$$v(P^-|1) \leq v(P^-|0) \leq \min\left\{1, \frac{v(P^-,1)(\mathcal{G}(P^+,1) - \mathcal{G}(P^-,1)) + \mathcal{G}(P^+,0) - \mathcal{G}(P^+,1)}{\mathcal{G}(P^+,0) - \mathcal{G}(P^-,0)}\right\}. \quad (85)$$

and for $v(P^-|1)$

$$\mathbb{1}\left\{\mathcal{G}(P^+,0) \leq \mathcal{G}((P^+,1)\right\}\bar{v} \leq v(P^-|1) \leq 1 - \mathbb{1}\left\{\mathcal{G}(P^-,0) \leq \mathcal{G}(P^-,1)\right\}(1+\bar{v}). \quad (86)$$

with

$$\bar{v} \equiv \frac{\mathcal{G}(P^+,1) - \mathcal{G}(P^+,0)}{\mathcal{G}(P^+,1) - \mathcal{G}(P^-,1) - \mathcal{G}(P^+,0) + \mathcal{G}(P^-,0)}.$$

Taking $v(P^-|1)$ as given, (85) gives bounds for $v(P^-|0)$ such that (83) is positive. Equation (86) gives bounds for $v(P^-|1)$ such that the bounds in (85) are non-empty. If $\mathcal{G}(P^-,0) \geq \mathcal{G}(P^-,1)$ and $\mathcal{G}(P^+,0) \geq \mathcal{G}(P^+,1)$, for any $v(P^-|1)$ between 0 and 1 there exists $v(P^-|0)$ such that (83) is positive. If $\mathcal{G}(P^-,0) \geq \mathcal{G}(P^-,1)$ and $\mathcal{G}(P^+,0) \leq \mathcal{G}(P^+,1)$, a $v(P^-|0)$ where (83) is positive exists if $v(P^-|1)$ is above $\bar{v}$. If $\mathcal{G}(P^-,0) \leq \mathcal{G}(P^-,1)$ and $\mathcal{G}(P^+,0) \geq \mathcal{G}(P^+,1)$, a $v(P^-|0)$ where (83) is positive exists if $v(P^-|1)$ is below $-\bar{v}$. ■

Proof of Statement 3.

To prove the third statement, we first find a sufficient condition for the coarsened propensity score nesting violation (61) to be negative for any A. We then show that it is possible to construct a process where this condition holds and the distilled nesting violation (63) is positive.

Consider again (61). Lemma 1 and Lemma 3 together imply

$$w(1|P^-)\mathcal{F}(P^-,1) \leq w(1|P^+)\mathcal{F}(P^+,1).$$

A sufficient condition for (61) to be negative for a given A is then

$$(1 - w(1|P^-))\mathcal{F}(P^-,0) \leq (1 - w(1|P^+))\mathcal{F}(P^+,0).$$

and a sufficient condition for (61) to be negative for all A where $\mathcal{F}(P^+,0) > 0$ is

$$\sup_A \left\{\frac{\mathcal{F}(P^-,0)}{\mathcal{F}(P^+,0)}\right\} \leq \frac{1 - w(1|P^+)}{1 - w(1|P^-)}. \quad (87)$$

Note that Lemma 3 implies the left-hand side is weakly below 1. Hence (87) will always hold if $w(1|P^+) = w(1|P^-)$. That is, if the mixture of $Z = 0$ and $Z = 1$ observations is sufficiently similar in P^- and P^+, then (61) will never be positive.

Now consider the distilled nesting violation (63). Assume that (78) holds. Lemmas 2, 3, 4 and 5 together imply

$$v(P^-|0)\mathcal{G}(P^-,0) \geq v(P^-|1)\mathcal{G}(P^-,1).$$

A sufficient condition for (63) to be positive at a given A is

$$v(P^+|0)\mathcal{G}(P^+,0) \geq v(P^+|1)\mathcal{G}(P^+,1).$$

A sufficient condition for (63) to be positive for some A where $\mathcal{G}(P^+,1) > 0$ is thus

$$\sup_A \left\{ \frac{\mathcal{G}(P^+,0)}{\mathcal{G}(P^+,1)} \right\} \geq \frac{v(P^+|1)}{v(P^+|0)}. \tag{88}$$

For any A where (78) holds, the left-hand side is bounded from below by 1. Equation (88) thus states that, if (78) holds for some A and the mixture of observations from P^- and P^+ is sufficiently similar for $Z = 0$ and $Z = 1$, then (63) is positive for that A.

A case where (87) and (88) are immediate is when there exists A such that $\mathcal{G}(P^+,0) > \mathcal{G}(P^+,1)$ and Z is an irrelevant instrument that is independent of X. Then

$$w(1|P^+) = w(1|P^-),$$
$$v(P^+|1) = v(P^+|0).$$

Hence, any violation can only be detected by conditioning on Z. ■

C Numerical Example

This section derives properties of the conditional joint densities of partial residuals and treatment assignment for the process introduced in Section 2.5.

Let $\tilde{\theta}_1$ and $\tilde{\theta}_0$ be the asymptotic bias of the estimates of θ_1 and θ_0

$$\tilde{\theta}_1 = \hat{\theta}_1^* - \theta_1,$$
$$\tilde{\theta}_0 = \hat{\theta}_0^* - \theta_0.$$

Using this notation , the asymptotic partial residuals are

$$U = \nu Z - \nu(1-Z) + D(\bar{U}_1 - X'\tilde{\theta}_1) + (1-D)(\bar{U}_0 - X'\tilde{\theta}_0).$$

Given process of Section 2.5, the following expressions can be derived for the joint density of $(U, D = 1)$ conditional on p and Z

$$f(U=u, D=1|p, Z=1) = \int_{-\Phi^{-1}(p)}^{\infty} \phi(U_D) \frac{1}{\sqrt{2\pi\sigma^2_{\hat{U}_1|U_D,Z=1}}} \exp\left(- \frac{\left(u - \nu - \mu_{\tilde{U}_1|U_D}(U_D) - \frac{\delta_1'\tilde{\theta}_1}{\delta_1'\delta_1} X'\delta_1\right)^2}{2\sigma^2_{\hat{U}_1|U_D,Z=1}} \right) dU_D,$$

$$f(U=u, D=1|p, Z=0) = \int_{-\Phi^{-1}(p)}^{\infty} \phi(U_D) \frac{1}{\sqrt{2\pi\sigma^2_{\hat{U}_1|U_D,Z=0}}} \exp\left(- \frac{\left(u + \nu - \mu_{\tilde{U}_1|U_D}(U_D) - \frac{\delta_0'\tilde{\theta}_1}{\delta_0'\delta_0} X'\delta_0\right)^2}{2\sigma^2_{\hat{U}_1|U_D,Z=0}} \right) dU_D.$$

where $\phi(\cdot)$ is the standard univariate normal density and

$$\mu_{\tilde{U}_1|U_D}(U_D) = \mu_1 + \sigma_1 \rho_{D,1} U_D,$$
$$\sigma^2_{\tilde{U}_1|U_D,Z=1} = \sigma_1^2(1-\rho_{D,1}^2) + \left(1 - \frac{(\tilde{\theta}_1'\delta_1)^2}{\tilde{\theta}_1'\tilde{\theta}_1\delta'\delta}\right)\tilde{\theta}_1'\tilde{\theta}_1,$$
$$\sigma^2_{\tilde{U}_1|U_D,Z=0} = \sigma_1^2(1-\rho_{D,1}^2) + \left(1 - \frac{(\tilde{\theta}_1'\delta_0)^2}{\tilde{\theta}_1'\tilde{\theta}_1\delta'\delta}\right)\tilde{\theta}_1'\tilde{\theta}_1,$$

with

$$X'\delta_1 = \Phi^{-1}(p) - \alpha, \tag{89}$$
$$X'\delta_0 = \Phi^{-1}(p) + \alpha. \tag{90}$$

The expressions for the conditional joint densities of $(U, D = 0)$ are analogous. If the asymptotic estimates are unbiased (i.e. $\tilde{\theta}_1 = 0$), then the propensity score only enters through the lower bound of each integral. The derivative of the joint density with respect to p is then weakly positive for all A and p and both $Z = 0$ and $Z = 1$. Condition (iv) of Proposition 4 must then hold for any non-overlapping intervals of p we consider. However, if $\tilde{\theta}_1 \neq 0$, there is an additional effect through (89) and (90), changing p affects $X'\delta_1$ and $X'\delta_0$. Intuitively, with biased estimates, partialling out does not remove the effect of covariates from the residuals. Changing propensity scores changes the distribution of covariates, which shifts the distribution of partial residuals.

$X'\delta_1$ and $X'\delta_0$ enter the density multiplied by $\tilde{\theta}_1'\delta_1$ and $\tilde{\theta}_1'\delta_0$ respectively. Rather than characterising $\tilde{\theta}_1$, we characterise these quantities. The following proposition holds

Proposition 7 *Consider the terms $\delta_1'\tilde{\theta}_1$, $\delta_0'\tilde{\theta}_1$, $\delta_1'\tilde{\theta}_0$ and $\delta_0'\tilde{\theta}_0$. Assume*
(i) The partially linear model is estimated as described in Section 3.1.2.
(ii) The propensity score function $p(X,Z)$ and the conditional expectations $E[X|p(X,Z)]$ and $E[Y|p(X,Z)]$ are consistently estimated.
(iii) There is no perfect multicollinearity among the components of $X - E[X|p(X,Z)]$.
(iv) The matrix E defined in (125) *and the matrix G defined in* (145) *are invertible.*
Then

1. *The following equalities hold*

$$\tilde{\theta}_1'\delta_1 = \tilde{\theta}_0'\delta_0,$$
$$\tilde{\theta}_0'\delta_1 = \tilde{\theta}_1'\delta_0.$$

2. *$\tilde{\theta}_1'\delta_1$ and $\tilde{\theta}_1'\delta_0$ have the form*

$$\tilde{\theta}_1'\delta_1 = \mathcal{H}_1(\alpha, \rho_\delta, \delta'\delta)\nu,$$
$$\tilde{\theta}_1'\delta_0 = \mathcal{H}_0(\alpha, \rho_\delta, \delta'\delta)\nu.$$

where $\mathcal{H}_1$ and $\mathcal{H}_0$ are functions. In the numerical exercise, the matrices introduced in assumption (iv) are always invertible. The first statement implies that the coefficients for $X'\delta_1$ and $X'\delta_0$ in the subdensities for $(U, D = 0)$ will mirror those in the subdensities for $(U, D = 1)$, so it is sufficient to only examine one set of subdensities. The second statement implies that, once we have fixed $\nu, \alpha, \delta'\delta$ and ρ_δ, the actual values of δ_0 and δ_1 and all other parameters are irrelevant.

C.1 Proof of Proposition 7

The proof proceeds as follows. First, we state several properties of density of the propensity score and the conditional density of X given the propensity score. These properties are then used to show a series of lemmas. Finally, the lemmas are used to show the two statements of Proposition 7.

C.1.1 Propensity Score Density

The propensity score function for this process is

$$p(X, Z) = \Phi(Z(X'\delta_1 + \alpha) + (1 - Z)(X'\delta_0 - \alpha)). \tag{91}$$

Given the distributions of X and Z, and this propensity score function, the joint density of p and Z is

$$f(p, Z = 0) = \frac{1}{2}\frac{1}{\sqrt{2\pi\delta'\delta}}\exp\left(-\frac{(\Phi^{-1}(p) + \alpha)^2}{2\delta'\delta}\right)\frac{1}{\phi(\Phi^{-1}(p))}, \tag{92}$$

$$f(p, Z = 1) = \frac{1}{2}\frac{1}{\sqrt{2\pi\delta'\delta}}\exp\left(-\frac{(\Phi^{-1}(p) - \alpha)^2}{2\delta'\delta}\right)\frac{1}{\phi(\Phi^{-1}(p))}. \tag{93}$$

Summing (92) and (93), the unconditional density of the propensity score is

$$f(p) = \frac{1}{\sqrt{2\pi\delta'\delta}}\exp\left(-\frac{\Phi^{-1}(p)^2 + \alpha^2}{2\delta'\delta}\right)\cosh\left(\frac{\Phi^{-1}(p)\alpha}{\delta'\delta}\right)\frac{1}{\phi(\Phi^{-1}(p))}. \tag{94}$$

Notice that this density satisfies

$$f(p) = f(1 - p), \tag{95}$$

which implies

$$\int_0^1 p^2 f(p)dp = \int_0^1 (1 - p)^2 f(p)dp. \tag{96}$$

Combining (93) and (94), the probability of $Z = 1$ conditional on the propensity score is

$$\Pr(Z = 1|p) = \frac{1}{2} + \frac{\tanh(\alpha\Phi^{-1}(p)/\delta'\delta)}{2}, \tag{97}$$

which satisfies

$$\Pr(Z = 1|p) = 1 - \Pr(Z = 1|1 - p). \tag{98}$$

The symmetry relations (95) and (98), and $\Pr(Z=0|p) = 1 - \Pr(Z=1|p)$ together imply

$$\int_0^1 p^2 \Pr(Z=1|p) f(p) dp = \int_0^1 (1-p)^2 \Pr(Z=0|p) f(p) dp, \tag{99}$$

$$\int_0^1 p^2 \Pr(Z=0|p) f(p) dp = \int_0^1 (1-p)^2 \Pr(Z=1|p) f(p) dp, \tag{100}$$

$$\int_0^1 p(1-p) \Pr(Z=1|p) f(p) dp = \int_0^1 p(1-p) \Pr(Z=0|p) f(p) dp, \tag{101}$$

$$\int_0^1 p \Pr(z=0|p) \Pr(z=1|p) f(p) dp = \int_0^1 (1-p) \Pr(z=0|p) \Pr(z=1|p) f(p) dp, \tag{102}$$

$$\int_0^1 p^2 \Pr(Z=0|p) \Pr(Z=1|p) f(p) dp = \int_0^1 (1-p)^2 \Pr(Z=0|p) \Pr(Z=1|p) f(p) dp. \tag{103}$$

Finally, as $\Phi^{-1}(p) = -\Phi^{-1}(1-p)$ and $\Phi^{-1}(p)^2 = \Phi^{-1}(1-p)^2$

$$\int_0^1 p(1-p) \Pr(Z=1|p) \Pr(Z=0|p) \Phi^{-1}(p) f(p) dp = 0, \tag{104}$$

$$\int_0^1 (1-p) \Pr(z=0|p) \Pr(z=1|p) \Phi^{-1}(p) f(p) dp = -\int_0^1 p \Pr(z=0|p) \Pr(z=1|p) \Phi^{-1}(p) f(p) dp, \tag{105}$$

$$\int_0^1 p^2 \Pr(Z=0|p) \Pr(Z=1|p) \Phi^{-1}(p) f(p) dp = -\int_0^1 (1-p)^2 \Pr(Z=0|p) \Pr(Z=1|p) \Phi^{-1}(p) f(p) dp, \tag{106}$$

$$\int_0^1 p^2 \Pr(Z=0|p) \Pr(Z=1|p) \Phi^{-1}(p)^2 f(p) dp = \int_0^1 (1-p)^2 \Pr(Z=0|p) \Pr(Z=1|p) \Phi^{-1}(p)^2 f(p) dp. \tag{107}$$

Notice that the only parameters which enter the propensity score density and conditional distribution of Z are α and $\delta'\delta$. As a consequence, these are the only parameters which affect the integrals above.

C.1.2 Conditional Density of X

Given the propensity score function (91), conditioning on $(p, Z=1)$ is equivalent to conditioning on $X'\delta_1 = \Phi^{-1}(p) - \alpha$, and conditioning on $(p, Z=0)$ is equivalent to conditioning on $X'\delta_0 = \Phi^{-1}(p) + \alpha$. Using that X is normally distributed

$$E[X|p, Z=1] = E[X|X'\delta_1 = \Phi^{-1}(p) - \alpha] = \frac{\delta_1}{\delta'\delta}(\Phi^{-1}(p) - \alpha), \tag{108}$$

$$E[X|p, Z=0] = E[X|X'\delta_0 = \Phi^{-1}(p) + \alpha] = \frac{\delta_0}{\delta'\delta}(\Phi^{-1}(p) + \alpha), \tag{109}$$

$$E[X|p] = \Pr(Z=1|p)\frac{\delta_1}{\delta'\delta}(\Phi^{-1}(p) - \alpha) + \Pr(Z=0|p)\frac{\delta_0}{\delta'\delta}(\Phi^{-1}(p) + \alpha). \tag{110}$$

Let

$$\text{Cov}(X - E[X|p,Z]|p,Z) \equiv E\left[(X - E[X|p,Z])(X - E[X|p,Z])'|p,Z\right],$$

then

$$\text{Cov}(X - E[X|p,Z]|p,Z=1) = \text{Cov}(X - E[X|p,Z]|X'\delta_1 = \Phi^{-1}(p) - \alpha) = I - \frac{\delta_1\delta_1'}{\delta'\delta}, \quad (111)$$

$$\text{Cov}(X - E[X|p,Z]|p,Z=0) = \text{Cov}(X - E[X|p,Z]|X'\delta_0 = \Phi^{-1}(p) + \alpha) = I - \frac{\delta_0\delta_0'}{\delta'\delta}. \quad (112)$$

Similarly, define

$$\text{Cov}(X - E[X|p]|p) \equiv E[(X - E[X|p])(X - E[X|p])'|p].$$

$\text{Cov}(X - E[X|p]|p)$ satisfies the decomposition

$$\text{Cov}(X - E[X|p]|p) = E[\text{Cov}(X - E[X|p,Z]|p,Z)|p] + E[\text{Cov}(E[X|p,Z] - E[X|p]|p,Z)|p].$$

Using (108), (109), (110), (111) and (112) and simplifying

$$\begin{aligned}\text{Cov}(X - E[X|p]|p) &= I - \Pr(Z=1|p)\frac{\delta_1\delta_1'}{\delta'\delta} - \Pr(Z=0|p)\frac{\delta_0\delta_0'}{\delta'\delta} \\ &+ \frac{\delta_1\delta_1'}{(\delta'\delta)^2}(\Phi^{-1}(p) - \alpha)^2 - \frac{\delta_0\delta_1' + \delta_1\delta_0'}{(\delta'\delta)^2}(\Phi^{-1}(p) - \alpha)(\Phi^{-1}(p) + \alpha) + \frac{\delta_0\delta_0'}{(\delta'\delta)^2}(\Phi^{-1}(p) + \alpha)^2. \quad (113)\end{aligned}$$

C.1.3 Lemmas

Let $\mathbf{X}$ be a column vector of length $2k_X$ which stacks $p(X - E[X|p])$ and $(1-p)(X - E[X|p])$. Let $\mathbf{Z} \equiv Z - E[Z|p]$.

Lemma 6

$$E[\mathbf{XZ}] = \begin{pmatrix} w_1\delta_1 + w_0\delta_0 \\ w_1\delta_0 + w_0\delta_1 \end{pmatrix},$$

where w_0 and w_1 are scalars that depend on α and $\delta'\delta$ only.

Lemma 7

$$E[\mathbf{XX}']^{-1} = \begin{pmatrix} \Omega_1 & \Omega_2 \\ \Omega_2' & \Omega_3 \end{pmatrix},$$

where Ω_1, Ω_2, and Ω_3 are $K \times K$ matrices.

Lemma 8 *The following equalities hold*

$$\begin{aligned}\delta_1'\Omega_1\delta_1 &= \delta_0'\Omega_3\delta_0, \\ \delta_0'\Omega_1\delta_0 &= \delta_1'\Omega_3\delta_1, \\ \delta_0'\Omega_1\delta_1 &= \delta_1'\Omega_3\delta_0, \\ \delta_1'\Omega_1\delta_0 &= \delta_0'\Omega_3\delta_1, \\ \delta_1'\Omega_2\delta_1 &= \delta_0'\Omega_2'\delta_0.\end{aligned}$$

Lemma 9 *The constants $\delta_i'\Omega_k\delta_j$ for $i,j=\{0,1\}$, $k=\{1,2,3\}$ depend on α, $\delta'\delta$ and ρ_δ only.*

Proof of Lemma 6.

$$
\begin{aligned}
E[\mathbf{XZ}] &= \begin{pmatrix} E[p(X-E[X|p])(Z-E[Z|p]] \\ E[(1-p)(X-E[X|p])(Z-E[Z|p]]\end{pmatrix}, \\
&= \begin{pmatrix} \int_0^1 p\Pr(z=0|p)\Pr(z=1|p)(E[X|p,Z=1]-E[X|p,Z=0])f(p)dp \\ \int_0^1 (1-p)\Pr(z=0|p)\Pr(z=1|p)(E[X|p,Z=1]-E[X|p,Z=0])f(p)dp\end{pmatrix}, \\
&= \frac{1}{\delta'\delta}\begin{pmatrix} \int_0^1 p\Pr(z=0|p)\Pr(z=1|p)(\delta_1(\Phi^{-1}(p)-\alpha)-\delta_0(\Phi^{-1}(p)+\alpha))f(p)dp \\ \int_0^1 (1-p)\Pr(z=0|p)\Pr(z=1|p)(\delta_1(\Phi^{-1}(p)-\alpha)-\delta_0(\Phi^{-1}(p)+\alpha))f(p)dp\end{pmatrix}, \\
&= \begin{pmatrix} w_1\delta_1+w_0\delta_0 \\ w_1\delta_0+w_0\delta_1\end{pmatrix}, \qquad (114)
\end{aligned}
$$

where w_1 and w_0 are the scalars

$$
\begin{aligned}
w_1 &= \frac{\int_0^1 p\Pr(z=0|p)\Pr(z=1|p)(\Phi^{-1}(p)-\alpha)f(p)dp}{\delta'\delta}, \\
w_0 &= -\frac{\int_0^1 p\Pr(z=0|p)\Pr(z=1|p)(\Phi^{-1}(p)+\alpha)f(p)dp}{\delta'\delta}.
\end{aligned}
$$

The first line uses the definitions of $\mathbf{X}$ and $\mathbf{Z}$. The second line uses the law of iterated expectations, and the Z is binary so $E[Z|p]=\Pr(Z=1|p)$. The third uses (108) and (109). Equations (102) and (105) are used in the fourth.

w_1 and w_0 are integrals of functions that depend on α and $\delta'\delta$ only. Hence, they themselves will depend on α and $\delta'\delta$ only. ■

C.1.4 Characterising $E[\mathbf{XX}']$

By definition

$$
E[\mathbf{XX}'] = \begin{pmatrix} E[p^2(X-E[X|p])(X-E[X|p])'] & E[p(1-p)(X-E[X|p])(X-E[X|p])'] \\ E[p(1-p)(X-E[X|p])(X-E[X|p])'] & E[(1-p)^2(X-E[X|p])(X-E[X|p])']\end{pmatrix}.
$$

All four components of this matrix can be written in the form $E[p^i(1-p)^j(X-E[X|p])(X-E[X|p])']$ where i and j take values $\{0,1,2\}$. Notice that

$$
\begin{aligned}
E[p^i(1-p)^j(X-E[X|p])(X-E[X|p])'] &= \int_0^1 p^i(1-p)^j E[(X-E[X|p])(X-E[X|p])'|p]f(p)dp, \\
&= \int_0^1 p^i(1-p)^j \mathrm{Cov}(X-E[X|p]|p)f(p)dp.
\end{aligned}
$$

Using (113) to substitute for $\text{Cov}(X - E[X|p]|p)$, we can obtain

$$E[\mathbf{X}\mathbf{X}'] = \begin{pmatrix} \Psi_1 & \Psi_2 \\ \Psi_2 & \Psi_3 \end{pmatrix}, \tag{115}$$

$$\Psi_1 = Ia_1 + \delta_1\delta_1'a_2 + \delta_0\delta_0'a_3 + (\delta_0\delta_1' + \delta_1\delta_0')a_4,$$
$$\Psi_3 = Ia_1 + \delta_1\delta_1'a_3 + \delta_0\delta_0'a_2 + (\delta_0\delta_1' + \delta_1\delta_0')a_4,$$
$$\Psi_2 = Ib_1 + (\delta_1\delta_1' + \delta_0\delta_0')b_2 + (\delta_0\delta_1' + \delta_1\delta_0')b_3,$$

with

$$a_1 = \int p^2 f(p)dp,$$
$$a_2 = -\frac{\int p^2 \Pr(Z=1|p)f(p)dp}{\delta'\delta} + \frac{\int p^2 \Pr(Z=1|p)\Pr(Z=0|p)(\Phi^{-1}(p)-\alpha)^2 f(p)dp}{(\delta'\delta)^2},$$
$$a_3 = -\frac{\int p^2 \Pr(Z=0|p)f(p)dp}{\delta'\delta} + \frac{\int p^2 \Pr(Z=1|p)\Pr(Z=0|p)(\Phi^{-1}(p)+\alpha)^2 f(p)dp}{(\delta'\delta)^2},$$
$$a_4 = \frac{\int p^2 \Pr(Z=1|p)\Pr(Z=0|p)(\alpha^2 - \Phi^{-1}(p)^2)f(p)dp}{(\delta'\delta)^2},$$
$$b_1 = \int p(1-p)f(p)dp,$$
$$b_2 = -\frac{\int p(1-p)\Pr(Z=1|p)f(p)dp}{\delta'\delta} + \frac{\int p(1-p)\Pr(Z=1|p)\Pr(Z=0|p)(\Phi^{-1}(p)^2+\alpha^2)f(p)dp}{(\delta'\delta)^2},$$
$$b_3 = \frac{\int p(1-p)\Pr(Z=1|p)\Pr(Z=0|p)(\alpha^2 - \Phi^{-1}(p)^2)f(p)dp}{(\delta'\delta)^2},$$

where (96), (99), (100), (101), (103), (104), (106), and (107) have all been imposed. By pre-multiplying the components of $E[\mathbf{X}\mathbf{X}']$ by δ_1 and δ_0, we can obtain the following

$$\delta_1'\Psi_1 = c_1\delta_1' + c_0\delta_0', \tag{116}$$
$$\delta_0'\Psi_1 = d_0\delta_1' + d_1\delta_0', \tag{117}$$
$$\delta_1'\Psi_3 = d_1\delta_1' + d_0\delta_0', \tag{118}$$
$$\delta_0'\Psi_3 = c_0\delta_1' + c_1\delta_0', \tag{119}$$
$$\delta_1'\Psi_2 = e_1\delta_1' + e_0\delta_0', \tag{120}$$
$$\delta_0'\Psi_2 = e_0\delta_1' + e_1\delta_0', \tag{121}$$

where

$$c_0 = a_3\rho_\delta\delta'\delta + a_4\delta'\delta, \qquad d_0 = a_2\rho_\delta\delta'\delta + a_4\delta'\delta, \qquad e_0 = b_2\rho_\delta\delta'\delta + b_3\delta'\delta,$$
$$c_1 = a_1 + a_2\delta'\delta + \rho_\delta\delta'\delta a_4, \qquad d_1 = a_1 + a_3\delta'\delta + \rho_\delta\delta'\delta a_4, \qquad e_1 = b_1 + b_2\delta'\delta + b_3\rho_\delta\delta'\delta.$$

Note that

$$c_0 + \rho_\delta c_1 = d_0 + \rho_\delta d_1. \tag{122}$$

Proof of Lemma 7.

$E[\mathbf{X}\mathbf{X}']$ is invertible by assumption (iii). Notice that the off-diagonal blocks in (115) are identical, and that each block is a symmetric matrix. Applying the formulas for block matrix inversion and imposing these properties gives

$$\begin{pmatrix} \Psi_1 & \Psi_2 \\ \Psi_2 & \Psi_3 \end{pmatrix}^{-1} = \begin{pmatrix} \Omega_1 & \Omega_2 \\ \Omega_2' & \Omega_3 \end{pmatrix}. \tag{123}$$

■

Proof of Lemma 8.

Equation (123) implies

$$\begin{pmatrix} \Psi_1\Omega_1 + \Psi_2\Omega_2' & \Psi_1\Omega_2 + \Psi_2\Omega_3 \\ \Psi_2\Omega_1 + \Psi_3\Omega_2' & \Psi_2\Omega_2 + \Psi_3\Omega_3 \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}. \tag{124}$$

Treating each block of (124) as a system of equations, consider pre and post-multiplying the top-left blocks by δ_1

$$\delta_1'\Psi_1\Omega_1\delta_1 + \delta_1'\Psi_2\Omega_2'\delta_1 = \delta'\delta.$$

Using (116) and (120), this becomes

$$c_1\delta_1'\Omega_1\delta_1 + c_0\delta_0'\Omega_1\delta_1 + e_1\delta_1'\Omega_2'\delta_1 + e_0\delta_0'\Omega_2'\delta_1 = \delta'\delta.$$

Repeat this with all four blocks and combinations of δ_1 and δ_0 to obtain sixteen conditions. These sixteen conditions can be written as the system of equations

$$E\begin{pmatrix} \delta_1'\Omega_1\delta_1 & \delta_1'\Omega_2\delta_1 & \delta_1'\Omega_1\delta_0 & \delta_1'\Omega_2\delta_0 \\ \delta_0'\Omega_1\delta_1 & \delta_0'\Omega_2\delta_1 & \delta_0'\Omega_1\delta_0 & \delta_0'\Omega_2\delta_0 \\ \delta_1'\Omega_2'\delta_1 & \delta_1'\Omega_3\delta_1 & \delta_1'\Omega_2'\delta_0 & \delta_1'\Omega_3\delta_0 \\ \delta_0'\Omega_2'\delta_1 & \delta_0'\Omega_3\delta_1 & \delta_0'\Omega_2'\delta_0 & \delta_0'\Omega_3\delta_0 \end{pmatrix} = \begin{pmatrix} \delta'\delta & 0 & \rho_\delta\delta'\delta & 0 \\ 0 & \delta'\delta & 0 & \rho_\delta\delta'\delta \\ \rho_\delta\delta'\delta & 0 & \delta'\delta & 0 \\ 0 & \rho_\delta\delta'\delta & 0 & \delta'\delta \end{pmatrix},$$

where

$$E \equiv \begin{pmatrix} E_1 & E_2 \\ PE_2P & PE_1P \end{pmatrix}, \quad E_1 = \begin{pmatrix} c_1 & c_0 \\ e_1 & e_0 \end{pmatrix}, \; E_2 = \begin{pmatrix} e_1 & e_0 \\ d_1 & d_0 \end{pmatrix}, \tag{125}$$

and P is the perturbation matrix

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

By assumption (iv), E is invertible. Let $F = E^{-1}$. Inverting E as a block matrix, and using that P is its own inverse

$$F = \begin{pmatrix} F_1 & F_2 \\ PF_2P & PF_1P \end{pmatrix}, \quad F_1 = \begin{pmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{pmatrix}, \; F_2 = \begin{pmatrix} f_{13} & f_{14} \\ f_{23} & f_{24} \end{pmatrix}. \tag{126}$$

Hence,

$$\begin{pmatrix} \delta_1'\Omega_1\delta_1 & \delta_1'\Omega_2\delta_1 & \delta_1'\Omega_1\delta_0 & \delta_1'\Omega_2\delta_0 \\ \delta_0'\Omega_1\delta_1 & \delta_0'\Omega_2\delta_1 & \delta_0'\Omega_1\delta_0 & \delta_0'\Omega_2\delta_0 \\ \delta_1'\Omega_2'\delta_1 & \delta_1'\Omega_3\delta_1 & \delta_1'\Omega_2'\delta_0 & \delta_1'\Omega_3\delta_0 \\ \delta_0'\Omega_2'\delta_1 & \delta_0'\Omega_3\delta_1 & \delta_0'\Omega_2'\delta_0 & \delta_0'\Omega_3\delta_0 \end{pmatrix} = \begin{pmatrix} F_1 & F_2 \\ PF_2P & PF_1P \end{pmatrix} \begin{pmatrix} \delta'\delta & 0 & \rho_\delta\delta'\delta & 0 \\ 0 & \delta'\delta & 0 & \rho_\delta\delta'\delta \\ \rho_\delta\delta'\delta & 0 & \delta'\delta & 0 \\ 0 & \rho_\delta\delta'\delta & 0 & \delta'\delta \end{pmatrix}.$$

From which we can obtain the relations stated in Lemma 8

$$\delta_1'\Omega_1\delta_1 = \delta_0'\Omega_3\delta_0 = f_{11}\delta'\delta + f_{13}\rho_\delta\delta'\delta, \tag{127}$$

$$\delta_0'\Omega_1\delta_0 = \delta_1'\Omega_3\delta_1 = f_{21}\rho_\delta\delta'\delta + f_{23}\delta'\delta, \tag{128}$$

$$\delta_0'\Omega_1\delta_1 = \delta_1'\Omega_3\delta_0 = f_{21}\delta'\delta + f_{23}\rho_\delta\delta'\delta, \tag{129}$$

$$\delta_1'\Omega_1\delta_0 = \delta_0'\Omega_3\delta_1 = f_{11}\rho_\delta\delta'\delta + f_{13}\delta'\delta, \tag{130}$$

$$\delta_1'\Omega_2\delta_1 = \delta_0'\Omega_2'\delta_0 = f_{12}\delta'\delta + f_{14}\rho_\delta\delta'\delta, \tag{131}$$

$$\delta_1'\Omega_2'\delta_1 = \delta_0'\Omega_2\delta_0 = f_{22}\rho_\delta\delta'\delta + f_{24}\delta'\delta, \tag{132}$$

$$\delta_0'\Omega_2'\delta_1 = \delta_1'\Omega_2\delta_0 = f_{12}\rho_\delta\delta'\delta + f_{14}\delta'\delta, \tag{133}$$

$$\delta_0'\Omega_2\delta_1 = \delta_1'\Omega_2'\delta_0 = f_{22}\delta'\delta + f_{24}\rho_\delta\delta'\delta. \tag{134}$$

■

Proof of Lemma 9.

The formulas for $\delta_i'\Omega_k\delta_j$ $i, j = \{0, 1\}$, $k = \{1, 2, 3\}$ above contain $\delta_\prime\delta$, ρ_δ, and the elements of the inverse of E. The elements of E are given by combinations of ρ_δ, $\delta'\delta$ and integrals of the propensity score density and the conditional distribution of Z. Recall that the propensity score density and the conditional distribution of Z depend only on α and $\delta'\delta$. Hence, E depends only on α, $\delta'\delta$ and ρ_δ, which implies its also inverse depends only on these parameters. Thus, the only parameters affecting each $\delta_i'\Omega_k\delta_j$ are α, $\delta'\delta$ and ρ_δ. ■

Proof of Statements 1 and 2.

Under assumption (i), estimates of θ_0 and θ_1 are obtained by regressing $Y - E[Y|p]$ on $p(X - E[X|p])$ and $(1-p)(X - E[X|p])$. The asymptotic limit of these estimates is

$$\begin{pmatrix} \hat{\theta}_0 \\ \hat{\theta}_1 \end{pmatrix} = \begin{pmatrix} \theta_0 \\ \theta_1 \end{pmatrix} + E[\mathbf{X}\mathbf{X}']^{-1}E[\mathbf{X}\mathbf{Z}]2\nu,$$

so that

$$\begin{pmatrix}\tilde{\theta}_1\\ \tilde{\theta}_0\end{pmatrix} = -E[\mathbf{X}\mathbf{X}']^{-1}E[\mathbf{X}\mathbf{Z}]2\nu.$$

Use Lemma 6 to substitute for $E[\mathbf{X}\mathbf{Z}]$, Lemma 7 to substitute for $E[\mathbf{X}\mathbf{X}']^{-1}$ and simplify to obtain

$$\begin{pmatrix}\tilde{\theta}_1\\ \tilde{\theta}_0\end{pmatrix} = -\begin{pmatrix}w_1\Omega_1\delta_1 + w_0\Omega_1\delta_0 + w_1\Omega_2\delta_0 + w_0\Omega_2\delta_1\\ w_1\Omega_2'\delta_1 + w_0\Omega_2'\delta_0 + w_1\Omega_3\delta_0 + w_0\Omega_3\delta_1\end{pmatrix}2\nu. \tag{135}$$

Pre-mutliply the two elements in (135) by δ_1' and δ_0' to obtain

$$\delta_1'\tilde{\theta}_1 = -(w_1\delta_1'\Omega_1\delta_1 + w_0\delta_1'\Omega_1\delta_0 + w_1\delta_1'\Omega_2\delta_0 + w_0\delta_1'\Omega_2\delta_1)2\nu, \tag{136}$$

$$\delta_1'\tilde{\theta}_0 = -(w_1\delta_1'\Omega_2'\delta_1 + w_0\delta_1'\Omega_2'\delta_0 + w_1\delta_1'\Omega_3\delta_0 + w_0\delta_1'\Omega_3\delta_1)2\nu, \tag{137}$$

$$\delta_0'\tilde{\theta}_1 = -(w_1\delta_0'\Omega_1\delta_1 + w_0\delta_0'\Omega_1\delta_0 + w_1\delta_0'\Omega_2\delta_0 + w_0\delta_0'\Omega_2\delta_1)2\nu, \tag{138}$$

$$\delta_0'\tilde{\theta}_0 = -(w_1\delta_0'\Omega_2'\delta_1 + w_0\delta_0'\Omega_2'\delta_0 + w_1\delta_0'\Omega_3\delta_0 + w_0\delta_0'\Omega_3\delta_1)2\nu. \tag{139}$$

Using (136) and (139)

$$\delta_1'\tilde{\theta}_1 - \delta_0'\tilde{\theta}_0 = -2(w_1(\delta_1'\Omega_1\delta_1 - \delta_0'\Omega_3\delta_0) + w_0(\delta_1'\Omega_1\delta_0 - \delta_0'\Omega_3\delta_1) + w_1(\delta_1'\Omega_2\delta_0 - \delta_0'\Omega_2'\delta_1) + w_0(\delta_1'\Omega_2\delta_1 - \delta_0'\Omega_2'\delta_0))\nu.$$

Impose the equalities of Lemma 8 and $\delta_1'\Omega_2\delta_0 = \delta_0'\Omega_2'\delta_1$, and this is zero. Repeating this process with (137) and (138) yields that $\delta_0'\tilde{\theta}_1 = \delta_1'\tilde{\theta}_0$. This proves the first statement of Proposition 7.

To see the second statement, consider (136). By Lemma 6, w_0 and w_1 depend only on α and $\delta'\delta$. By Lemma 9, $\delta_1'\Omega_1\delta_1$, $\delta_1'\Omega_1\delta_0$, $\delta_1'\Omega_2\delta_0$ and $\delta_1'\Omega_2\delta_1$ depend only on α, $\delta'\delta$ and ρ_δ. Hence, we can write

$$\delta_1'\tilde{\theta}_1 = \mathcal{H}_1(\alpha, \rho_\delta, \delta'\delta)\nu,$$

with

$$\mathcal{H}_1(\alpha, \rho_\delta, \delta'\delta) \equiv -(w_1\delta_1'\Omega_1\delta_1 + w_0\delta_1'\Omega_1\delta_0 + w_1\delta_1'\Omega_2\delta_0 + w_0\delta_1'\Omega_2\delta_1)2.$$

The same argument can be applied to (137) to obtain the second identity. ■

C.1.5 Additional Results

For (129) and (130) to both hold, it must be that

$$f_{21} + f_{23}\rho_\delta = f_{11}\rho_\delta + f_{13}.$$

Similarly, for (131) and (132) to both hold, we require

$$f_{12} + f_{14}\rho_\delta = f_{22}\rho_\delta + f_{24}.$$

As F is the inverse of E

$$\begin{pmatrix} F_1 & F_2 \\ PF_2P & PF_1P \end{pmatrix} \begin{pmatrix} E_1 & E_2 \\ PE_2P & PE_1P \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}.$$

Post-multiply both sides by the matrix with columns $(\rho_\delta, 1, 0, 0)$ and $(0, 0, \rho_\delta, 1)$ to obtain

$$\begin{pmatrix} F_1 & F_2 \\ PF_2P & PF_1P \end{pmatrix} \begin{pmatrix} G_1 \\ G_2 \end{pmatrix} = \begin{pmatrix} \rho_\delta & 0 \\ 1 & 0 \\ 0 & \rho_\delta \\ 0 & 1 \end{pmatrix}, \quad G_1 = \begin{pmatrix} g_1 & g_2 \\ g_2 & g_1 \end{pmatrix}, \; G_2 = \begin{pmatrix} g_4 & g_5 \\ g_5 & g_3 \end{pmatrix}, \tag{140}$$

with

$$\begin{aligned} g_1 &= \rho_\delta c_1 + c_0 = \rho_\delta d_1 + d_0, & g_3 &= \rho_\delta c_0 + c_1, \\ g_2 &= \rho_\delta e_1 + e_0, & g_4 &= \rho_\delta d_0 + d_1, \\ & & g_5 &= \rho_\delta e_0 + e_1. \end{aligned}$$

Where the g_1 uses (122). Notice that

$$G_1 P = P G_1, \tag{141}$$

$$G_1 = P G_1 P = G_1^T, \tag{142}$$

$$G_2 = G_2^T. \tag{143}$$

The system (140) can be rearranged to

$$G \begin{pmatrix} F_1^T \\ F_2^T \end{pmatrix} = \begin{pmatrix} \rho_\delta & 1 \\ 0 & 0 \\ 0 & 0 \\ 1 & \rho_\delta \end{pmatrix}, \tag{144}$$

where

$$G \equiv \begin{pmatrix} G_1 & G_2 \\ G_2 P & G_1 P \end{pmatrix}. \tag{145}$$

By assumption (iv), G is invertible. Let $G^{-1} = H$. Applying the formulas for block matrix inversion

$$H = \begin{pmatrix} H_1 & H_2 \\ PH_2 & PH_1 \end{pmatrix}.$$

Furthermore, using (141), (142) and (143), it is possible to show

$$H_1^T = PH_1P,$$
$$H_2^T = PH_2P,$$

which implies that H_1 and H_2 are constant diagonal matrices

$$H_1 = \begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{11} \end{pmatrix}, \; H_2 = \begin{pmatrix} h_{13} & h_{14} \\ h_{23} & h_{13} \end{pmatrix}.$$

Multiplying both sides of (144) by H

$$\begin{pmatrix} F_1^T \\ F_2^T \end{pmatrix} = \begin{pmatrix} H_1 & H_2 \\ PH_2 & PH_1 \end{pmatrix} \begin{pmatrix} \rho_\delta & 1 \\ 0 & 0 \\ 0 & 0 \\ 1 & \rho_\delta \end{pmatrix},$$

which yields the following formulas

$$\begin{aligned} f_{11} &= \rho_\delta h_{11} + h_{14}, & f_{21} &= h_{11} + \rho_\delta h_{14}, \\ f_{12} &= \rho_\delta h_{21} + h_{13}, & f_{22} &= h_{21} + \rho_\delta h_{13}, \\ f_{13} &= \rho_\delta h_{23} + h_{11}, & f_{23} &= h_{23} + \rho_\delta h_{11}, \\ f_{14} &= \rho_\delta h_{13} + h_{12}, & f_{24} &= h_{13} + \rho_\delta h_{12}. \end{aligned}$$

Using these formulas

$$\begin{aligned} f_{21} + \rho_\delta f_{23} = \rho_\delta f_{11} + f_{13} = (1 + \rho_\delta^2) h_{11} + \rho_\delta (h_{14} + h_{23}), \\ f_{12} + \rho_\delta f_{14} = \rho_\delta f_{22} + f_{24} = (1 + \rho_\delta^2) h_{13} + \rho_\delta (h_{12} + h_{21}). \end{aligned}$$

D Modified Distillation Algorithm

As in section 2.6, assume the sample is sorted in ascending order by propensity score, so that observation $i = 1$ has the lowest propensity score and $i = N$ the highest, with ties broken by placing observations with $Z = 0$ above observations with $Z = 1$. Maintain the definitions of n_0, n_1, and Δ_j of section 2.6, and let $J^- = \{j : p_j \in P^-\}$ and Let $J^+ = \{j : p_j \in P^+\}$.

Define the following 'reaction' functions

$$R_{d_1}(d_0) = \max_{j \in J^-} \left\{ n_1 \frac{n_0 - d_0}{n_0 - d_0 - \sum_{i=1}^{j} (1 - Z_i)} \left(\Delta_j - \frac{d_0}{n_0 - d_0} \frac{\sum_{i=1}^{j} (1 - Z_i)}{n_0} \right) \right\} \tag{146}$$

$$R_{d_0}(d_1) = \max_{j \in J^+} \left\{ n_0 \frac{n_1 - d_1}{\sum_{i=1}^{j} Z_i - d_1} \left(\Delta_j - \frac{d_1}{n_1 - d_1} \frac{n_1 - \sum_{i=1}^{j} Z_i}{n_1} \right) \right\} \tag{147}$$

Given that d_0 observations with $(Z_i = 0, p_i \in P^+)$ are trimmed, (146) returns the number of observations with $(Z_i = 1, p_i \in P^-)$ that must be trimmed for first-order stochastic dominance to

hold. Given that d_1 observations with $(Z_i = 1, p_i \in P^-)$ are trimmed, (147) returns the number of observations with $(Z_i = 0, p_i \in P^+)$ that must be trimmed for first-order stochastic dominance to hold.

The modified algorithm is as follows

1. Set $S_{1i} = 1 \; \forall \; i$
2. Calculate Δ_j for $j = 1, ..., N$. If $\max(\Delta_j) \leq 0$, stop. If $\max(\Delta_j) > 0$, continue to steps 3, 4, and 5.
3. Obtain d_1 and d_0 as follows
   (a) Calculate
   $$\begin{aligned} d_1^{\max} &= R_{d_1}(0) \\ d_0^{\max} &= R_{d_0}(0) \\ d_1^{\min} &= \max\{R_{d_1}(d_0^{\max}), 0\} \\ d_0^{\min} &= \max\{R_{d_0}(d_1^{\max}), 0\} \end{aligned}$$
   (b) If $d_1^{\min} > 0$ and $d_0^{\min} > 0$ or $d_1^{\min} = 0$ and $d_0^{\min} = 0$, obtain all d_0, d_1 where $R_{d_1}(d_0) = R_{d_0}(d_1)$
   (c) Evaluate $d_0 + d_1$ at $(d_0^{\min}, d_1^{\max})$, $(d_0^{\max}, d_1^{\min})$ and all d_0, d_1 where $R_{d_1}(d_0) = R_{d_0}(d_1)$, Select the combination that attains the minimum.
   (d) Obtain j^- as the lowest $j \in J^-$ where $R_{d_1}(d_0) = d_1$ and j^+ as the highest $j \in J^+$ where $R_{d_0}(d_1) = d_0$.
4. Loop over $j = 1, ..., j^-$. If $Z_j = 1$
   (a) Calculate
   $$\delta_{1j} = \frac{\sum_{i=1}^{j} S_{1,i} Z_i}{n_1 - d_1} - \frac{\sum_{i=1}^{j} (1 - Z_i)}{n_0 - d_0}$$
   (b) If $\delta_{1j} > 0$, set $S_{1j} = 0$
5. Loop over $j = N, ..., j^+ + 1$. If $Z_j = 0$
   (a) Calculate
   $$\delta_0 = \frac{\sum_{i=j}^{N} S_{1,i} (1 - Z_i)}{n_0 - d_0} - \frac{\sum_{i=j}^{N} Z_i}{n_1 - d_1}$$
   (b) If $\delta_{0j} > 0$, set $S_{1j} = 0$

Steps 1 and 2 are identical to the algorithm of Section 2.6. In step 3, d_0 and d_1 are jointly determined such that d_1 is the smallest number of observations with $(Z_i = 1, p_i \in P^-)$ that must be trimmed given that d_0 observations with $(Z_i = 0, p_i \in P^+)$ are trimmed, d_0 is the smallest number of observations with $(Z_i = 0, p_i \in P^+)$ that must be trimmed given that d_1 observations with $(Z_i = 1, p_i \in P^-)$ are trimmed, and the sum $d_0 + d_1$ is minimised. Step 4 corresponds to step 3b of the algorithm of Section 2.6, with the modification that the trimmed distribution for $Z = 0$ is used. Step 5 is identical to step 4b of the algorithm of Section 2.6.

Figure 20 illustrates the modified algorithm using the same artificial sample as Figure 3. Panel a plots $R_{d_0}(d_1)$ and $R_{d_1}(d_0)$ for this sample. The algorithm of Section 2.6 leads to $d_1 = 6, d_0 = 11$. In comparison, the modified algorithm selects $d_1 = 4, d_0 = 12$, so we trim two less observations with $Z = 1$ at the cost of trimming an additional observation with $Z = 0$. Panels b and c show how steps 4 and 5 choose which points to trim. Panel d plots the resulting trimmed conditional propensity score distributions.

[Figure 20 about here.]

E Test Statistic Proof

E.1 Notation and General Assumptions

Let n denote the sample size.

Assumption 2 *(i)* $\{(Y_i, X_i, Z_i, D_i) : i = 1, ..., n\}$ *is an iid sample.*

(ii) There exists a deterministic sequence $\hat{\lambda}_n = n_1/n \to \lambda$ *as* $n \to \infty$ *with* $\lambda \in (0, 1)$.

(iii) $E[||X_i||^2] < \infty$

(iv) The conditional distribution of U *given* (X, Z, D) *has bounded density with respect to Lebesgue measure.*

Let $(\beta^*, \delta_p^*, \delta_q^*)$ be the true parameter values, and $(\hat{\beta}, \hat{\delta}_p, \hat{\delta}_q)$ be estimates.

Assumption 3 *(i) The parameter estimators satisfy*

$$\sqrt{n}(\hat{\beta} - \beta^*) = n^{-1/2}\Gamma_\beta \sum_{i=1}^{n} \psi_\beta(Y_i, X_i, Z_i, D_i; \beta^*, \delta_p^*) + o_p(1)$$

$$\sqrt{n}(\hat{\delta}_p - \delta_p^*) = n^{-1/2}\Gamma_{\delta_p} \sum_{i=1}^{n} \psi_{\delta_p}(X_i, Z_i, D_i; \delta_p^*) + o_p(1)$$

$$\sqrt{n}(\hat{\delta}_q - \delta_q^*) = n^{-1/2}\Gamma_{\delta_q} \sum_{i=1}^{n} \psi_{\delta_q}(X_i, Z_i; \delta_q^*, \delta_p^*) + o_p(1)$$

where Γ_β, Γ_{δ_p} *and* Γ_{δ_q} *are non-stochastic matrices.*

(ii) The functions $\psi_\beta(y,x,z,d;\beta,\delta_p) : \mathbb{R}\times\mathbb{R}^{k_X}\times\{0,1\}^2\times\mathbf{B}\times\mathbf{D}_p \to \mathbb{R}^{2k_X}$, $\psi_{\delta_p}(x,z,d;\delta_p) : \mathbb{R}^{k_X}\times\{0,1\}^2\times\mathbf{D}_p\to\mathbb{R}^{k_p}$ *and* $\psi_{\delta_q}(x,z;\delta_q,\delta_p):\mathbb{R}^{k_X}\times\{0,1\}\times\mathbf{D}_q\times\mathbf{D}_q\to\mathbb{R}^{k_q}$ *are measurable and satisfy*

a. $E[\psi_\beta(Y_i,X_i,Z_i,D_i;\beta^*,\delta_p^*)]=0, E[\psi_{\delta_p}(X_i,Z_i,D_i;\delta_p^*)]=0, E[\psi_{\delta_q}(X_i,Z_i;\delta_q^*,\delta_p^*)]=0$

b. $E[||\psi_\beta(Y_i,X_i,Z_i,D_i;\beta^*,\delta_p^*)||^{2+\zeta}]<\infty$, $E[||\psi_{\delta_p}(X_i,Z_i,D_i;\delta_p^*)||^{2+\zeta}]<\infty, E[||\psi_{\delta_q}(X_i,Z_i;\delta_q^*,\delta_p^*)||^{2+\zeta}]$

for some $\zeta>0$.

Use the shorthand notation

$$\begin{aligned}U_i(\beta)&=Y_i-DX_i^T\beta_1-(1-D)X_i^T\beta_0\\ p_i(\delta_p)&=p(X_i,Z_i;\delta_p)\\ q_i(\delta_p,\delta_q)&=q(X_i,Z_i;\delta_p,\delta_q)\end{aligned}$$

Let $\mathcal{A}$ be a set of closed connected intervals with generic element $A=[a^-,a^+]$ where $a^-\in\mathbb{R}$, $a^+\in\mathbb{R}$, and $a^-\le a^+$.

E.2 Statement of Theorems

E.2.1 Nesting Inequalities Assumptions and Theorem

We can consider tests of the nesting inequalities with propensity score intervals B drawn from a set $\mathcal{B}$ constructed as follows.

Assumption 4 *Let* $\tilde{\mathcal{B}}=\{b_1,...,b_J\}$ *be a collection of* $J<\infty$ *points with* $b_j\in[0,1]$ *for* $j=1,...,J$. *Any* $B\in\mathcal{B}$ *can be written as* $[b^-,b^+]$ *with* $b^-\in\tilde{\mathcal{B}}$, $b^+\in\tilde{\mathcal{B}}$ *and* $b^-\le b^+$.

Define

$$\begin{aligned}H_n(B;\delta_p)&=\frac{\sum_{i=1}^n\mathbb{1}\{b^-<p_i(\delta_p)\le b^+\}}{n},\\ H(B;\delta_p)&=E[H_n(B;\delta_p)]=\Pr(p(\delta_p)\in B),\end{aligned}$$

similarly

$$\begin{aligned}H_n(A,B,d;\beta,\delta_p)&=\frac{\sum_{i=1}^n\mathbb{1}\{a^-<U_i(\beta)\le a^+,D_i=d,b^-<p_i(\delta_p)\le b^+\}}{n},\\ H(A,B,d;\beta,\delta_p)&=E[H_n(A,B,d;\beta,\delta_p)]=\Pr(U(\beta)\in A,D=d,p(\delta_p)\in B).\end{aligned}$$

We impose that mean-value expansions of $H(B;\delta_p)$ and $H(A,B,d;\beta,\delta_p)$ are possible

Assumption 5 *(i) $H(B;\delta_p)$ is differentiable in δ_p on a neighborhood $\mathbf{D}_p^*$ of δ_p^* for any $B \in \mathcal{B}$. $H(A,B,d;\beta,\delta_p)$ is differentiable in (β,δ_p) on a neighborhood $\mathbf{B}^* \times \mathbf{D}_p^*$ of (β_d^*,δ_p^*) for any $(A,B,d) \in \mathcal{A}\times\mathcal{B}\times\{0,1\}$.*

(ii) For all sequences of positive constants $\{\tau_n : N \geq 1\}$ such that $\tau_n \to 0$

$$\sup_{B\in[0,1]} \sup_{\delta_p:||\delta_p-\delta_p^*||\leq\tau_n} \left|\left|\frac{\partial H(B;\delta_p)}{\partial\delta_p} - \Delta_{\delta_p}(B)\right|\right| \to 0$$

$$\sup_{A\in\mathcal{A},B\in[0,1],d\in\{0,1\}} \sup_{(\beta,\delta_p):||(\beta,\delta_p)-(\beta^*,\delta_p^*)||\leq\tau_n} \left|\left|\frac{\partial H(A,B,d;\beta,\delta_p)}{\partial(\beta,\delta_p)} - (\Delta_\beta(A,B,d),\Delta_{\delta_p}(A,B,d))\right|\right| \to 0$$

where

$$\Delta_{\delta_p}(B) \coloneqq \left.\frac{\partial H(B;\delta_p)}{\partial\delta_p}\right|_{\delta_p=\delta_p^*}$$
$$\Delta_\beta(A,B,d) \coloneqq \left.\frac{\partial H(A,B,d;\beta,\delta_p)}{\partial\beta}\right|_{\beta=\beta^*,\delta_p=\delta_p^*}$$
$$\Delta_{\delta_p}(A,B,d) \coloneqq \left.\frac{\partial H(A,B,d;\beta,\delta_p)}{\partial\delta_p}\right|_{\beta=\beta^*,\delta_p=\delta_p^*}$$

(iii) $\sup_{B\in\mathcal{B}}||\Delta_{\delta_p}(B)|| < \infty$, $\sup_{A\in\mathcal{A},B\in\mathcal{B},d\in\{0,1\}}||\Delta_\beta(A,B,d)|| < \infty$ *and* $\sup_{A\in\mathcal{A},B\in\mathcal{B},d\in\{0,1\}}||\Delta_{\delta_p}(A,B,d)|| < \infty$.

Let $\theta \coloneqq (\beta,\delta_p)$, $\theta^* = (\beta^*,\delta_p^*)$, and $\hat{\theta} = (\hat{\beta}_1,\hat{\delta}_p)$. Fix intervals $A \in \mathcal{A}$ and $(B,B') \in \mathcal{B}^2$ with the lower bound of B (weakly) exceeding the upper bound of B'. Write

$$\mathcal{N}_n(A,B,B',d;\hat{\theta}) = \sqrt{nH_n(B;\hat{\delta}_p)H_n(B';\hat{\delta}_p)}\left(\frac{H_n(A,B,d;\hat{\beta},\hat{\delta}_p)}{H_n(B;\hat{\delta}_p)} - \frac{H_n(A,B',d;\hat{\beta},\hat{\delta}_p)}{H_n(B';\hat{\delta}_p)}\right) \quad (148)$$

The nesting statistic is

$$\mathcal{N}_n(B,B';\hat{\theta}) = \max\left\{\sup_A \mathcal{N}_n(A,B,B',0;\hat{\theta}),\ \sup_A \mathcal{N}_n(A,B',B,1;\hat{\theta})\right\} \quad (149)$$

In addition, define

$$\mathcal{N}(A,B,B',d;\theta^*) = \left(\frac{H(A,B,d;\beta^*,\delta_p^*)}{H(B;\delta_p^*)} - \frac{H(A,B',d;\beta^*,\delta_p^*)}{H(B';\delta_p^*)}\right) \quad (150)$$

and

$$\mathcal{N}(B,B';\theta) = \max\left\{\sup_A \mathcal{N}(A,B,B',0;\theta^*),\ \sup_A \mathcal{N}(A,B',B,1;\theta^*)\right\} \quad (151)$$

Theorem 3 *Suppose that Assumptions 1 - 5 hold. Then, under the null we have*

$$\mathcal{N}_n(B, B'; \hat{\theta}) \implies \begin{cases} \max\limits_{d\in\{0,1\}} \sup\limits_{A\in\mathcal{C}_{B,B',d}} a^T \varepsilon(A, B, B', d; \theta^*) & \mathcal{N}(B, B'; \theta^*) = 0 \\ -\infty & \mathcal{N}(B, B'; \theta^*) < 0 \end{cases}$$

where $\mathcal{C}_{B,B',d}$ is a contact set where the inequalities are binding, and $\varepsilon(A, B, B', d; \theta^*)$ is a mean zero Gaussian process

$$\varepsilon(A, B, B', d; \theta^*) = (\varepsilon(A, B, d; \beta^*, \delta_p^*), \varepsilon_\beta, \varepsilon_{\delta_p})$$

with covariance

$$\Sigma(A, B, B', d; \theta^*) = \lim_{n\to\infty} E\left[\begin{pmatrix} \nu_n(A, B, d; \beta^*, \delta_p^*) \\ \nu_n(A, B', d; \beta^*, \delta_p^*) \\ \nu_n(B; \delta_p^*) \\ \nu_n(B'; \delta_p^*) \\ \sqrt{n}\bar{\psi}_{n\beta}(\beta^*, \delta_p^*) \\ \sqrt{n}\bar{\psi}_{n\delta_p}(\delta_p^*) \end{pmatrix} \begin{pmatrix} \nu_n(A, B, d; \beta^*, \delta_p^*) \\ \nu_n(A, B', d; \beta^*, \delta_p^*) \\ \nu_n(B; \delta_p^*) \\ \nu_n(B'; \delta_p^*) \\ \sqrt{n}\bar{\psi}_{n\beta}(\beta^*, \delta_p^*) \\ \sqrt{n}\bar{\psi}_{n\delta_p}(\delta_p^*) \end{pmatrix}^T\right]$$

where

$$\begin{aligned} \nu_n(A, B, d; \beta, \delta_p) &= \sqrt{n}(H_n(A, B, d; \beta, \delta_p) - H(A, B, d; \beta, \delta_p)) \\ \nu_n(B; \delta_p) &= \sqrt{n}(H_n(B; \delta_p) - H(B; \delta_p)) \\ \bar{\psi}_{n\beta}(\theta^*) &= \frac{\sum_{i=1}^n \psi_\beta(Y_i, X_i, Z_i, D_i; \beta^*, \delta_p^*)}{n} \\ \bar{\psi}_{n\delta_p}(\theta^*) &= \frac{\sum_{i=1}^n \psi_{\delta_p}(X_i, Z_i, D_i; \delta_p^*)}{n} \end{aligned}$$

and $a = (a_1, a_2, a_3, a_4, a_5, a_6)$ with elements

$$\begin{aligned} a_1 &= \left(\frac{H(B'; \delta_p^*)}{H(B; \delta_p^*)}\right)^{1/2} \\ a_2 &= -\frac{1}{a_1} \\ a_3 &= -a_1 \frac{H(A, B, 1; \beta^*, \delta_p^*)}{H(B; \delta_p^*)} \\ a_4 &= a_2 \frac{H(A, B', 1; \beta^*, \delta_p^*)}{H(B'; \delta_p^*)} \\ a_5 &= \left(a_1 \Delta_\beta(A, B, 1)^T + a_2 \Delta_\beta(A, B', 1)^T\right) \Gamma_{\beta_d}^* \\ a_6 &= (a_1 \Delta_\delta(A, B, 1)^T + a_3 \Delta_\delta(B)^T) \Gamma_\delta^* + (a_2 \Delta_\delta(A, B', 1)^T + a_4 \Delta_\delta(B')^T) \Gamma_\delta^* \end{aligned}$$

E.2.2 Index Sufficiency Assumptions and Theorem

For index sufficiency, we impose the following additional assumption

Assumption 6

(i) X *contains at least one continuous variable.*

(ii) $\Pr(S_2 = 1)$ *is estimated by*

$$\bar{S}_{2n}(\delta_p, \delta_q) = \frac{\sum_{i=1}^{n} \mathbb{1}\{\iota \leq q_i(\delta_p, \delta_q) \leq 1 - \iota\}}{n}$$

(iii) There exists some $\epsilon > 0$ *such that* $\bar{S}_{2n}(\hat{\delta}_p, \hat{\delta}_q) > \epsilon$.

(iv) There exists an interval $\mathcal{P} \subset [0,1]$ *of measure* $\Pr(p(\delta_p^*) \in \mathcal{P}) > \epsilon$ *such that* $\iota < q(\delta_p^*, \delta_q^*) < 1 - \iota$ *for all* $p \in \mathcal{P}$.

For simplicity, we consider trimming with a single fixed trimming constant. With minimal modifications to the current proofs, the trimming constant can be allowed to vary with sample size and converge to some constant, subject to the restriction that the trimming constant converges at a faster rate than the estimators. Similarly, the trimming function can be made asymmetric. We impose the practical restriction that we would not test index sufficiency when there are no observations to do so. The final condition imposes that index sufficiency is testable for an oracle with knowledge of the true parameters values and access to an asymptotic sample.

As before, let $\theta = (\beta, \delta_p, \delta_q)$ be a vector of (arbitrary) parameter values. Let

$$G_n(A, d, z; \theta) = \frac{1}{n} \sum_{i=1}^{n} \frac{\mathbb{1}\{U_i(\beta) \in A, D_i = d, Z_i = z\} S_{2i}(\delta_p, \delta_q)}{q_i(\delta_p, \delta_q)^z (1 - q_i(\delta_p, \delta_q))^{1-z}}$$

$$G(A, d, z; \theta) = E\left[G_n(A, d, z; \theta)\right]$$

$G_i(A, d, z; \theta)$ is the empirical distribution of (U, D, Z) reweighted by the index sufficiency weighting function. In addition, let

$$\bar{S}_2(\delta_p, \delta_q) = E[\bar{S}_{2n}(\delta_p, \delta_q)] = \Pr(\iota \leq q(\delta_p, \delta_q) \leq 1 - \iota)$$

We impose that mean-value expansions of $G(A, d, z; \theta)$ and $\bar{S}_2(\delta_p, \delta_q)$ are possible

Assumption 7 *(i)* $\bar{S}_2(\delta_p, \delta_q)$ *is differentiable in* (δ_p, δ_q) *on a neighbourhood* $\mathbf{D}_p^* \times \mathbf{D}_q^*$ *of* (δ_p^*, δ_q^*). $G(A, d, z; \theta)$ *is differentiable in* θ *on a neighborhood* $\mathbf{D}_p^* \times \mathbf{D}_q^* \times \mathbf{B}^*$ *of* θ^* *for any* $(A, d, z) \in \mathcal{A} \times [0,1] \times [0,1]$.

(ii) For all sequences of positive constants $\{\tau_n : n \geq 1\}$ *such that* $\tau_n \to 0$

$$\sup_{(\delta_p,\delta_q):||(\delta_p,\delta_q)-(\delta_p^*,\delta_q^*)||\leq\tau_n} \left|\left| \frac{\partial \bar{S}_2(\delta_p,\delta_q)}{\partial(\delta_p,\delta_q)} - \Delta_\delta \right|\right| \to 0$$

$$\sup_{A\in\mathcal{A},d\in\{0,1\},z\in\{0,1\}} \sup_{\theta:||\theta-\theta^*||\leq\tau_n} \left|\left| \frac{\partial G(A,d,z;\theta)}{\partial\theta} - \Delta_\theta(A,d,z) \right|\right| \to 0$$

where $\Delta_\delta = (\Delta_{\delta_p}, \Delta_{\delta_q})$ *with*

$$\Delta_{\delta_p} \coloneqq \left.\frac{\partial \bar{S}_2(\delta_p,\delta_q)}{\partial\delta_p}\right|_{(\delta_p,\delta_q)=(\delta_p^*,\delta_q^*)}$$

$$\Delta_{\delta_q} \coloneqq \left.\frac{\partial \bar{S}_2(\delta_p,\delta_q)}{\partial\delta_q}\right|_{(\delta_p,\delta_q)=(\delta_p^*,\delta_q^*)}$$

and $\Delta_\theta(A,d,z) = (\Delta_\beta(A,d,z), \Delta_{\delta_p}(A,d,z), \Delta_{\delta_q}(A,d,z))$ *with*

$$\Delta_\beta(A,d,z) \coloneqq \left.\frac{\partial G(A,d,z;\theta)}{\partial\beta}\right|_{\theta=\theta^*}$$

$$\Delta_{\delta_p}(A,d,z) \coloneqq \left.\frac{\partial G(A,d,z;\theta)}{\partial\delta_p}\right|_{\theta=\theta^*}$$

$$\Delta_{\delta_q}(A,d,z) \coloneqq \left.\frac{\partial G(A,d,z;\theta)}{\partial\delta_q}\right|_{\theta=\theta^*}$$

(iii) $||\Delta_{\delta_p}|| < \infty$, $||\Delta_{\delta_q}|| < \infty$, $\sup_{A\in\mathcal{A},d\in\{0,1\},z\in\{0,1\}} ||\Delta_\beta(A,d,z)|| < \infty$, $\sup_{A\in\mathcal{A},d\in\{0,1\},z\in\{0,1\}} ||\Delta_{\delta_p}(A,d,z)|| < \infty$, *and* $\sup_{A\in\mathcal{A},d\in\{0,1\},z\in\{0,1\}} ||\Delta_{\delta_q}(A,d,z)|| < \infty$

For a given $(A,d) \in \mathcal{A}\times\{0,1\}$, the index sufficiency test statistic is

$$\sqrt{n}\mathcal{I}_n(A,d;\hat{\theta}) = \sqrt{n}|I_n^1(A,d;\hat{\theta}) - I_n^0(A,d;\hat{\theta})|$$

with

$$I_n^z(A,d;\theta) = \frac{G_n(A,d,z;\theta)}{\bar{S}_{2n}(\delta_p,\delta_q)}$$

The overall test statistic can be written as

$$\mathcal{I}_n(\hat{\theta}) = \sup_{A\in\mathcal{A},d\in\{0,1\}} \mathcal{I}_n(A,d;\hat{\theta})$$

Theorem 4 *Suppose Assumptions 1 - 3 and 6 - 7 hold. Then, under the null we have*

$$\mathcal{I}_n(\hat{\theta}) \implies \max_{d\in\{0,1\}} \sup_{A\in\mathcal{A}} a^T\varepsilon(A,d;\theta^*)$$

where $\varepsilon(A, d; \theta^*)$ *is a mean zero Gaussian process*

$$\varepsilon(A, d; \theta^*) = (\varepsilon(A, d; \theta^*), \varepsilon_\beta, \varepsilon_{\delta_p}, \varepsilon_{\delta_q})$$

with covariance

$$\Sigma(A, d; \theta^*) = \lim_{n\to\infty} E\left[\begin{pmatrix} \nu_n^1(A, d; \theta^*) \\ \nu_n^0(A, d; \theta^*) \\ \sqrt{n}\bar{\psi}_{n\beta}(\beta^*, \delta_p^*) \\ \sqrt{n}\bar{\psi}_{n\delta_p}(\delta_p^*) \\ \sqrt{n}\bar{\psi}_{n\delta_q}(\delta_q^*, \delta_p^*) \end{pmatrix} \begin{pmatrix} \nu_n^1(A, d; \theta^*) \\ \nu_n^0(A, d; \theta^*) \\ \sqrt{n}\bar{\psi}_{n\beta}(\beta^*, \delta_p^*) \\ \sqrt{n}\bar{\psi}_{n\delta_p}(\delta_p^*) \\ \sqrt{n}\bar{\psi}_{n\delta_q}(\delta_q^*, \delta_p^*) \end{pmatrix}^T\right]$$

where

$$\begin{aligned}
\nu_n^z(A, d; \theta) &= \sqrt{n}(G_n(A, d, z; \theta) - G(A, d, z; \theta))) \\
\bar{\psi}_{n\beta}(\theta^*) &= \frac{\sum_{i=1}^n \psi_\beta(Y_i, X_i, Z_i, D_i; \beta^*, \delta_p^*)}{n} \\
\bar{\psi}_{n\delta_p}(\theta^*) &= \frac{\sum_{i=1}^n \psi_{\delta_p}(X_i, Z_i, D_i; \delta_p^*)}{n} \\
\bar{\psi}_{n\delta_q}(\theta^*) &= \frac{\sum_{i=1}^n \psi_{\delta_q}(X_i, Z_i; \delta_q^*, \delta_p^*)}{n}
\end{aligned}$$

and $a = (a_1, a_2, a_3, a_4, a_5)$ *with elements*

$$\begin{aligned}
a_1 &= 1 \\
a_2 &= -1 \\
a_3 &= (\Delta_\beta(A, d, 1) - \Delta_\beta(A, d, 0))\Gamma_\beta \\
a_4 &= (\Delta_{\delta_p}(A, d, 1) - \Delta_{\delta_p}(A, d, 0))\Gamma_{\delta_p} \\
a_5 &= (\Delta_{\delta_q}(A, d, 1) - \Delta_{\delta_q}(A, d, 0))\Gamma_{\delta_q}
\end{aligned}$$

E.2.3 Distilled Test Assumptions and Theorem

The joint empirical distributions of $(U(\beta), D)$ given $Z = 0$ and $Z = 1$ can be written as

$$\begin{aligned}
P_n(A, d; \beta) &= \frac{\sum_{i=1}^n \mathbb{1}\{U(\beta) \in A, D = d, Z = 1\}}{n} \\
Q_n(A, d; \beta) &= \frac{\sum_{i=1}^n \mathbb{1}\{U(\beta) \in A, D = d, Z = 0\}}{n}
\end{aligned}$$

Let $P(A, d; \beta)$ and $Q(A, d; \beta)$ denote their population counterparts.

$$\begin{aligned}
P(A, d; \beta) &= E[P_n(A, d; \beta)] \\
Q(A, d; \beta) &= E[Q_n(A, d; \beta)]
\end{aligned}$$

The following assumption facilitates mean-value expansions of $P(A, d; \beta)$ and $Q(A, d; \beta)$.

Assumption 8

(i) $P(A,d;\beta)$ *and* $Q(A,d;\beta)$ *are differentiable in* β *on a neighborhood* $\mathbf{B}^*$ *of* β^* *for any* $(A,d) \in \mathcal{A} \times [0,1]$.

(ii) For all sequences of positive constants $\{\tau_n : n \geq 1\}$ *such that* $\tau_n \to 0$

$$\sup_{A\in\mathcal{A},d\in\{0,1\}} \sup_{\beta:||\beta-\beta^*||\leq\tau_n} \left|\left|\frac{\partial P(A,d;\beta)}{\partial\beta} - \Delta_\beta^P(A,d)\right|\right| \to 0$$
$$\sup_{A\in\mathcal{A},d\in\{0,1\}} \sup_{\beta:||\beta-\beta^*||\leq\tau_n} \left|\left|\frac{\partial Q(A,d;\beta)}{\partial\beta} - \Delta_\beta^Q(A,d)\right|\right| \to 0$$

with

$$\Delta_\beta^P(A,d) \coloneqq \left.\frac{\partial P(A,d;\beta)}{\partial\beta}\right|_{\beta=\beta^*}$$
$$\Delta_\beta^Q(A,d) \coloneqq \left.\frac{\partial Q(A,d;\beta)}{\partial\beta}\right|_{\beta=\beta^*}$$

(iii) $\sup_{(A,d)\in\mathcal{A}\times\{0,1\}} ||\Delta_\beta^P(A,d)|| < \infty$, *and* $\sup_{(A,d)\in\mathcal{A}\times\{0,1\}} ||\Delta_\beta^Q(A,d)|| < \infty$

The distilled test statistic without trimming can then be written as

$$\mathcal{D}_n(\hat{\beta}) = \sqrt{n\hat{\lambda}_n(1-\hat{\lambda}_n)}\max\left(\sup_A\left(\frac{P_n(A,0;\hat{\beta})}{\hat{\lambda}_n} - \frac{Q_n(A,0;\hat{\beta})}{1-\hat{\lambda}_n}\right), \sup_A\left(\frac{Q_n(A,1;\hat{\beta})}{1-\hat{\lambda}_n} - \frac{P_n(A,1;\hat{\beta})}{\hat{\lambda}_n}\right)\right)$$

In addition, define

$$\mathcal{D}^*(\beta^*) = \max\left(\sup_A\left(\frac{P(A,0;\beta^*)}{\lambda} - \frac{Q(A,0;\beta^*)}{1-\lambda}\right), \sup_A\left(\frac{Q(A,1;\beta^*)}{1-\lambda} - \frac{P(A,1;\beta^*)}{\lambda}\right)\right)$$

Theorem 5 *Suppose Assumptions 1 - 3 and 8 hold. Then, under the null*

$$\mathcal{D}_n(\hat{\beta}) \implies \begin{cases} \max\left(\sup_{A\in\mathcal{C}_0} a_0^T\varepsilon(A,0;\beta^*), \sup_{A\in\mathcal{C}_1} a_1^T\varepsilon(A,1;\beta^*)\right) & \mathcal{D}^*(\beta^*) = 0 \\ -\infty & \mathcal{D}^*(\beta^*) < 0 \end{cases}$$

where $\mathcal{C}_d$ *is a contact set where the inequalities are binding, and* $\varepsilon(A,d;\beta^*)$ *is a mean zero Gaussian process*

$$\varepsilon(A,d;\beta^*) = (\varepsilon(A,d;\beta^*), \varepsilon_\beta)$$

with covariance

$$\Sigma(A,d;\beta^*) = \lim_{n\to\infty} E\left[\begin{pmatrix} \nu_n^P(A,d;\beta^*) \\ \nu_n^Q(A,d;\beta^*) \\ \sqrt{n}\bar{\psi}_{n\beta}(\beta^*,\delta_p^*) \end{pmatrix}\begin{pmatrix} \nu_n^P(A,d;\beta^*) \\ \nu_n^Q(A,d;\beta^*) \\ \sqrt{n}\bar{\psi}_{n\beta}(\beta^*,\delta_p^*) \end{pmatrix}^T\right]$$

where

$$\nu_n^P(A,d;\beta) = \sqrt{n}(P_n(A,d;\beta) - P(A,d;\beta))$$
$$\nu_n^Q(A,d;\beta) = \sqrt{n}(Q_n(A,d;\beta) - Q(A,d;\beta))$$
$$\bar{\psi}_{n\beta}(\theta^*) = \frac{\sum_{i=1}^n \psi_\beta(Y_i, X_i, Z_i, D_i; \beta^*, \delta_p^*)}{n}$$

and $a_d = (a_{d1}, a_{d2}, a_{d3})$ *with*

$$a_0 = \begin{pmatrix} 1/\lambda \\ -1/(1-\lambda) \\ (\Delta_\beta^P(A,0)/\lambda - \Delta_\beta^Q(A,0)/(1-\lambda))\Gamma_\beta \end{pmatrix} \quad a_1 = \begin{pmatrix} -1/\lambda \\ 1/(1-\lambda) \\ (\Delta_\beta^Q(A,1)/(1-\lambda) - \Delta_\beta^P(A,1)/\lambda)\Gamma_\beta \end{pmatrix}$$

E.3 Proofs

E.3.1 Notation

For a class of functions $\mathcal{F}$, some $\varepsilon > 0$ and a pseudometric d, let $N(\varepsilon, d, \mathcal{F})$ be the covering number of $\mathcal{F}$, and $N_{[]}(\varepsilon, d, \mathcal{F})$ be the bracketing number. Where relevant, let $\bar{F}$ denote a measurable envelope of $\mathcal{F}$. Let $\mathcal{Q}$ be the set of all possible joint distributions over (Y, D, X, Z), and $Q \in \mathcal{Q}$ be a generic element. η and ϵ denote constants whose value can be chosen arbitrarily.

E.3.2 Proof of Theorem 3

Fix $A \in \mathcal{A}$, $(B, B') \in \mathcal{B}$, and $d \in \{0, 1\}$. Consider the first term in (148). This can be decomposed as follows

$$\begin{aligned}
&\sqrt{nH_n(B;\hat{\delta}_p)H_n(B';\hat{\delta}_p)}\frac{H_n(A,B,d;\hat{\beta},\hat{\delta}_p)}{H_n(B;\hat{\delta}_p)} \\
&= \left(\frac{H_n(B';\hat{\delta}_p)}{H_n(B;\hat{\delta}_p)}\right)^{1/2}\left(\nu_n(A,B,d;\hat{\beta},\hat{\delta}_p) + \sqrt{n}(H(A,B,d;\hat{\beta},\hat{\delta}_p) - H(A,B,d;\beta^*,\delta_p^*))\right) \\
&+ \left(\frac{H_n(B';\hat{\delta}_p)}{H_n(B;\hat{\delta}_p)}\right)^{1/2}\frac{H(A,B,d;\beta^*,\delta_p^*)}{H(B;\delta_p^*)}\left(\sqrt{n}(H(B;\delta_p^*) - H(B;\hat{\delta}_p)) - \nu_n(B;\hat{\delta}_p))\right) \\
&+ \sqrt{H_n(B;\hat{\delta}_p)H_n(B';\hat{\delta}_p)}\sqrt{n}\left(\frac{H(A,B,d;\beta^*,\delta_p^*)}{H(B;\delta_p^*)}\right)
\end{aligned}$$

We establish the convergence of each element in a series of lemmas. Lemma 15 shows

$$\left(\frac{H_n(B';\hat{\delta}_p)}{H_n(B;\hat{\delta}_p)}\right)^{1/2} \xrightarrow{p} \left(\frac{H(B';\delta_p^*)}{H(B;\delta_p^*)}\right)^{1/2}$$

Lemma 18 gives

$$\nu_n(B;\hat{\delta}_p) \xrightarrow{p} \nu_n(B;\delta_p^*)$$

From Lemma 12,

$$H(B;\hat{\delta}_p) - H(B;\delta_p^*) \xrightarrow{p} \Delta_{\delta_p}(B)^T \Gamma_{\delta_p} \bar{\psi}_{\delta n}(\delta_p^*)$$

$$H(A,B,d;\hat{\beta},\hat{\delta}_p) - H(A,B,d;\beta^*,\delta_p^*) \xrightarrow{p} \Delta_{\delta_p}(A,B,d)^T \Gamma_{\delta_p} \bar{\psi}_{\delta_p n}(\delta_p^*) + \Delta_\beta(A,B,d)^T \Gamma_\beta \bar{\psi}_\beta(\beta^*,\delta_p^*)$$

Lemma 21 shows

$$\nu_n(A,B,d;\hat{\beta},\hat{\delta}_p) \xrightarrow{p} \nu_n(A,B,d;\beta^*,\delta_p^*)$$

Repeat with the remaining term from equation (148) and collect terms to obtain

$$\begin{aligned}
&\sqrt{nH_n(B;\hat{\delta}_p)H_n(B';\hat{\delta}_p)}\left(\frac{H_n(A,B,d;\hat{\beta},\hat{\delta}_p)}{H_n(B;\hat{\delta}_p)} - \frac{H_n(A,B',d;\hat{\beta},\hat{\delta}_p)}{H_n(B';\hat{\delta}_p)}\right) \\
&\xrightarrow{p} \left(\frac{H(B';\delta_p^*)}{H(B;\delta_p^*)}\right)^{1/2} \left(\nu_n(A,B,d;\beta^*,\delta_p^*) + \sqrt{n}\Delta_{\delta_p}(A,B,d)^T \Gamma_{\delta_p} \bar{\psi}_{\delta_p n}(\delta_p^*) + \sqrt{n}\Delta_\beta(A,B,d)^T \Gamma_\beta \bar{\psi}_\beta(\beta^*,\delta_p^*)\right) \\
&+ \left(\frac{H(B';\delta_p^*)}{H(B;\delta_p^*)}\right)^{1/2} \frac{H(A,B,d;\beta^*,\delta_p^*)}{H(B;\delta_p^*)} \left(\sqrt{n}\Delta_{\delta_p}(B)^T \Gamma_{\delta_p} \bar{\psi}_{\delta n}(\delta_p^*) - \nu_n(B;\delta_p^*)\right)\Big) \\
&- \left(\frac{H(B;\delta_p^*)}{H(B';\delta_p^*)}\right)^{1/2} \left(\nu_n(A,B',d;\beta^*,\delta_p^*) + \sqrt{n}\Delta_{\delta_p}(A,B',d)^T \Gamma_{\delta_p} \bar{\psi}_{\delta_p n}(\delta_p^*) + \sqrt{n}\Delta_\beta(A,B',d)^T \Gamma_\beta \bar{\psi}_\beta(\beta^*,\delta_p^*)\right) \\
&- \left(\frac{H(B;\delta_p^*)}{H(B';\delta_p^*)}\right)^{1/2} \frac{H(A,B',d;\beta^*,\delta_p^*)}{H(B';\delta_p^*)} \left(\sqrt{n}\Delta_{\delta_p}(B')^T \Gamma_{\delta_p} \bar{\psi}_{\delta n}(\delta_p^*) - \nu_n(B';\delta_p^*)\right)\Big) \\
&+ \sqrt{H(B;\hat{\delta}_p)H(B';\hat{\delta}_p)}\sqrt{n}\left(\frac{H(A,B,d;\beta^*,\delta_p^*)}{H(B;\delta_p^*)} - \frac{H(A,B',d;\beta^*,\delta_p^*)}{H(B';\delta_p^*)}\right)
\end{aligned}$$

Which simplifies to

$$a^T \upsilon(A,B,B',d;\theta^*) + \sqrt{H(B;\hat{\delta}_p)H(B';\hat{\delta}_p)}\sqrt{n}\left(\frac{H(A,B,d;\beta^*,\delta_p^*)}{H(B;\delta_p^*)} - \frac{H(A,B',d;\beta^*,\delta_p^*)}{H(B';\delta_p^*)}\right)$$

with a as defined in the statement of Theorem 3 and

$$\upsilon(A,B,B',d;\theta^*) = (\nu_n(A,B,d;\beta^*,\delta_p^*), \nu_n(A,B',d;\beta^*,\delta_p^*), \nu_n(B;\delta_p^*), \nu_n(B';\delta_p^*), \sqrt{n}\bar{\psi}_{n\beta}(\beta^*,\delta_p^*), \sqrt{n}\bar{\psi}_{n\delta_p}(\delta_p^*))$$

Then we may write

$$\mathcal{N}(A,B,B',d;\theta) \xrightarrow{p} e_1(A,B,B',d;\theta^*) + C e_2(A,B,B',d;\theta^*)$$

with

$$\begin{aligned}
e_1(A,B,B',d;\theta^*) &= a^T \upsilon(A,B,B',d;\theta^*) \\
C &= \sqrt{P(B;\delta_p^*)P(B';\delta_p^*)} \\
e_2(A,B,B',d;\theta^*) &= \sqrt{n}\left(\frac{P(A,B,d;\beta^*,\delta_p^*)}{P(B;\delta_p^*)} - \frac{P(A,B',d;\beta^*,\delta_p^*)}{P(B';\delta_p^*)}\right)
\end{aligned}$$

Note that, under the null, $e_2(A, B, B', d; \theta^*) \leq 0$ for all $A \in \mathcal{A}$, $e_2(A, B, B', d; \theta^*) = 0$ only for $A \in \mathcal{C}_{B,B',d}$, and $e_2(A, B, B', d; \theta^*) \to -\infty$ as $n \to \infty$ for $A \notin \mathcal{C}_{B,B',d}$. The remaining steps duplicate the arguments of Klecan et al. (1991) and Linton et al. (2005). They are shown for completeness. We wish to establish that

$$\sup_{A \in \mathcal{A}} \mathcal{N}(A, B, B', d; \theta^*) = \sup_{A \in C_{B,B',d}} e_1(A, B, B', d; \theta^*)$$

By definition $e_2(A, B, B', d; \theta^*) = 0$ for $A \in C_{B,B',d}$, and hence

$$\sup_{A \in C_{B,B',d}} \mathcal{N}(A, B, B', d; \theta^*) = \sup_{A \in C_{B,B',d}} e_1(A, B, B', d; \theta^*)$$

As $C_{B,B',d} \subset \mathcal{A}$, for any $\epsilon > 0$

$$\Pr\left(\sup_{A \in C_{B,B',d}} \mathcal{N}(A, B, B', d; \theta^*) \leq \epsilon\right) \geq \Pr\left(\sup_{A \in \mathcal{A}} \mathcal{N}(A, B, B', d; \theta^*) \leq \epsilon\right) \tag{152}$$

Let $\rho(A, A')$ be the pseudometric of Lemma 22. Fix some arbitrary $\eta > 0$ and $\lambda > 0$. Define the following sets

$$\mathcal{D}_{B,B',d} = \{A \in \mathcal{A} | A \notin \mathcal{C}_{B,B',d}, \rho(A, A') \leq \eta \text{ for some } A' \in \mathcal{A}\}$$
$$\mathcal{E}_{B,B',d} = \{A \in \mathcal{A} | A \notin \mathcal{C}_{B,B',d} \cup \mathcal{D}_{B,B',d}\}$$

Then

$$\begin{aligned}
&\Pr\left(\sup_{A \in C_{B,B',d}} \mathcal{N}(A, B, B', d; \theta^*) \leq \epsilon\right) \\
&= \Pr\left(\sup_{A \in C_{B,B',d}} \mathcal{N}(A, B, B', d; \theta^*) \leq \epsilon, \sup_{A \in \mathcal{A}} \mathcal{N}(A, B, B', d; \theta^*) \leq \epsilon + \lambda\right) \\
&+ \Pr\left(\sup_{A \in C_{B,B',d}} \mathcal{N}(A, B, B', d; \theta^*) \leq \epsilon, \sup_{A \in \mathcal{A}} \mathcal{N}(A, B, B', d; \theta^*) > \epsilon + \lambda\right) \\
&\leq \Pr\left(\sup_{A \in \mathcal{A}} \mathcal{N}(A, B, B', d; \theta^*) \leq \epsilon + \lambda\right) \\
&+ \Pr\left(\sup_{A \in C_{B,B',d}} \mathcal{N}(A, B, B', d; \theta^*) \leq \epsilon, \sup_{A \in \mathcal{D}_{B,B',d}} \mathcal{N}(A, B, B', d; \theta^*) > \epsilon + \lambda\right) \\
&+ \Pr\left(\sup_{A \in \mathcal{E}_{B,B',d}} \mathcal{N}(A, B, B', d; \theta^*) > \epsilon + \lambda\right)
\end{aligned} \tag{153}$$

where the inequality holds by the union bound, and because $C_{B,B',d}, \mathcal{D}_{B,B',d}$ and $\mathcal{E}_{B,B',d}$ form a partition of $\mathcal{A}$.

By Lemma 22 and Assumptions 2 - 5, given $\delta > 0$ and λ there exists some η such that

$$\Pr\left(\sup_{\rho(A,A')\leq\eta,\ A'\in C_{B,B',d}} |e_1(A,B,B',d;\theta^*) - e_1(A',B,B',d;\theta^*)| > \lambda\right) \leq \delta$$

It follows that

$$\begin{aligned}
&\Pr\left(\sup_{A\in C_{B,B',d}} \mathcal{N}(A,B,B',d;\theta^*) \leq \epsilon, \sup_{A\in \mathcal{D}_{B,B',d}} \mathcal{N}(A,B,B',d;\theta^*) > \epsilon + \lambda\right) \\
&\leq \Pr\left(\sup_{A\in C_{B,B',d}} e_1(A,B,B',d;\theta^*) \leq \epsilon, \sup_{A\in \mathcal{D}_{B,B',d}} e_1(A,B,B',d;\theta^*) > \epsilon + \lambda\right) \\
&\leq \Pr\left(\sup_{\rho(A,A')\leq\eta,\ A'\in C_{B,B',d}} |e_1(A,B,B',d;\theta^*) - e_1(A',B,B',d;\theta^*)| > \lambda\right) \\
&\leq \delta
\end{aligned} \tag{154}$$

Lemma 22 also implies that

$$\sup_{A\in\mathcal{A}} |e_1(A,B,B',d;\theta^*)| = O_p(1)$$

and hence, for a given δ, there exists some κ such that $\Pr(\sup_{A\in\mathcal{A}} |e_1(A,B,B',d;\theta^*)| > \epsilon+\lambda+\kappa) < \delta$. As $e_2(A,B,B',d;\theta^*)$ diverges to $-\infty$, there exists some N such that for all $n > N$ we have $Ce_2(A,B,B',d;\theta^*) < -\kappa$. Then, for $n > N$

$$\begin{aligned}
&\Pr\left(\sup_{A\in\mathcal{E}_{B,B',d}} \mathcal{N}(A,B,B',d;\theta^*) > \epsilon + \lambda\right) \\
&= \Pr\left(\sup_{A\in\mathcal{E}_{B,B',d}} e_1(A,B,B',d;\theta^*) + Ce_2(A,B,B',d;\theta^*) > \epsilon + \lambda\right) \\
&\leq \Pr\left(\sup_{A\in\mathcal{E}_{B,B',d}} e_1(A,B,B',d;\theta^*) > \epsilon + \lambda + \kappa\right) \\
&\leq \Pr\left(\sup_{A\in\mathcal{C}_{B,B',d}} e_1(A,B,B',d;\theta^*) > \epsilon + \lambda + \kappa\right) \\
&\leq \delta
\end{aligned} \tag{155}$$

Using (154) and (155) in (153) yields

$$\Pr\left(\sup_{A\in C_{B,B',d}} \mathcal{N}(A,B,B',d;\theta^*) \leq \epsilon\right) \leq \Pr\left(\sup_{A\in\mathcal{A}} \mathcal{N}(A,B,B',d;\theta^*) \leq \epsilon + \lambda\right) + 2\delta \tag{156}$$

To obtain the final result, combine this with (152) and let λ and δ go to zero.

E.3.3 Proof of Theorem 4

Fix some $A \in \mathcal{A}$ and $d \in \{0,1\}$. Note that

$$\begin{aligned}
E\left[G_n(A,d,1;\theta^*)\right] &= E\left[\frac{S_{2i}(\delta_p^*,\delta_q^*)}{q_i(\delta_p^*,\delta_q^*)} E\left[\mathbb{1}\{U_i(\beta^*)\in A, D_i = d, Z_i = 1\}|p(\delta_p^*) = p\right]\right] \\
&= E\left[\frac{S_{2i}(\delta_p^*,\delta_q^*)}{q_i(\delta_p^*,\delta_q^*)} \Pr\left(U_i(\beta^*)\in A, D_i = d, Z_i = 1|p(\delta_p^*) = p\right)\right] \\
&= E\left[\frac{S_{2i}(\delta_p^*,\delta_q^*)}{q_i(\delta_p^*,\delta_q^*)} \Pr\left(U_i(\beta^*)\in A, D_i = d|p(\delta_p^*) = p\right) \Pr\left(Z_i = 1|p(\delta_p^*) = p\right)\right] \\
&= \Pr\left(U_i(\beta^*)\in A, D_i = d|S_2(\delta_p^*,\delta_q^*) = 1\right)
\end{aligned}$$

where the penultimate line uses that, under the null hypothesis, Z is independent of (U, D) conditional on p. Following the same steps yields $E\left[G_n(A,d,0;\theta^*)\right] = \Pr\left(U_i(\beta^*)\in A, D_i = d|S_2(\delta_p^*,\delta_q^*) = 1\right)$.

Consider $I_n^1(A,d;\theta)$. We can write

$$\begin{aligned}
\sqrt{n} I_n^1(A,d;\hat{\theta}) = \sqrt{n}\frac{1}{\bar{S}_{2n}(\hat{\delta}_p,\hat{\delta}_q)}\left(G_n(A,d,1;\hat{\theta}) - G(A,d,1;\hat{\theta})\right) \\
+ \sqrt{n}\frac{1}{\bar{S}_{2n}(\hat{\delta}_p,\hat{\delta}_q)}\left(G(A,d,1;\hat{\theta}) - G(A,d,1;\theta^*)\right) \\
+ \sqrt{n}\frac{\Pr\left(U_i(\beta^*)\in A, D_i = d|S_2(\delta_p^*,\delta_q^*) = 1\right)}{\bar{S}_{2n}(\hat{\delta}_p,\hat{\delta}_q)}
\end{aligned}$$

As with Theorem 3, we deal with this expression term-by-term. Lemma 25 shows that $1/\bar{S}_{2n}(\hat{\delta}_p,\hat{\delta}_q) \xrightarrow{p} 1/\bar{S}_2(\delta_p^*,\delta_q^*)$. Lemma 29 gives that the first empirical process converges to $\nu_n^1(A,d;\theta^*)$. Lemma 12 performs a mean-value expansion of the second term. Repeat the process with $\sqrt{n} I_n^0(A,d,\hat{\theta})$ and subtract to obtain

$$\begin{aligned}
&\sqrt{\frac{n_0}{n}\frac{n_1}{n}}\sqrt{n}\left(I_n^1(A,d,\hat{\theta}) - I_n^0(A,d,\hat{\theta})\right) = \sqrt{(1-\hat{\lambda}_n)\hat{\lambda}_n}\left(\frac{1}{\bar{S}_2(\delta_p^*,\delta_q^*)}\left(\nu_n^1(A,d;\theta^*) - \nu_n^0(A,d;\theta^*)\right.\right. \\
&+(\Delta_\beta(A,d,1) - \Delta_\beta(A,d,0))\Gamma_\beta\xi_\beta + (\Delta_{\delta_p}(A,d,1) - \Delta_{\delta_p}(A,d,0))\Gamma_{\delta_p}\xi_{\delta_p} \\
&\left.\left. + (\Delta_{\delta_q}(A,d,1) - \Delta_{\delta_q}(A,d,0))\Gamma_{\delta_q}\xi_{\delta_q}\ \right)\right) + o_p(1)
\end{aligned}$$

Lemma 30 shows weak convergence of the term in brackets. Note that, by Assumption 2 (ii), $\sqrt{(1-\hat{\lambda}_n)\hat{\lambda}_n}$ converges deterministically to $\sqrt{(1-\lambda)\lambda}$.

E.3.4 Proof of Theorem 5

Fix some $A \in \mathcal{A}$. We can write

$$\begin{aligned}
\sqrt{n} P_n(A,d;\hat{\beta}) &= \sqrt{n}(P_n(A,d;\hat{\beta}) - P(A,d;\hat{\beta}) + P(A,d;\hat{\beta}) - P(A,d;\beta^*)) \\
&= \nu_n(A,d,1,\beta^*) + \Delta_\beta(A,d,1)^T\Gamma_\beta\sqrt{n}\bar{\psi}_{n\beta}(\beta^*,\delta_p^*) + o_p(1)
\end{aligned}$$

where we apply Lemmas 31 and 12. Similarly,

$$\sqrt{n}Q_n(A,d,\hat{\beta}) = \nu_n(A,d,0,\beta^*) + \Delta_\beta(A,d,0)^T\Gamma_\beta\sqrt{n}\bar{\psi}_{n\beta}(\beta^*,\delta_p^*) + o_p(1)$$

Hence,

$$\frac{\sqrt{n}P_n(A,d;\hat{\beta})}{\hat{\lambda}_n} - \frac{\sqrt{n}Q_n(A,d,\hat{\beta})}{(1-\hat{\lambda}_n)} = \frac{\nu_n(A,d,1,\beta^*)}{\lambda} - \frac{\nu_n(A,d,0,\beta^*)}{1-\lambda} + \left(\frac{\Delta_\beta(A,d,1)}{\lambda} - \frac{\Delta_\beta(A,d,0)}{1-\lambda}\right)^T \Gamma_\beta\sqrt{n}\bar{\psi}_{n\beta}(\beta^*,\delta_p^*) + o_p(1),$$

and weak convergence follows by Lemma 32. Define constants appropriately to obtain the statement. The contact set can be shown by the same argument as Theorem 3.

E.3.5 General Lemmas

Lemma 10 *Suppose Assumption 1 holds. The following are VC-classes with finite measurable envelopes*

(i) $\mathcal{F} \coloneqq \{\mathbb{1}\{D = d\} : d \in \{0,1\}\}$

(ii) $\mathcal{F} \coloneqq \{\mathbb{1}\{Z = z\} : z \in \{0,1\}\}$

(iii) $\mathcal{F} \coloneqq \{\mathbb{1}\{p(\delta_p) \leq b\} : b \in [0,1]\}$

(iv) $\mathcal{F} \coloneqq \{\mathbb{1}\{p(\delta_p) \leq b\} : \delta_p \in \mathbf{D}_p\}$

(v) $\mathcal{F} \coloneqq \{\mathbb{1}\{p(\delta_p) \leq b\} : \delta_p \in \mathbf{D}_p, b \in [0,1]\}$

(vi) $\mathcal{F} \coloneqq \{\mathbb{1}\{U(\beta) \leq a\} : a \in \mathbb{R}\}$

(vii) $\mathcal{F} \coloneqq \{\mathbb{1}\{U(\beta) \leq a\} : \beta \in \mathbf{B}, a \in \mathbb{R}\}$

(viii) $\mathcal{G} = \mathcal{F}_1 \times \mathcal{F}_2$ *where* $\mathcal{F}_1$ *and* $\mathcal{F}_2$ *are any of the function classes above.*

For any function class described above, $\sup_{Q\in\mathcal{Q}} N(\varepsilon, L_r(Q), \mathcal{F}) = O(\varepsilon^{-rk})$ *for some* k.

As all the classes above are indicators, they possess the trivial envelope 1. Claims (i) - (iii) and (vi) are immediate. For (iv), Assumption 1, allows us to write

$$\mathbb{1}\{p(\delta_p) \leq b\} = \mathbb{1}\{0 \leq \varphi_p^{-1}(b) - \kappa_p(X,Z)^T\delta_p\}.$$

The function class $\{\varphi_p^{-1}(b) - \kappa_p(X,Z)^T\delta_p : \delta_p \in \mathbf{D}_p\}$ is a finite dimensional vector space of measurable functions. Lemmas 2.6.16 and 2.6.20 of Vaart and Wellner (2023) yield the result. Claim (v) is shown in the same manner. For (vii), write

$$\mathbb{1}\{U(\beta) \leq a, D = d\} = \mathbb{1}\{Y - X^T\beta_d \leq a, D = d\}$$

and again apply Lemmas 2.6.16 and 2.6.20 of Vaart and Wellner (2023). Claim (viii) holds by Lemma 2.6.20 (iii) of Vaart and Wellner (2023): as each of the function classes generates a VC-class of sets, their products generate the intersection of VC-classes of sets. The final statement of the Lemma holds by Theorem 2.6.7 of Vaart and Wellner (2023).

Lemma 11 *Define the following classes of indicator functions*

$$\mathcal{F} \coloneqq \{\mathbb{1}\{p(\delta_p) \leq b\} : \delta_p \in \mathbf{D}_p, b \in [0,1]\}$$
$$\mathcal{G} \coloneqq \{\mathbb{1}\{U(\beta) \leq a, D = d\} : \beta \in \mathbf{B}, a \in \mathbb{R}, d \in \{0,1\}\}$$

(i) $\mathcal{F}$ and $\mathcal{G}$ satisfy the assumptions of Theorem 1 of Andrews (1994a).

(ii) Any function class constructed by the (finite) addition, multiplication or some combination of these operations of elements of $\mathcal{F}$ and $\mathcal{G}$ also satisfies the conditions of Theorem 1 of Andrews (1994b).

For claim (i), we need to verify Assumptions A, B, and C of Andrews (1994b). Assumption C is immediate with iid data. Assumption B holds as both classes have finite envelopes and the data is iid. Assumption A is satisfied by the uniform covering numbers shown in Lemma 10.

For claim (ii), Assumption C is immediate given Assumption 2. Assumption B holds as any function class formed in the manner described must have a measurable envelope. Assumption A follows by Theorem 3 of Andrews (1994b), which states that if the entropy condition holds for $\mathcal{F}$ and $\mathcal{G}$, then it also holds for function classes formed by addition and multiplication.

Lemma 12 *Suppose Assumptions 3 and 5 hold. Then*

(i) $\sqrt{n} \sup_{B \in \mathcal{B}} \left|\left| H(B, \hat{\delta}_p) - H(B, \delta_p^*) - \Delta_{\delta_p}(B)^T \Gamma_{\delta_p} \bar{\psi}_{\delta_p n}(\theta^*) \right|\right| = o_p(1).$

(ii) $\sqrt{n} \sup_{A \in \mathcal{A}, B \in \mathcal{B}, d \in \{0,1\}} \left|\left| H(A, B, d; \hat{\beta}, \hat{\delta}_p) - H(A, B, d; \beta^*, \delta^*) - \Delta_\beta(A, B, d)^T \Gamma_\beta \bar{\psi}_{\beta n}(\theta^*) - \Delta_{\delta_p}(A, B, d)^T \Gamma_\delta \bar{\psi}_{\delta n}(\theta^*) \right|\right| = o_p(1)$

Suppose Assumptions 3 and 7 hold. Then

(iii) $\sqrt{n} \sup_{A \in \mathcal{A}, d \in \{0,1\}, z \in \{0,1\}} \left|\left| G(A, d, z; \hat{\theta}) - G(A, d, z; \theta^*) - \Delta_\beta(A, d, z)^T \Gamma_\beta \bar{\psi}_{\beta n}(\theta^*) - \Delta_{\delta_p}(A, B, d)^T \Gamma_{\delta_p} \bar{\psi}_{\delta_p n}(\theta^*) - \Delta_{\delta_q}(A, B, d)^T \Gamma_{\delta_q} \bar{\psi}_{\delta_q n}(\theta^*) \right|\right| = o_p(1)$

Suppose Assumptions 3 and 8 hold. Then

(iv) $\sqrt{n} \sup_{(A,d,z) \in \mathcal{A} \times \{0,1\}^2} \left|\left| P(A, d, z; \hat{\beta}) - P(A, d, z; \beta^*) - \Delta_\beta(A, d, z)^T \Gamma_\beta \bar{\psi}_{\beta n}(\theta^*) \right|\right| = o_p(1)$

We show claim (i). The remaining claims are shown in the same manner. Choose $0 < \epsilon < \eta$.

$$
\begin{aligned}
&\Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|H(B;\hat{\delta}_p) - H(B;\delta_p^*) - \Delta_{\delta_p}(B)^T\Gamma_{\delta_p}\bar{\psi}_{n\delta_p}(\delta_p^*)\right|\right| > \eta\right) \\
&\leq \Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|H(B;\hat{\delta}_p) - H(B;\delta_p^*) - \Delta_{\delta_p}(B)^T\Gamma_{\delta_p}\bar{\psi}_{n\delta_p}(\delta_p^*)\right|\right| > \eta, \delta_p \in \mathbf{D}_p^*\right) + o(1) \\
&= \Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|\frac{\partial H(B,\tilde{\delta}_p)}{\partial\delta_p}(\hat{\delta}_p - \delta_p^*) - \Delta_{\delta_p}(B)^T\Gamma_{\delta_p}\bar{\psi}_{n\delta_p}(\delta_p^*)\right|\right| > \eta\right) + o(1) \\
&\leq \Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|\left(\frac{\partial H(B,\tilde{\delta}_p)}{\partial\delta_p} - \Delta_{\delta_p}(B)^T\right)(\hat{\delta}_p - \delta_p^*)\right|\right| + \sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|\Delta_{\delta_p}(B)^T(\hat{\delta}_p - \delta_p^* - \Gamma_{\delta_p}\bar{\psi}_{n\delta_p}(\delta_p^*))\right|\right| > \eta\right) + o(1) \\
&\leq \Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|\left(\frac{\partial H(B,\tilde{\delta}_p)}{\partial\delta_p} - \Delta_{\delta_p}(B)^T\right)(\hat{\delta}_p - \delta_p^*)\right|\right| \geq \epsilon\right) \\
&\quad + \Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|\Delta_{\delta_p}(B)^T(\hat{\delta}_p - \delta_p^* - \Gamma_{\delta_p}\bar{\psi}_{n\delta_p}(\delta_p^*))\right|\right| > \eta - \epsilon\right) + o(1) \\
&\leq \Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|\frac{\partial H(B,\tilde{\delta}_p)}{\partial\delta_p} - \Delta_{\delta_p}(B)^T\right|\right|\left|\left|\hat{\delta}_p - \delta_p^*\right|\right| \geq \epsilon\right) \\
&\quad + \Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|\Delta_{\delta_p}(B)\right|\right|\left|\left|\hat{\delta}_p - \delta_p^* - \Gamma_{\delta_p}\bar{\psi}_{n\delta_p}(\delta_p^*)\right|\right| > \eta - \epsilon\right) + o(1) \\
&\leq \Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\left|\left|\frac{\partial H(B,\tilde{\delta}_p)}{\partial\delta_p} - \Delta_{\delta_p}(B)^T\right|\right|\left|\left|\hat{\delta}_p - \delta_p^*\right|\right| \geq \epsilon\right) + o(1) \\
&\leq \Pr\left(\sqrt{n}\sup_{B\in\mathcal{B}}\sup_{\delta_p:||\delta_p - \delta_p*||\leq\xi_n}\left|\left|\frac{\partial H(B,\delta_p)}{\partial\delta_p} - \Delta_{\delta_p}(B)^T\right|\right| \geq \frac{\epsilon}{\xi_n}\right) + o(1) \\
&= o(1)
\end{aligned}
$$

The first inequality uses that $\Pr\left(\hat{\delta}_p \notin \mathbf{D}_p^*\right) = o(1)$. The first equality follows from Assumption 5 (i). Here $\tilde{\delta}_p$ lies between $\hat{\delta}_p$ and δ_p^*, and may differ between rows. The second inequality uses the triangle inequality. The third inequality holds by the union bound, and the fourth by the Cauchy-Schwarz inequality. The fifth inequality follows as Assumption 3 implies that $\sqrt{n}(\hat{\delta}_p - \delta_p^* - \Gamma_{\delta_p}\bar{\psi}_{n\delta_p}(\delta_p^*))$ is $o_p(1)$, and Assumption 5 that $\sup_{B\in[0,1]}\left|\left|\Delta_{\delta_p}(B)\right|\right|$ is finite, so their product is $o_p(1)$. To obtain the final inequality, use that Assumption 3 implies that $\hat{\delta}_p - \delta_p^*$ is $o_p(1)$. Hence, we can construct a sequence $\{\xi_n : n \geq 1\}$ such that the vector $(\hat{\delta}_p - \delta_p^*)$ is $o_p(\xi_n)$ and $\xi_n = o(1)$. The inequality then holds by the union bound. The final equality holds by Assumption 5.

E.3.6 Lemmas for the Nesting Inequalities

Lemma 13 *Suppose Assumptions 3 and 5 hold. Then*

$$\max_{B\in\mathcal{B}}\left|H(B;\hat{\delta}_p) - H(B;\delta_p^*)\right| = o_p(1) \tag{157}$$

Assumption 3 implies that $\hat{\delta}_p - \delta_p^* = o_p(1)$. Hence, we can construct an $o(1)$ sequence ξ_n such that $\Pr(||\hat{\delta}_p - \delta_p^*|| > \xi_n) = o(1)$. Choose $\eta > \epsilon > 0$

$$
\begin{aligned}
&\Pr\left(\max_{B\in\mathcal{B}} |H(B;\hat{\delta}_p) - H(B;\delta_p^*)| > \eta\right) \\
&\leq \Pr\left(\max_{B\in\mathcal{B}} ||H(B;\hat{\delta}_p) - H(B;\delta_p^*)|| > \eta, \hat{\delta}_p \in \mathbf{D}_p^*\right) + o(1) \\
&\leq \Pr\left(\max_{B\in\mathcal{B}} \left|\left|\frac{\partial H(B;\tilde{\delta}_p)}{\partial \delta_p}\right|\right| \left|\left|\hat{\delta}_p - \delta_p^*\right|\right| > \eta\right) + o(1) \\
&\leq \Pr\left(\max_{B\in\mathcal{B}} \left|\left|\frac{\partial H(B;\tilde{\delta}_p)}{\partial \delta_p}\right|\right| \left|\left|\hat{\delta}_p - \delta_p^*\right|\right| > \eta, ||\hat{\delta}_p - \delta_p^*|| \leq \xi_n\right) + o(1) \\
&\leq \Pr\left(\max_{B\in\mathcal{B}} \sup_{\delta:||\delta_p - \delta_p^*||\leq\xi_n} \left|\left|\frac{\partial H(B;\delta_p)}{\partial \delta_p}\right|\right| > \frac{\eta}{\xi_n}\right) + o(1) \\
&\leq \Pr\left(\max_{B\in\mathcal{B}} \sup_{\delta:||\delta_p - \delta_p^*||\leq\xi_n} \left|\left|\frac{\partial H(B;\delta_p)}{\partial \delta_p} - \Delta_{\delta_p}(B)\right|\right| + \max_{B\in\mathcal{B}} \left|\left|\Delta_{\delta_p}(B)\right|\right| > \frac{\eta}{\xi_n}\right) + o(1) \\
&\leq \Pr\left(\max_{B\in\mathcal{B}} \sup_{\delta:||\delta_p - \delta_p^*||\leq\xi_n} \left|\left|\frac{\partial H(B;\delta_p)}{\partial \delta_p} - \Delta_{\delta_p}(B)\right|\right| > \frac{\eta-\epsilon}{\xi_n}\right) + \Pr\left(\max_{B\in\mathcal{B}} ||\Delta_{\delta_p}(B)|| > \frac{\epsilon}{\xi_n}\right) + o(1) \\
&= o(1)
\end{aligned}
$$

The first inequality follows from the union bound and $\Pr\left(\hat{\delta}_p \notin \mathbf{D}_p^*\right) = o(1)$. The second uses that $H(B;\delta_p)$ is differentiable in δ for $\delta \in \mathbf{D}_p^*$. $\tilde{\delta}_p$ lies between $\hat{\delta}_p$ and δ_p^*, and can differ between rows. The third inequality applies the union bound, and uses $\Pr\left(||\hat{\delta}_p - \delta_p^*|| > \xi_n\right) = o(1)$. The final line follows from Assumption 5.

Lemma 14 *Suppose Assumptions 1 -5 hold.*

$$\max_{B\in\mathcal{B}} \left|H_n(B;\hat{\delta}_p) - H(B;\delta_p^*)\right| = o_p(1)$$

By definition

$$
\begin{aligned}
\max_{B\in\mathcal{B}} \left|H_n(B;\hat{\delta}_p) - H(B;\delta_p^*)\right| &\leq \max_{B\in\mathcal{B}} \left|H_n(B;\hat{\delta}_p) - H(B;\hat{\delta}_p)\right| + \max_{B\in\mathcal{B}} \left|H(B;\hat{\delta}_p) - H(B;\delta_p^*)\right| \\
&= \max_{B\in\mathcal{B}} \left|\frac{\sum_{i=1}^n \mathbb{1}\{b^- < p_i(\hat{\delta}) \leq b^+\}}{n} - \Pr(p(b^- < \hat{\delta}_p \leq b^+)\right| + o_p(1) \\
&\leq 2 \sup_{(\delta_p,b)\in\mathbf{D}_p\times[0,1]} \left|\frac{\sum_{i=1}^n \mathbb{1}\{p_i(\hat{\delta}) \leq b\}}{n} - \Pr(\hat{\delta}_p \leq b)\right| + o_p(1) \\
&= o_p(1) + o_p(1).
\end{aligned}
$$

The second line uses Lemma 13 The last line holds as Lemma 10 and Theorem 19.13 of van der Vaart (1998) imply that the function class $\{\mathbb{1}\{p(\delta_p) \leq b\} : \delta_p \in \mathbf{D}_p, b \in [0,1]\}$ is Glivenko-Cantelli.

Lemma 15 *Suppose Assumptions 2 and 4 hold*

$$\sup_{B,B'}\left|\left(\frac{H_n(B';\hat{\delta_p})}{H_n(B;\hat{\delta_p})}\right)^{1/2}-\left(\frac{H(B';\delta_p^*)}{H(B;\delta_p^*)}\right)^{1/2}\right|=o_p(1)$$

Choose $\eta>\epsilon>0$

$$\begin{aligned}
&\Pr\left(\sup_{B,B'}\left|\frac{H_n(B';\hat{\delta_p})}{H_n(B;\hat{\delta_p})}-\frac{H(B';\delta_p^*)}{H(B;\delta_p^*)}\right|>\eta\right)\\
&=\Pr\left(\sup_{B,B'}\left|\frac{H_n(B';\hat{\delta_p})}{H_n(B;\hat{\delta_p})}-\frac{H(B';\delta_p^*)}{H_n(B;\hat{\delta_p})}+\frac{H(B';\delta_p^*)}{H_n(B;\hat{\delta_p})}-\frac{H(B';\delta_p^*)}{H(B;\delta_p^*)}\right|>\eta\right)\\
&\leq\Pr\left(\sup_{B,B'}\left(\frac{\left|H_n(B';\hat{\delta_p})-H(B';\delta_p^*)\right|}{H_n(B;\hat{\delta_p})}+H(B';\delta_p^*)\frac{\left|H_n(B;\hat{\delta_p})-H(B;\delta_p^*)\right|}{H_n(B;\hat{\delta_p})H(B;\delta_p^*)}\right)>\eta\right)\\
&\leq\Pr\left(\frac{\sup_{B'}\left|H_n(B';\hat{\delta_p})-H(B';\delta_p^*)\right|}{\inf_B H_n(B;\hat{\delta_p})}+\frac{\sup_B\left|H_n(B;\hat{\delta_p})-H(B;\delta_p^*)\right|}{\inf_B H_n(B;\hat{\delta_p})H(B;\delta_p^*)}>\eta\right)\\
&\leq\Pr\left(\sup_B\left|H_n(B;\hat{\delta_p})-H(B;\delta_p^*)\right|>\epsilon\right)\\
&\quad+\Pr\left(\frac{\sup_B\left|H_n(B;\hat{\delta_p})-H(B;\delta_p^*)\right|}{\inf_B H_n(B;\hat{\delta_p})}+\frac{\sup_B\left|H_n(B;\hat{\delta_p})-H(B;\delta_p^*)\right|}{\inf_B H_n(B;\hat{\delta_p})H(B;\delta_p^*)}>\eta,\sup_B\left|H_n(B;\hat{\delta_p})-H(B;\delta_p^*)\right|\leq\epsilon\right)\\
&\leq\Pr\left(\sup_B\left|H_n(B;\hat{\delta_p})-H(B;\delta_p^*)\right|>\epsilon\right)+\Pr\left(\frac{\epsilon}{\inf_B H(B;\delta_p^*)-\epsilon}\left(1+\frac{1}{\inf_B H(B;\delta_p^*)}\right)>\eta\right)
\end{aligned}$$

Where we use that $\inf_B H_n(B;\hat{\delta_p})=\inf_B H(B;\delta_p^*)-\sup_B|H(B;\delta_p^*)-H_n(B;\hat{\delta_p})|$. The first term is $o(1)$ by Lemma 14. Assumption 4 implies that $\inf_B H(B;\delta_p^*)>0$. Hence, the second term can be made 0 by choosing ϵ appropriately.

Lemma 16 *Consider the pseudometric*

$$\rho_p((B,\delta_p),(B',\delta_p')):=\left(E\left[\left(\mathbb{1}\{p_i(\delta_p)\in B\}-\mathbb{1}\{p_i(\delta_p')\in B'\}\right)^2\right]\right)^{1/2}\tag{158}$$

Suppose Assumption 1 holds, then

$$\max_{B\in\mathcal{B}}\rho_p((B,\hat{\delta}_p),(B,\delta_p^*))=o(1)$$

As $\mathbb{1}\{p(\delta_p) \in B\} = \mathbb{1}\{p(\delta_p) \leq b^+\} - \mathbb{1}\{p(\delta_p) \leq b^-\}$,

$$\begin{aligned}
\max_{B\in\mathcal{B}} \rho_p((\hat{\delta}_p, B), (\delta_p^*, B))^2 &= \max_{B\in\mathcal{B}} E\left[(\mathbb{1}\{p_i(\hat{\delta}_p) \in B\} - \mathbb{1}\{p_i(\delta_p^*) \in B\})^2\right] \\
&\leq C \max_{(b^-,b^+)\in\{b_j\}^2} \Big(E\left[(\mathbb{1}\{p_i(\hat{\delta}_p) \leq b^-\} - \mathbb{1}\{p_i(\delta_p^*) \leq b^-\})^2\right] \\
&\qquad + E\left[(\mathbb{1}\{p_i(\hat{\delta}_p) \leq b^+\} - \mathbb{1}\{p_i(\delta_p^*) \leq b^+\})^2\right]\Big) \\
&\leq C \max_{b\in\{b_j\}} E\left[(\mathbb{1}\{p_i(\hat{\delta}_p) \leq b\} - \mathbb{1}\{p_i(\delta_p^*) \leq b\})^2\right]
\end{aligned}$$

Choose $\epsilon > 0$.

$$\begin{aligned}
E\left[\Big(\mathbb{1}\{p(\hat{\delta}_p) \leq b\} - \mathbb{1}\{p(\delta_p^*) \leq b\}\Big)^2\right] &\leq E\left[\mathbb{1}\{|p(\hat{\delta}_p) - p(\delta_p^*)| \geq |b - p(\delta_p^*)|\}\right] \\
&\leq E\left[\mathbb{1}\{C_p\, ||\hat{\delta}_p - \delta_p^*|| \geq |b - p(\delta_p^*)|\}\right] \\
&\leq E\left[\mathbb{1}\{||\hat{\delta}_p - \delta_p^*|| \geq \epsilon/C_p\}\right] + E\left[\mathbb{1}\{|b - p(\delta_p^*)| \leq \epsilon\}\right] \\
&= \Pr(||\hat{\delta}_p - \delta_p^*|| \geq \epsilon/C_p) + \Pr(b - \epsilon \leq p(\delta_p^*) \leq b + \epsilon)
\end{aligned}$$

The first inequality holds as $\Big(\mathbb{1}\{p(\hat{\delta}_p) \leq b\} - \mathbb{1}\{p(\delta_p^*) \leq b\}\Big)^2 = 1$ implies either $0 \leq b - p(\delta_p^*) < p(\hat{\delta}_p) - p(\delta_p^*)$ or $p(\hat{\delta}_p) - p(\delta_p^*) \leq b - p(\delta_p^*) < 0$. The second inequality follows from Assumption 1. For the third,

$$\begin{aligned}
\mathbb{1}\{C_p\, ||\hat{\delta}_p - \delta_p^*|| \geq |b - p(\delta_p^*)|\} &= \mathbb{1}\{C_p\, ||\hat{\delta}_p - \delta_p^*|| \geq |b - p(\delta_p^*)|\}(\mathbb{1}\{||\hat{\delta}_p - \delta_p^*|| \geq \epsilon/C_p\} + \mathbb{1}\{||\hat{\delta}_p - \delta_p^*|| < \epsilon/C_p\}) \\
&\leq \mathbb{1}\{||\hat{\delta}_p - \delta_p^*|| \geq \epsilon/C_p\} + \mathbb{1}\{|b - p(\delta_p^*)| \leq \epsilon\}
\end{aligned}$$

Consistency of $\hat{\delta}_p$ implies that $\Pr(||\hat{\delta}_p - \delta_p^*|| \geq \epsilon/C_p) = o(1)$. By Assumption 4, b is not a mass point, so ϵ can be chosen to make $\Pr(b - \epsilon \leq p(\delta_p^*) \leq b + \epsilon)$ arbitrarily small. Hence, the expression as a whole is $o(1)$.

Lemma 17 *Suppose Assumptions 1 and 2 hold. Then for each $\epsilon > 0$ there exists $\eta > 0$ such that*

$$\limsup_{n\to\infty} \left|\left| \sup_{\rho_p((\delta_p,B),(\delta_p',B'))<\eta} |\nu_n(B;\delta_p) - \nu_n(B';\delta_p')| \right|\right| < \epsilon \tag{159}$$

As $\mathbb{1}\{p(\delta_p) \in B\} = \mathbb{1}\{p(\delta_p) \leq b^+\} - \mathbb{1}\{p(\delta_p) \leq b^-\}$ the result follows by Lemma 11.

Lemma 18 *Suppose Assumptions 1 holds.*

$$\sup_{B\in\mathcal{B}} |\nu_n(B;\hat{\delta}_p) - \nu_n(B;\delta_p^*)| = o_p(1) \tag{160}$$

This follows from Lemma 16 and a standard argument which can be found, for example, in Andrews (1994a).

$$\begin{aligned}
&\Pr\left(\max_{B\in\mathcal{B}}|v_n(B,\hat{\beta})-v_n(B,\beta^*)|>\varepsilon\right)\\
&=\Pr\left(\max_{B\in\mathcal{B}}|v_n(B,\hat{\beta})-v_n(B,\beta^*)|>\varepsilon,\max_{B\in\mathcal{B}}\rho_p((B,\hat{\beta}),(B,\beta^*))>\eta\right)\\
&\quad+\Pr\left(\max_{B\in\mathcal{B}}|v_n(B,\hat{\beta})-v_n(B,\beta^*)|>\varepsilon,\max_{B\in\mathcal{B}}\rho_p((B,\hat{\beta}),(B,\beta^*))\leq\eta\right)\\
&\leq\Pr\left(\max_{B\in\mathcal{B}}\rho_p((B,\hat{\beta}),(B,\beta^*))>\eta\right)+\Pr\left(\sup_{\rho_p((B,\hat{\beta}),(B',\beta^*))\leq\eta}|v_n(B,\hat{\beta})-v_n(B',\beta^*)|>\varepsilon\right)\\
&=o(1).
\end{aligned}$$

Lemma 19 *Suppose Assumptions 1 and 2 hold. Then for each $\epsilon>0$ there exists $\eta>0$ such that*

$$\limsup_{n\to\infty}\left|\left|\sup_{\rho^*((\beta,\delta_p),(\beta',\delta'_p))<\eta}|\nu_n(A,B,d;\beta,\delta_p)-\nu_n(A,B,d;\beta',\delta'_p)|\right|\right|_q<\epsilon \tag{161}$$

where

$$\begin{aligned}
&\rho((\beta,\delta_p),(\beta',\delta'_p))=\\
&\qquad\left(E\left[(\mathbb{1}\{U_i(\beta)\in A,p_i(\delta_p)\in B,D_i=d\}-\mathbb{1}\{U_i(\beta')\in A,p_i(\delta'_p)\in B,D_i=d\})^2\right]\right)^{1/2}
\end{aligned} \tag{162}$$

Write

$$\begin{aligned}
&\mathbb{1}\{U_i(\beta)\in A,p_i(\delta_p)\in B,D_i=d\}=\\
&\qquad\left(\mathbb{1}\{U_i(\beta)\leq a^+,D_i=d\}-\mathbb{1}\{U_i(\beta)\leq a^-,D_i=d\}\right)\times\left(\mathbb{1}\{p(\delta_p)\leq b^+\}-\mathbb{1}\{p(\delta_p)\leq b^-\}\right).
\end{aligned}$$

As this function is formed by adding and multiplying members of $\mathcal{F}$ and $\mathcal{G}$, the result is immediate from Lemma (11).

Lemma 20 *Consider the pseudometrics*

$$\begin{aligned}
&\rho_u((A,d,\beta),(A',d',\beta'))=\left(E\left[(\mathbb{1}\{U_i(\beta)\in A,D_i=d\}-\mathbb{1}\{U_i(\beta')\in A',D_i=d'\})^2\right]\right)^{1/2},\\
&\rho_u((A,d,z,\beta),(A',d',z',\beta'))=\left(E\left[(\mathbb{1}\{U_i(\beta)\in A,D_i=d,Z_i=z\}-\mathbb{1}\{U_i(\beta')\in A',D_i=d',Z_i=z'\})^2\right]\right)^{1/2}.
\end{aligned}$$

Suppose Assumptions 2 and 3 hold. Then

$$\sup_{(A,d)\in\mathcal{A}\times\{0,1\}}\rho_u((A,d,\hat{\beta}),(A,d,\beta^*))=o_p(1),$$

$$\sup_{(A,d,z)\in\mathcal{A}\times\{0,1\}^2}\rho_u((A,d,z,\hat{\beta}),(A,d,z,\beta^*))=o_p(1).$$

Notice that

$$\begin{aligned}\rho_u((A,d,z,\hat{\beta}),(A,d,z,\beta^*)) &= \left(E\left[\mathbb{1}\{D_i=d,Z_i=z\}(\mathbb{1}\{U(\hat{\beta})\in A\}-\mathbb{1}\{U(\beta^*)\in A\})^2\right]\right)^{1/2},\\ &\leq \left(E\left[\mathbb{1}\{D_i=d\}(\mathbb{1}\{U(\hat{\beta})\in A\}-\mathbb{1}\{U(\beta^*)\in A\})^2\right]\right)^{1/2},\\ &= \rho_u((A,d,\hat{\beta}),(A,d,\beta^*)),\end{aligned}$$

and therefore the first claim of the lemma implies the second. Consider $\rho_u((A,d,\hat{\beta}),(A,d,\beta^*))$

$$\begin{aligned}\rho_u((A,d,\hat{\beta}),(A,d,\beta^*))^2 &= E\left[(\mathbb{1}\{U(\hat{\beta})\in A,D=d\}-\mathbb{1}\{U(\beta^*)\in A,D=d\})^2\right]\\ &= E[\mathbb{1}\{D=d\}(\mathbb{1}\{U_d-X^T(\hat{\beta}_d-\beta_d^*)\in A\}-\mathbb{1}\{U_d\in A\})^2]\\ &\leq E[\mathbb{1}\{D=d\}(\mathbb{1}\{U_d-X^T(\hat{\beta}_d-\beta_d^*)\leq a^+\}-\mathbb{1}\{U_d\leq a^+\}\\ &\qquad -(\mathbb{1}\{U_d-X^T(\hat{\beta}_d-\beta_d^*)\leq a^-\}-\mathbb{1}\{U_d\leq a^-\}))^2]\end{aligned}$$

$$\therefore \sup_{(A,d)\in\mathcal{A}\times\{0,1\}}\rho_u((A,d,\hat{\beta}),(A,d,\beta^*))^2 \leq C\max_{d\in\{0,1\}}\sup_a E[\mathbb{1}\{D=d\}(\mathbb{1}\{U_d-X^T(\hat{\beta}_d-\beta_d^*)\leq a\}-\mathbb{1}\{U_d\leq a\})^2]$$

The remaining steps are essentially identical to Lemma 2 of Linton et al. (2005). For $d=1$,

$$\begin{aligned}\sup_a\rho_u((A,1,\hat{\beta}),(A,1,\beta^*))^2 &\leq C\sup_a E\left[D\mathbb{1}\{|U_1-a|\leq|X^T(\beta_1^*-\hat{\beta}_1)|\}\right]\\ &= C\sup_a E\left[p(\delta_p^*)E[\mathbb{1}\{|U_1-a|\leq|X^T(\beta_1^*-\hat{\beta}_1)|\}|X,Z,D=1]\right]\\ &\leq CE\left[p(\delta_p^*)|X^T(\beta_1^*-\hat{\beta}_1)|\right]\\ &\leq C||\beta_1^*-\hat{\beta}_1||\end{aligned}$$

The second line uses the law of iterated expectations and that $p(\delta_p^*) \coloneqq \Pr(D=1|X,Z)$. The third line uses that the density of U_1 conditional on $(X,Z,D=1)$ is bounded from Assumption 2. The final inequality uses $|X^T(\beta_1'-\beta_1)| = ||X^T(\beta_1'-\beta_1)|| \leq ||X||\,||\beta_1'-\beta_1||$, and that $E[p(\delta_p^*)||X||] \leq E[||X||] \leq \infty$. Repeating with $d=0$ gives a bound $C||\beta_0^*-\hat{\beta}_0||$. By Assumption 3, $\beta^*-\hat{\beta}$ is $o_p(1)$, which gives the final result.

Lemma 21 *Suppose Assumptions 1 - 4 hold. Then*

$$\sup_{(A,B,d)\in\mathcal{A}\times\mathcal{B}\times\{0,1\}}|v_n(A,B,d;\hat{\beta},\hat{\delta}_p)-v_n(A,B,d;\beta^*,\delta_p^*)| \xrightarrow{p} 0 \tag{163}$$

Using the pseudometric of Lemma 19

$$\begin{aligned}
\sup_{A\in\mathcal{A},B\in\mathcal{B},d\in\{0,1\}} \rho((\hat{\beta},\hat{\delta}),(\beta^*,\delta^*))^2 &= \sup_{A\in\mathcal{A},B\in\mathcal{B},d\in\{0,1\}} E\Big[(\mathbb{1}\{U(\hat{\beta})\in A, D=d\}\mathbb{1}\{p(\hat{\delta}_p)\in B\} \\
&\quad -\mathbb{1}\{U(\beta^*)\in A, D=d\}\mathbb{1}\{p(\delta_p^*)\in B\})^2\Big] \\
&\leq C \sup_{A\in\mathcal{A},d\in\{0,1\}} E\Big[(\mathbb{1}\{U(\hat{\beta})\in A, D=d\} - \mathbb{1}\{U(\beta^*)\in A, D=d\})^2\Big] \\
&\quad + C\sup_{B\in\mathcal{B}} E\Big[(\mathbb{1}\{p(\hat{\delta}_p)\in B\} - \mathbb{1}\{p(\delta_p^*)\in B\})^2\Big] \\
&\leq C \sup_{(A,d)\in\mathcal{A}\times\{0,1\}} \rho_u((A,d,\hat{\beta}),(A,d,\beta^*)) + C\sup_{B\in\mathcal{B}} \rho_p((B,\hat{\delta}_p),(B,\delta_p^*)) \\
&= o_p(1) + o(1)
\end{aligned}$$

where the final line uses Lemma 16 and 20. The remainder of the proof is identical to Lemma 18.

Lemma 22 *Suppose Assumptions 1 - 4 hold. Then, for any $d\in\{0,1\}$ we have*

$$\big(\nu_n(\cdot,\cdot,d;\beta^*,\delta_p^*),\ \sqrt{n}\bar{\psi}_{n\beta}(\beta^*,\delta_p^*),\ \sqrt{n}\bar{\psi}_{n\delta_p}(\delta_p^*)\big) \implies \big(\varepsilon(\cdot,\cdot,d;\beta^*,\delta_p^*),\varepsilon_\beta,\ \varepsilon_{\delta_p}\big)$$

In addition, the sample paths of $\varepsilon(\cdot,\cdot,d;\beta^*,\delta_p^*)$ are uniformly continuous with respect to the pseudometric ρ on $\mathcal{A}^2\times\mathcal{B}^2$ with probability one, where

$$\rho((A,B),(A',B')) = \Big\{E\Big[\big(\mathbb{1}\{U(\beta^*)\in A, D=d, p(\delta_p^*)\in B\} - \mathbb{1}\{U(\beta^*)\in A', D=d, p(\delta_p^*)\in B'\}\big)^2\Big]\Big\}^{1/2} \tag{164}$$

We verify the conditions of Theorem 10.2 of Pollard (1990). These are (i) total boundedness of the pseudometric space (U,p) (ii) stochastic equicontinuity of $\{\nu_n(\cdot,\cdot,d;\beta^*,\delta_p^*) : n\geq 1\}$ and (iii) finite dimensional convergence.

(i) and (ii) follow from Lemma 10 and 11. Condition (iii) requires that $\big(\nu_n(A_1,B_1,d;\beta^*,\delta_p^*)$, $\ldots,\nu_n(A_J,B_J,d;\beta^*,\delta_p^*)$ $\sqrt{n}\bar{\psi}_{n\beta}(\beta^*,\delta_p^*),\ \sqrt{n}\bar{\psi}_{n\delta_p}(\delta_p^*)\big)$ converges in distribution to $\big(\varepsilon(A_1,B_1,d;\beta^*,\delta_p^*),\ldots,\varepsilon(A_J,B_J$ By assumption 2, the data is iid. Furthermore, all variables have finite variance. Hence, we can apply the Lindeberg-Levy central limit theorem and the Cramer-Wold device.

E.3.7 Index Sufficiency Lemmas

Lemma 23 *Suppose Assumption 1 holds. Then, For any $(x,z)\in\mathcal{X}\times\{0,1\}$ and any $(\delta_p,\delta_q)\in \mathbf{D}_p\times\mathbf{D}_q$*

$$\sup_{(x,z)\in\mathcal{X}\times\{0,1\}} |q(\delta_p,\delta_q) - q(\delta_p',\delta_q')| \leq C||\delta_p-\delta_p'|| + C||\delta_q-\delta_q'||$$

Fix some $(x, z) \in \mathcal{X} \times \{0, 1\}$. By Assumption 1

$$\begin{aligned} q(\delta_p, \delta_q) - q(\delta'_p, \delta'_q) &= q(\delta_p, \delta_q) - q(\delta_p, \delta'_q) + q(\delta_p, \delta'_q) - q(\delta'_p, \delta'_q) \\ &\leq C_q ||\delta_q - \delta'_q|| + C_q ||p(\delta_p) - p(\delta'_p)|| \\ &\leq C_q ||\delta_q - \delta'_q|| + C_q C_p ||\delta_p - \delta'_p|| \end{aligned}$$

Define constants appropriately. As this holds for any (x, z), the result follows.

Lemma 24 *Suppose Assumptions 1, 2, and 6 hold. Let $\mathcal{F}$ be the class of indicator functions $\mathcal{F} \coloneqq \{\mathbb{1}\{\iota \leq q(\delta_p, \delta_q) \leq 1 - \iota\} : \delta_p \in \mathbf{D}_p, \delta_q \in \mathbf{D}_q\}$. $\mathcal{F}$ is a Glivenko-Cantelli class of functions.*

Assume that the joint distribution of (X, Z) admits a continuous density with $0 \leq \ell < \infty$ mass points. There are 2^ℓ possible combinations of values for $q(\delta_p, \delta_q)$ for all mass points. The lth combination defines a subspace $(\mathbf{D}_p \times \mathbf{D}_q)_l$ that is consistent with the values of $q(\delta_p, \delta_q)$ at mass points.

Form an ε-cover such that for any $(\delta_p, \delta_q) \in (\mathbf{D}_p \times \mathbf{D}_q)_l$ there exists $(\delta_{pj}, \delta_{qj})$ with $||\delta_{pj} - \delta_p|| \leq C\varepsilon$ and $||\delta_{qj} - \delta_q|| \leq C\varepsilon$. This cover will consist of $O(\varepsilon^{-(k_p + k_q)})$ points. For a given (δ_p, δ_q), let $(\delta_{pj}, \delta_{qj})$ be the closest point from the ε-net. Set the lower and upper brackets to be

$$\begin{aligned} l(\delta_p, \delta_q) &= \mathbb{1}\{\iota + C\varepsilon \leq q(\delta_{pj}, \delta_{qj}) \leq 1 - \iota - C\varepsilon\} \\ u(\delta_p, \delta_q) &= \mathbb{1}\{\iota - C\varepsilon \leq q(\delta_{pj}, \delta_{qj}) \leq 1 - \iota + C\varepsilon\} \end{aligned}$$

The lower bracket satisfies

$$l(\delta_p, \delta_q) = \mathbb{1}\{\iota + C\varepsilon \leq q(\delta_{pj}, \delta_{qj})\} + \mathbb{1}\{q(\delta_{pj}, \delta_{qj}) \leq 1 - \iota - C\varepsilon\}$$

By Lemma 23 and the construction of the ε-net,

$$\begin{aligned} &\mathbb{1}\{\iota + C\varepsilon \leq q(\delta_{pj}, \delta_{qj})\} \leq \mathbb{1}\{\iota + C\varepsilon \leq q(\delta_{pj}, \delta_{qj}) + C\varepsilon\} \leq \mathbb{1}\{\iota \leq q(\delta_p, \delta_q)\} \\ &\mathbb{1}\{q(\delta_{pj}, \delta_{qj}) \leq 1 - \iota - C\varepsilon\} \leq \mathbb{1}\{q(\delta_p, \delta_q) - C\varepsilon \leq 1 - \iota - C\varepsilon\} \leq \mathbb{1}\{q(\delta_p, \delta_q) \leq 1 - \iota\} \end{aligned}$$

and hence

$$l(\delta_p, \delta_q) \leq \mathbb{1}\{\iota \leq q(\delta_p, \delta_q)\} + \mathbb{1}\{q(\delta_p, \delta_q) \leq 1 - \iota\} = \mathbb{1}\{\iota \leq q(\delta_p, \delta_q) \leq 1 - \iota\}$$

The upper bracket can be shown to bound the function from above by the same steps. The $L_1(P)$ width of this bracket is

$$\begin{aligned} &E[|\mathbb{1}\{\iota - C\varepsilon \leq q(\delta_{pj}, \delta_{qj}) \leq 1 - \iota + C\varepsilon\} - \mathbb{1}\{\iota + C\varepsilon \leq q(\delta_{pj}, \delta_{qj}) \leq 1 - \iota - C\varepsilon\}|] \\ &\leq \Pr(\iota - C\varepsilon \leq q(\delta_{pj}, \delta_{qj}) \leq \iota + C\varepsilon) + \Pr(1 - \iota - C\varepsilon \leq q(\delta_{pj}, \delta_{qj}) \leq 1 - \iota + C\varepsilon) \end{aligned}$$

which reduces to $C\varepsilon$. Repeating this for all 2^ℓ possible combinations gives that the bracketing number is $O(\varepsilon^{-(k_p+k_q)})$. If the distribution of (X, Z) consists of 2^ℓ mass points, then the $q(\delta_p, \delta_q)$ can be perfectly approximated by 2^ℓ functions. In either case, the bracketing entropy is finite, which gives the result.

Lemma 25 *Suppose Assumptions 1 - 3 and 6 - 7 hold. Then*

$$\frac{1}{\bar{S}_{2n}(\hat{\delta}_p, \hat{\delta}_q)} - \frac{1}{\bar{S}_2(\delta_p^*, \delta_q^*)} = o_p(1)$$

$$\begin{aligned}
\left| \frac{1}{\bar{S}_{2n}(\hat{\delta}_p, \hat{\delta}_q)} - \frac{1}{\bar{S}_2(\delta_p^*, \delta_q^*)} \right| &= \left| \frac{\bar{S}_{2n}(\hat{\delta}_p, \hat{\delta}_q) - \bar{S}_2(\delta_p^*, \delta_q^*)}{\bar{S}_{2n}(\hat{\delta}_p, \hat{\delta}_q)\bar{S}_2(\delta_p^*, \delta_q^*)} \right| \\
&\leq C|\bar{S}_{2n}(\hat{\delta}_p, \hat{\delta}_q) - \bar{S}_2(\delta_p^*, \delta_q^*)| \\
&\leq C|\bar{S}_{2n}(\hat{\delta}_p, \hat{\delta}_q) - \bar{S}_2(\hat{\delta}_p, \hat{\delta}_q)| + C|\bar{S}_2(\hat{\delta}_p, \hat{\delta}_q) - \bar{S}_2(\delta_p^*, \delta_q^*)| \\
&\leq \sup_{\delta_p \in \mathbf{D}_p, \delta_q \in \mathbf{D}_q} C|\bar{S}_{2n}(\delta_p, \delta_q) - \bar{S}_2(\delta_p, \delta_q)| + C|\bar{S}_2(\hat{\delta}_p, \hat{\delta}_q) - \bar{S}_2(\delta_p^*, \delta_q^*)| \\
&= o_p(1) + o_p(1)
\end{aligned}$$

Here $C = 1/\epsilon^2$, with ϵ defined in Assumption 6. In the last line, the first term is $o_p(1)$ by Lemma 24. For the second, Assumptions 3 and 7, and a mean value expansion give $\bar{S}_2(\hat{\delta}_p, \hat{\delta}_q) - \bar{S}_2(\delta_p^*, \delta_q^*) = o_p(1)$.

Lemma 26 *Let (δ_p, δ_q) and (δ_p', δ_q') be two points in $\mathbf{D}_p \times \mathbf{D}_q$.*

$$E\left[\left(\frac{S_2(\delta_p, \delta_q)}{q(\delta_p, \delta_q)} - \frac{S_2(\delta_p', \delta_q'))}{q(\delta_p', \delta_q'))}\right)^2\right] \leq C \sup_{(x,z)\in\mathcal{X}\times\{0,1\}} |q(\delta_p, \delta_q) - q(\delta_p', \delta_q')| + C \sup_{(x,z)\in\mathcal{X}\times\{0,1\}} |q(\delta_p, \delta_q) - q(\delta_p', \delta_q')|^2$$

We have

$$\begin{aligned}
E\left[\left(\frac{S_2(\delta_p, \delta_q)}{q(\delta_p, \delta_q)} - \frac{S_2(\delta_p', \delta_q')}{q(\delta_p', \delta_q')}\right)^2\right] &\leq C \Pr(S_2(\delta_p, \delta_q) \neq S_2(\delta_p', \delta_q')) \\
+ E\left[\left(\frac{1}{q(\delta_p, \delta_q)} - \frac{1}{q(\delta_p', \delta_q')}\right)^2 \middle| S_2(\delta_p, \delta_q) = 1, S_2(\delta_p', \delta_q') = 1\right]
\end{aligned}$$

where $C = 1/\iota$. The first term is the probability an observation is included with parameters (δ_p, δ_q) but trimmed with parameters (δ_p', δ_q'), or vice versa. The second term is the difference in weights

for observations included for both (δ_p, δ_q) and (δ'_p, δ'_q). For the first term

$$\begin{aligned}
\Pr(S_2(\delta_p,\delta_q) \neq S_2(\delta'_p,\delta'_q)) &\leq \Pr\left(|\iota - q(\delta_p,\delta_q)| \leq |q(\delta'_p,\delta'_q) - q(\delta_p,\delta_q)|\right) \\
&\quad + \Pr\left(|1-\iota - q(\delta_p,\delta_q)| \leq |q(\delta'_p,\delta'_q) - q(\delta_p,\delta_q)|\right), \\
&\leq \Pr\left(|\iota - q(\delta_p,\delta_q)| \leq \sup_{(x,z)\in\mathcal{X}\times\{0,1\}} |q(\delta'_p,\delta'_q) - q(\delta_p,\delta_q)|\right) \\
&\quad + \Pr\left(|1-\iota - q(\delta_p,\delta_q)| \leq \sup_{(x,z)\in\mathcal{X}\times\{0,1\}} |q(\delta'_p,\delta'_q) - q(\delta_p,\delta_q)|\right), \\
&\leq C \sup_{(x,z)\in\mathcal{X}\times\{0,1\}} |q(\delta'_p,\delta'_q) - q(\delta_p,\delta_q)|.
\end{aligned}$$

For the last inequality, note that $q(\delta_p, \delta_q)$ must be 0 or 1 at any mass point. Hence, for a given C, there is at most a finite mass of observations in the intervals $\iota - C \leq q(\delta_p,\delta_q) \leq \iota + C$ and $1 - \iota - C \leq q(\delta_p,\delta_q) \leq 1 - \iota + C$.

For the second term,

$$\begin{aligned}
&E\left[\left(\frac{1}{q(\delta_p,\delta_q)} - \frac{1}{q(\delta'_p,\delta'_q)}\right)^2 \middle| S_2(\delta_p,\delta_q) = 1, S_2(\delta'_p,\delta'_q) = 1\right] \\
&\leq CE\left[\left(q(\delta'_p,\delta'_q) - q(\delta_p,\delta_q)\right)^2 \middle| S_2(\delta_p,\delta_q) = 1, S_2(\delta'_p,\delta'_q) = 1\right] \\
&\leq C \sup_{(x,z)\in\mathcal{X}\times\{0,1\}} |q(\delta'_p,\delta'_q) - q(\delta_p,\delta_q)|^2
\end{aligned}$$

where the second inequality uses that $S_2(\delta_p,\delta_q) = 1$ implies $q(\delta_p,\delta_q) \geq \iota$.

Lemma 27 *Let $\mathcal{F}$ be the class of functions $\mathcal{F} \coloneqq \{S_2(q(\delta_p,\delta_q))/q(\delta_p,\delta_q) : (\delta_p,\delta_q) \in \mathbf{D}_p \times \mathbf{D}_q\}$. $\sup_{Q\in\mathcal{Q}} N(\varepsilon(QF^2)^{1/2}, L_2(P), \mathcal{F}) = O(\varepsilon^{-2(k_p+k_q)})$ for $\varepsilon \in (0,1)$.*

Note first that the measurable envelope of $\mathcal{F}$ is $1/\iota$. Consider any Q such that the joint distribution of (X,Z) is continuous with a finite number ℓ of mass points, so that there are 2^ℓ possible combinations of values of $S_2(\delta_p,\delta_q)$. For a given combination, we can define a subspace of $\mathbf{D}_p \times \mathbf{D}_q$ such that for any (δ'_p,δ'_q) in this subspace $S_2(\delta_p,\delta_q) = S_2(\delta'_p,\delta'_q)$ for all mass points. Fix some $\varepsilon \in (0,1)$. Form a ε^2-cover of the subspace such that for any (δ_p,δ_q) there exists $(\delta_{pj},\delta_{qj})$ such that $||\delta_{pj} - \delta_p|| \leq C\varepsilon^2$ and $||\delta_{qj} - \delta_q|| \leq C\varepsilon^2$. This cover will consist of $O(\varepsilon^{-2(k_p+k_q)})$ points. By Lemma 23, within this cover there will be some element j such that

$$\sup_{(x,z)\in\mathcal{X}\times\{0,1\}} ||q(\delta_p,\delta_q) - q(\delta_{pj},\delta_{qj})|| \leq C\varepsilon^2$$

By construction, $S_2(q(\delta_p,\delta_q)) \neq S_2(q(\delta_{pj},\delta_{qj}))$ over at most an interval in which $q(\delta_p,\delta_q)$ is continuously distributed. Applying Lemma 26, the squared $L_2(P)$ distance is bounded by $C\varepsilon^2 + C\varepsilon^4 \leq C\varepsilon^2$.

Hence, the $L_2(P)$ distance is bounded by ε. The same argument applies for Q where the distribution of (X, Z) is consists of a finite number of mass points only. For Q where the distribution has no mass points, the argument applies with the modification that we need only a single ε-net.

Lemma 28 *For each $\epsilon > 0$ there exists $\xi > 0$ such that*

$$\limsup_{n\to\infty} \left|\left| \sup_{\rho_d^*(\theta,\theta')<\xi} |v_n^z(A, d; \theta) - v_n^z(A, d, z; \theta')| \right|\right|_q < \epsilon \tag{165}$$

where

$$\rho_d^*(\theta, \theta') = \left(E\left[\left(\frac{(\mathbb{1}\{U_i(\beta) \in A, D_i = d, Z_i = z\} S_{2i}(q(\delta_p, \delta_q))}{q(\delta_p, \delta_q)^z (1 - q(\delta_p, \delta_q)^z)^{1-z}} - \frac{\mathbb{1}\{U_i(\beta') \in A, D_i = d, Z_i = z\} S_{2i}(q(\delta'_p, \delta'_q)}{q(\delta'_p, \delta'_q)^z (1 - q(\delta'_p, \delta'_q))^{1-z}} \right)^2 \right] \right)^{1/2} \tag{166}$$

This can be shown in the same manner as Lemmas 17 and 19. Set $z = 1$ and let

$$\mathcal{F} \coloneqq \left\{ \mathbb{1}\{U_i(\beta) \in A, D_i = d\} \mathbb{1}\{Z_i = z\} \frac{S_{2i}(q(\delta_p, \delta_q))}{q(\delta_p, \delta_q)} : (\beta, \delta_p, \delta_q) \in \mathbf{B} \times \mathbf{D}_p \times \mathbf{D}_q \right\}$$

The conditions of Theorem 1 of Andrews (1994b) were verified for $\mathbb{1}\{U_i(\beta) \in A, D_i = d\}$ in Lemma 11. $\mathbb{1}\{Z_i = z\}$ is a simple indicator, so the conditions are immediate. $\frac{S_{2i}(q(\delta_p,\delta_q))}{q(\delta_p,\delta_q)}$ is equipped with the measurable envelope ι^{-1}. Together with the covering number established in Lemma 27, we have that the conditions of Theorem 1 hold. The result follows by Theorem 3 of Andrews (1994b). The result for $z = 0$ can be shown by the same steps.

Lemma 29 *Suppose Assumptions 1 - 3 hold. Then*

$$\sup_A |\nu_n^z(A, d; \hat{\theta}) - \nu_n^z(A, d; \theta^*)| \xrightarrow{p} 0 \tag{167}$$

The argument here is the same as Lemma 21. Consider the square of the pseudo-metric from Lemma 28

$$E\left[\left(\frac{\mathbb{1}\{U_i(\hat{\beta}) \in A, D_i = d, Z_i = 1\} S_2(q(\hat{\delta}_p, \hat{\delta}_q))}{q(\hat{\delta}_p, \hat{\delta}_q)} - \frac{\mathbb{1}\{U_i(\beta^*) \in A, D_i = d, Z_i = 1\} S_2(q(\delta_p^*, \delta_q^*))}{q(\delta_p^*, \delta_q^*)} \right)^2 \right]$$

$$\leq CE\left[\left(\mathbb{1}\{U_i(\beta^*) \in A, D_i = d\} - \mathbb{1}\{U_i(\hat{\beta}) \in A, D_i = d\} \right)^2 \right] + CE\left[\left(\frac{S_2(q(\delta_p^*, \delta_q^*))}{q(\delta_p^*, \delta_q^*)} - \frac{S_2(q(\hat{\delta}_p, \hat{\delta}_q))}{q(\hat{\delta}_p, \hat{\delta}_q)} \right)^2 \right]$$

The first term is $o_p(1)$ by Lemma 20. For the second, it follows from Lemmas 23 and 26 that

$$E\left[\left(\frac{S_2(q(\delta_p^*, \delta_q^*))}{q(\delta_p^*, \delta_q^*)} - \frac{S_2(q(\hat{\delta}_p, \hat{\delta}_q))}{q(\hat{\delta}_p, \hat{\delta}_q)} \right)^2 \right] \leq C||\hat{\delta}_p - \delta_p^*|| + C||\hat{\delta}_q - \delta_q^*|| \xrightarrow{p} 0$$

The final result then follows from Lemma 28 by the same argument as Lemma 18.

Lemma 30 *Suppose Assumptions 1 - 3 and 6 - 7 hold. Then, for any $d \in \{0,1\}$ and $z \in \{0,1\}$ we have*

$$\left(\nu_n^z(\cdot, d, z; \theta^*),\ \sqrt{n}\bar{\psi}_{n\beta}(\beta^*, \delta_p^*),\ \sqrt{n}\bar{\psi}_{n\delta_p}(\delta_p^*),\ \sqrt{n}\bar{\psi}_{n\delta_p}(\delta_p^*, \delta_q^*)\right) \implies \left(\varepsilon(\cdot, \cdot, d; \beta^*, \delta_p^*), \varepsilon_\beta,\ \varepsilon_{\delta_p}, \varepsilon_{\delta_q}\right)$$

In addition, the sample paths of ν_n are uniformly continuous with respect to the pseudometric ρ_d on $\mathcal{A} \times \mathcal{A}$ with probability one, where

$$\rho(A, A') = \left\{ E\left[(\mathbb{1}\{U(\beta^*) \in A\} - \mathbb{1}\{U(\beta^*) \in A'\})^2 \left(\mathbb{1}\{D = d, Z = z\} \frac{S_{2i}(\delta_p^*, \delta_q^*)}{q(\delta_p^*, \delta_q^*)} \right)^2 \right] \right\}^{1/2} \tag{168}$$

Define the function class

$$\mathcal{F} := \left\{ (\mathbb{1}\{U(\beta^*) \in A, D = d, Z = z\} \frac{S_2(\delta_p^*, \delta_q^*)}{q(\delta_p^*, \delta_q^*)} : A \in \mathcal{A} \right\}$$

The measurable envelope of this class is $1/\iota$. For any A, A', the pseudometric distance is bounded by $C|\mathbb{1}\{U(\beta^*) \in A\} - \mathbb{1}\{U(\beta^*) \in A'\}|$ with $C = 1/\iota$. It follows from Lemma 10 that the covering number is uniformly bounded uniformly. Hence, stochastic equicontinuity and total boundedness are satisfied. Finite dimensional convergence holds by the Cramer-Wold device and the Lindeberg-Levy central limit theorem as the data is iid and all variables have finite variance.

E.3.8 Distilled Test Lemmas

Lemma 31 *Suppose Assumptions 1 and 2 hold*

(i) For each $\epsilon > 0$ there exists $\eta > 0$ such that

$$\limsup_{n \to \infty} \left|\left| \sup_{\rho((A,d,z,\beta),(A',d',z',\beta')) < \eta} |\nu_n(A, d, z; \beta) - \nu_n(A', d', z'; \beta')| \right|\right| < \epsilon$$

where

$$\rho((A, d, z, \beta), (A', d', z', \beta')) = \left(E\left[(\mathbb{1}\{U_i(\beta) \in A, D = d, Z = z\} - \mathbb{1}\{U_i(\beta') \in A', D = d', Z = z'\})^2 \right] \right)^{1/2}$$

(ii) $$\sup_{A \in \mathcal{A}, d \in \{0,1\}, z \in \{0,1\}} |\nu_n(A, d, z; \hat{\beta}) - \nu_n(A, d, z; \beta^*)| = o_p(1)$$

For claim (i), note that $\mathbb{1}\{U(\beta) \in A, D = d, Z = z\}$ can be written as $\mathbb{1}\{D = d\}\mathbb{1}\{Z = z\}(\mathbb{1}\{U(\beta) \leq a^+\} - \mathbb{1}\{U(\beta) \leq a^-\})$ and apply Lemma 11. Claim (ii) is show by applying Lemma 20, and following the method of Lemma 18.

Lemma 32 *Suppose Assumptions 1 - 3 and 8 hold. Then, for any* $(d, z) \in \{0, 1\}^2$ *we have*

$$\left(\nu_n(\cdot, d, z; \beta^*),\ \sqrt{n}\bar{\psi}_{n\beta}(\beta^*, \delta_p^*)\right) \implies \left(\varepsilon(\cdot, \cdot, d; \beta^*, \delta_p^*), \varepsilon_\beta\right)$$

In addition, the sample paths of ν_n *are uniformly continuous with respect to the pseudometric* ρ_d *on* $\mathcal{A} \times \mathcal{A}$ *with probability one, where*

$$\rho(A, A') = \left\{E\left[(\mathbb{1}\{U(\beta^*) \in A, D = d, Z = z\} - \mathbb{1}\{U(\beta^*) \in A', D = d, Z = z\})^2\right]\right\}^{1/2}$$

Define the function class $\mathcal{F} \coloneqq \{\mathbb{1}\{U(\beta^*) \in A, D = d, Z = z\} : A \in \mathcal{A}\}$. By Lemma 10, this class is equiped with a measurable envelope and has a finite covering number. Hence, stochastic equicontinuity and total boundedness are satisfied. Finite dimensional convergence holds by the Cramer-Wold device and the Lindebery-Levy central limit theorem as the data is iid and all variables have finite variance.

F Test Procedure for a Multivalued Instrument

The test statistic and procedure of sections 3.1 and 3.1.2 can be extended to the case of a single multi-valued instrument. This test procedure forms pairs of neighbouring instrument values and jointly tests the nesting inequalities and index sufficiency for all pairs. In the case of a continuous instrument, this instrument should first be discretised before calculating the test statistic.

Let $Z \in \{z_1, z_2, ..., z_K\}$, let n_k be the number of observations with $Z = z_k$, and let $\lambda_{z_k} = \Pr(Z = z_k)$. Form $K - 1$ pairs of neighbouring instrument values (z_1, z_2), (z_2, z_3),...,(z_{K-1}, z_L). For $k = 1, ..., K - 1$, let (S_{1k}, S_{2k}) be a pair of sample inclusion indicators corresponding to the subsample containing only observations with $Z \in \{z_k, z_{k+1}\}$. S_{1k} can be constructed by applying a distillation procedure to each (z_k, z_{k+1}) subsample. The construction of S_{2k} differs slightly from the headline test procedure in that it must now be based on the value of the two inverse probability weighting terms $\Pr(Z = z_k|p(X_i, Z_i))$ and $\Pr(Z = z_{k+1}|p(X_i, Z_i))$. For example $S_{2j} = 1$ if $\Pr(Z = z_k|p(X_i, Z_i)) > 0.05$ and $\Pr(Z = z_{k+1}|p(X_i, Z_i)) > 0.05$.

For the subsample of observations with $Z \in \{z_k, z_{k+1}\}$, denote the empirical conditional distributions of (U, D, X) given $Z = z_{k+1}$ and $Z = z_k$ by $P_{n_{k+1}}$ and Q_{n_k}.

Define

$$f_i^{z_k,S_{1k}}(A,d) = \frac{1\{U_i \in A, D_i = d\}S_{1ik}}{\Pr(S_{1k}=1|Z=z_k)}, \tag{169}$$

$$f_i^{z_{k+1},S_{1k}}(A,d) = \frac{1\{U_i \in A, D_i = d\}S_{1ik}}{\Pr(S_{1k}=1|Z=z_{k+1})}, \tag{170}$$

$$g_i^{z_k,S_{2k}}(A,d) = \frac{1\{U_i \in A, D_i = d\}\lambda_{z_k}S_{2ik}}{\Pr(Z=z_k|p(X_i,Z_i))\Pr(S_{2k}=1|Z=z_k)}, \tag{171}$$

$$g_i^{z_{k+1},S_{2k}}(A,d) = \frac{1\{U_i \in A, D_i = d\}\lambda_{z_{k+1}}S_{2ik}}{\Pr(Z=z_k|p(X_i,Z_i))\Pr(S_{2k}=1|Z=z_{k+1})}. \tag{172}$$

The test statistic is

$$\begin{aligned}
T &\equiv \max_{k\in\{1,\ldots,K-1\}} T_k \\
T_k &\equiv \max\left\{\sup_{A\in\mathcal{C}(\mathcal{Y}),d\in\{0,1\}} T_{1k}(A,d), \sup_{A\in\mathcal{C}(\mathcal{Y}),d\in\{0,1\}} T_{2k}(A,d), \sup_{A\in\mathcal{C}(\mathcal{Y}),d\in\{0,1\}} -T_{2k}(A,d)\right\}, \text{ where} \\
T_{1k}(A,d) &= \sqrt{\frac{n_{k+1}n_k}{n_{k+1}+n_k}}\Big[\left(Q_{n_k}f_i^{z_k,S_{1k}}(A,1) - P_{n_{k+1}}f_i^{z_{k+1},S_{1k}}(A,1)\right)d \\
&\quad + (P_{n_{k+1}}f_i^{z_{k+1},S_{1k}}(A,0) - Q_{n_k}f_i^{z_k,S_{1k}}(A,0))(1-d)\Big] \\
T_{2k}(A,d) &= \sqrt{\frac{n_{k+1}n_k}{n_{k+1}+n_k}}\left(Q_{n_k}g_i^{z_k,S_{2k}}(A,d) - P_{n_{k+1}}g_i^{z_{k+1},S_{2k}}(A,d)\right).
\end{aligned}$$

For the bootstrap, we again consider resampling with replacement. A bootstrap analogue of T under a least favorable null is

$$\begin{aligned}
\hat{T} &= \max_{k\in\{1,\ldots,K-1\}} \hat{T}_k \\
\hat{T}_k &\equiv \max\left\{\sup_{A\in\mathcal{C}(\mathcal{Y}),d\in\{0,1\}} \hat{T}_{1k}(A,d), \sup_{A\in\mathcal{C}(\mathcal{Y}),d\in\{0,1\}} \hat{T}_{2k}(A,d), \sup_{A\in\mathcal{C}(\mathcal{Y}),d\in\{0,1\}} -\hat{T}_{2k}(A,d)\right\}, \text{ where} \\
\hat{T}_{1k}(A,d) &= \sqrt{\frac{n_{k+1}n_k}{n_{k+1}+n_k}}\Big[\Big(\hat{Q}_{\hat{n}_k}\hat{f}_i^{z_k,\hat{S}_{1k}}(A,1) - Q_{n_k}f_i^{z_k,S_{1k}}(A,1) - (\hat{P}_{\hat{n}_{k+1}}\hat{f}_i^{z_{k+1},\hat{S}_{1k}}(A,1) - P_{n_{k+1}}f_i^{z_{k+1},S_{1k}}(A,1))\Big)d \\
&\quad + (\hat{P}_{\hat{n}_{k+1}}\hat{f}_i^{z_{k+1},\hat{S}_{1k}}(A,0) - P_{n_{k+1}}f_i^{z_{k+1},S_{1k}}(A,0) - (\hat{Q}_{\hat{n}_k}\hat{f}_i^{z_k,\hat{S}_{1k}}(A,0) - Q_{n_k}f_i^{z_k,S_{1k}}(A,0)))(1-d)\Big] \\
\hat{T}_{2k}(A,d) &= \sqrt{\frac{n_{k+1}n_k}{n_{k+1}+n_k}}\left(\hat{Q}_{\hat{n}_k}\hat{g}_i^{z_k,\hat{S}_{2k}}(A,d) - Q_{n_k}g_i^{z_k,S_{2k}}(A,d) - (\hat{P}_{\hat{n}_{k+1}}\hat{g}_i^{z_{k+1},\hat{S}_{2k}}(A,d) - P_{n_{k+1}}g_i^{z_{k+1},S_{2k}}(A,d))\right).
\end{aligned}$$

where $\hat{n}_k$ is the number of observations in the bootstrap sample with $Z = z_k$, $\hat{P}_{\hat{n}_{k+1}}$ and $\hat{Q}_{\hat{n}_k}$ are the bootstrap sample empirical conditional distributions of (U,D,X) given $Z = z_k$ and $Z = z_{k+1}$, $\hat{f}_i^{z_k,\hat{S}_{1k}}(A,D)$, $\hat{f}_i^{z_{k+1},\hat{S}_{1k}}(A,D)$, $\hat{g}_i^{z_k,\hat{S}_{2k}}(A,d)$ and $\hat{g}_i^{z_{k+1},\hat{S}_{2k}}(A,d)$ are as defined in (169) and (172) but calculated with the bootstrap sample, and $\hat{S}_{1k}$ and $\hat{S}_{2k}$ are inclusion indicators for the bootstrap

sample. Any quantity which is estimated in the main sample is re-estimated in the bootstrap sample.

Estimation can be performed as described in Section 3.1.2. In the case of a continuous instrument, the instrument can be used as is for the estimation of the propensity score and partial residuals (steps 1-3), but should be discretised before proceeding to step 4, and the discretised instrument used thereafter.

G An MTE Model for Multivalued Treatment

Let Y, X and Z be as defined in Section 2.1, and let $D \in \mathcal{D}$ be a ordered multivalued treatment variable. With a discrete ordered treatment $\mathcal{D}$ can be normalised to $\{0, 1, ..., D^{\max}\}$. The set of potential outcomes is $\{Y_{dz}\}_{d\in\mathcal{D},z\in\mathcal{Z}}$, with Y_{dz} being the outcome that would be observed if (D, Z) was exogeneously set to (d, z). The observed outcome is given by

$$Y = Y_{0z} + \sum_{d\geq 1} \mathbb{1}\{D \geq d\}(Y_{dz} - Y_{d-1,z})$$

We can write a general model of selection with an ordered discrete treatment as

$$D = \sum_{d\geq 1} \mathbb{1}\{\nu_d(X, Z, U_d) \geq 0\}$$

with

$$\nu_d(X, Z, U_d) \geq \nu_{d-1}(X, Z, U_{d-1}) \text{ for all } d \geq 1$$

The identifying assumptions of Section 2.1 are modified to

Identification Assumptions with a Multivalued Treatment:

(A1 MV) *Instrument exclusion restriction*: Potential outcomes do not depend on the instrument in the sense $Y_{dz} = Y_d$ for all $z \in \mathcal{Z}$ and $d \in \mathcal{D}$ with probability one.

(A2 MV) *Random assingnment*: $(\{Y_{dz}\}_{d\in\mathcal{D},z\in\mathcal{Z}}, \{U_d\}_{d\in\mathcal{D}})$ are statistically independent of Z conditional on X, and all $\{U_d\}_{d\geq 1}$ are statistically independent of X.

(A3 MV) *Instrument monotonicity*: For every z, $z' \in \mathcal{Z}$, $v_d(x, z, u_d) \geq v_d(x, z', u_d)$ holds for all $u_d \in \mathcal{U}_d$ and $d \geq 1$ or $v_d(x, z, u_d) \leq v_d(x, z', u_d)$ holds for all $u_d \in \mathcal{U}_d$ and $d \geq 1$

These assumptions imply a heterogeneous treatment model of the form

$$\begin{cases} Y_d = m_d(X, \tilde{U}_d); \\ Y = Y_0 + \sum_{d\geq 1} \mathbb{1}\{D \geq d\}(Y_d - Y_{d-1}); \\ D = \sum_{d\geq 1} \mathbb{1}\{\nu(X, Z) - V_{d-1} \geq 0\}, \end{cases} \tag{173}$$

with

$$(\{U_d\}_{d\in\mathcal{D}}, \{V_d\}_{d\in\mathcal{D}}) \perp Z|X \text{ and } \{V_d\}_{d\in\mathcal{D}} \perp X. \tag{174}$$

$\{\bar{U}_d\}_{d\in\mathcal{D}}$ is a set of unobserved individual potential outcome heterogeneities. $\{V_d\}_{d\in\mathcal{D}}$ is a set of individual selection heterogeneities that satisfy $V_d \geq V_{d-1}$ and $V_{D^{\max}} = \infty$.

To obtain semiparametric testable implications, we impose versions of Assumptions 4 and 5

(A4 MV) *Linear Functional Form*: The regression equations for potential outcomes have the parametric form

$$Y_d = \alpha_d + X'\theta_d + \tilde{U}_d. \tag{175}$$

(A5 MV) *Strong Exogeneity*: Let $(\{\tilde{U}_d\}_{d\in\mathcal{D}}$ be the residuals of the potential outcome regression equations. $(\{\tilde{U}_d\}_{d\in\mathcal{D}}, \{V_d\}_{d\in\mathcal{D}})$ is independent of (X, Z).

G.1 Testable Implications

Consider first the index sufficiency restriction (41). The event $(U \in A, D = d)$ is equivalent to $(U_d \in A, V_{d-1} \leq v(X,Z) < V_d)$. By Assumption 4 (MV), all $\{U_d\}$ and $\{V_d\}$ are independent of (X, Z). Hence,

$$\begin{aligned}\Pr(U \in A, D = d|X = x, Z = z) &= \Pr(V_{d-1} \leq v(x,z), v(x,z) < V_d, U_d \in A) \\ &= \Pr(U \in A, D = d|v(x,z)).\end{aligned}$$

For the nesting inequalities (42), fix $v' \leq v$. For treatment intensity $1 < d < D^{\max}$

$$\begin{aligned}&\Pr(U \in A, D = d|v) - Pr(U \in A, D = d|v') \\ &= \Pr(U_d \in A, V_{d-1} \leq v < V_d) - \Pr(U_d \in A, V_{d-1} \leq v' < V_d) \\ &= \Pr(U_d \in A, v' \leq V_{d-1} < v) - \Pr(U_d \in A, v' \leq V_d < v)\end{aligned}$$

which is of ambiguous sign. Consider the highest treatment intensity $D^{\max}$. As $V_{D^{\max}} = \infty$

$$\Pr(U \in A, D = D^{\max}|v) - Pr(U \in A, D = D^{\max}|v') = \Pr(U_{D^{\max}} \in A, v' \leq V_{D^{\max}-1} < v) \geq 0$$

In addition, $d = 0$ implies that $v(x,z) < V_0$, so

$$\begin{aligned}&\Pr(U \in A, D = 0|v) - Pr(U \in A, D = 0|v') \\ &= \Pr(U_0 \in A, v < V_0) - \Pr(U_d \in A, v' < V_0) \\ &= -\Pr(U_0 \in A, v' \leq V_0 < v) \leq 0\end{aligned}$$

H Additional Monte Carlo Size Results

This section contains additional Monte Carlo results for the case of a valid but relevant instrument. The data generating process here is identical to the size process of Section 4, with the exception that we set

$$\alpha_0 = \Phi^{-1}(0.45),$$
$$\alpha_1 = \Phi^{-1}(0.55),$$

as opposed to $\alpha_0 = \alpha_1 = 0$. The case of an irrelevant instrument corresponds to a least favourable null for the test of nesting inequalities as all inequalities are binding, whereas with a relevant instrument they are slack. However, changing from an irrelevant to a relevant instrument does not affect the test size of index sufficiency in this manner, as index sufficiency is a test of equalities. Rejection rates for this process are presented in Table 9. With a relevant instrument, rejection rates are driven by the test of index sufficiency, although overall they are similar to the case of an irrelevant instrument.

[Table 7 about here.]

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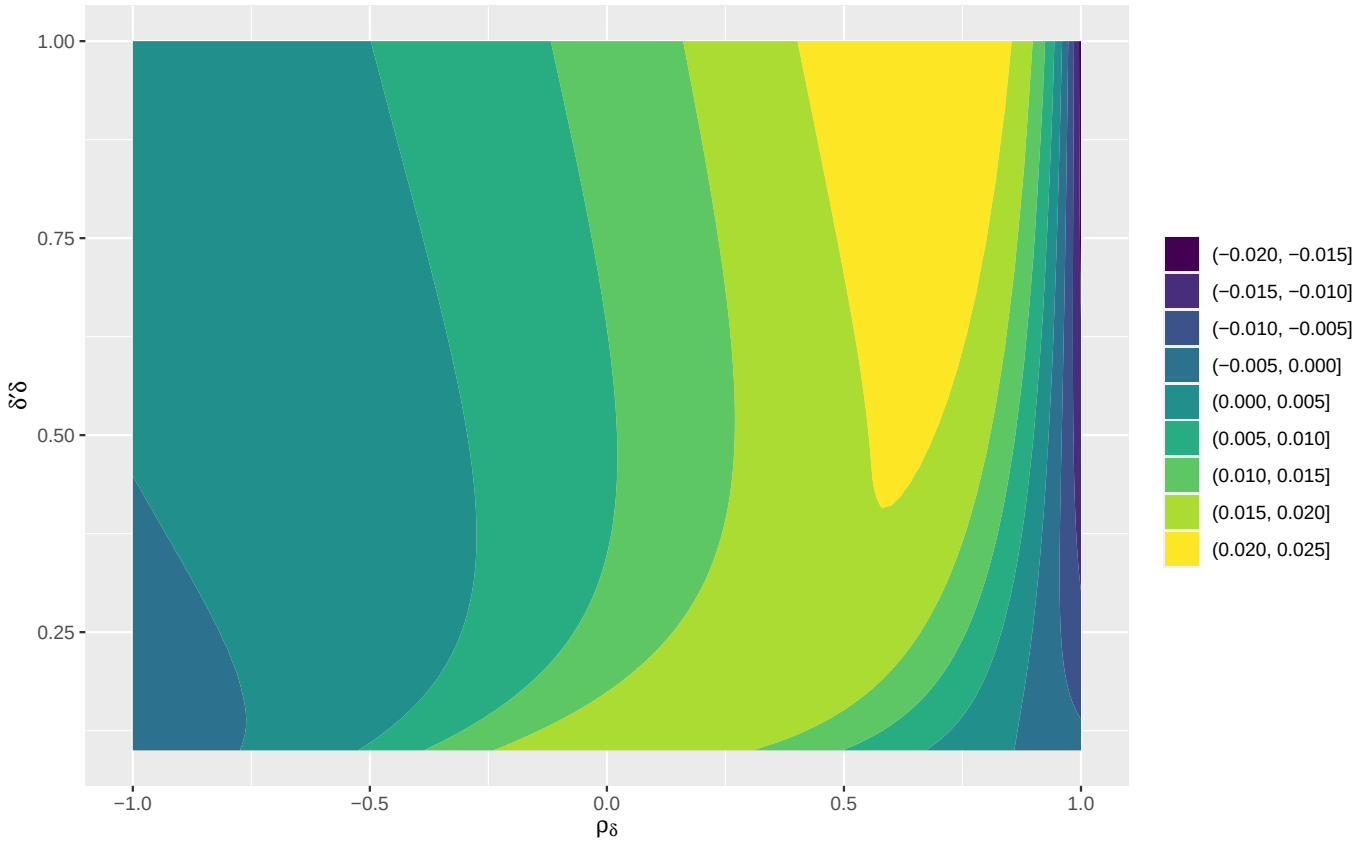


Figure 1: Contour plot of the difference between the violation of the nesting equality with a distilled sample and the violation of the nesting equality with a coarsened propensity score. A positive value implies that the violation with a distilled sample is larger than the violation with a coarsened propensity score. A negative value implies that the violation with a coarsened propensity score is larger that the violation with a distilled sample.

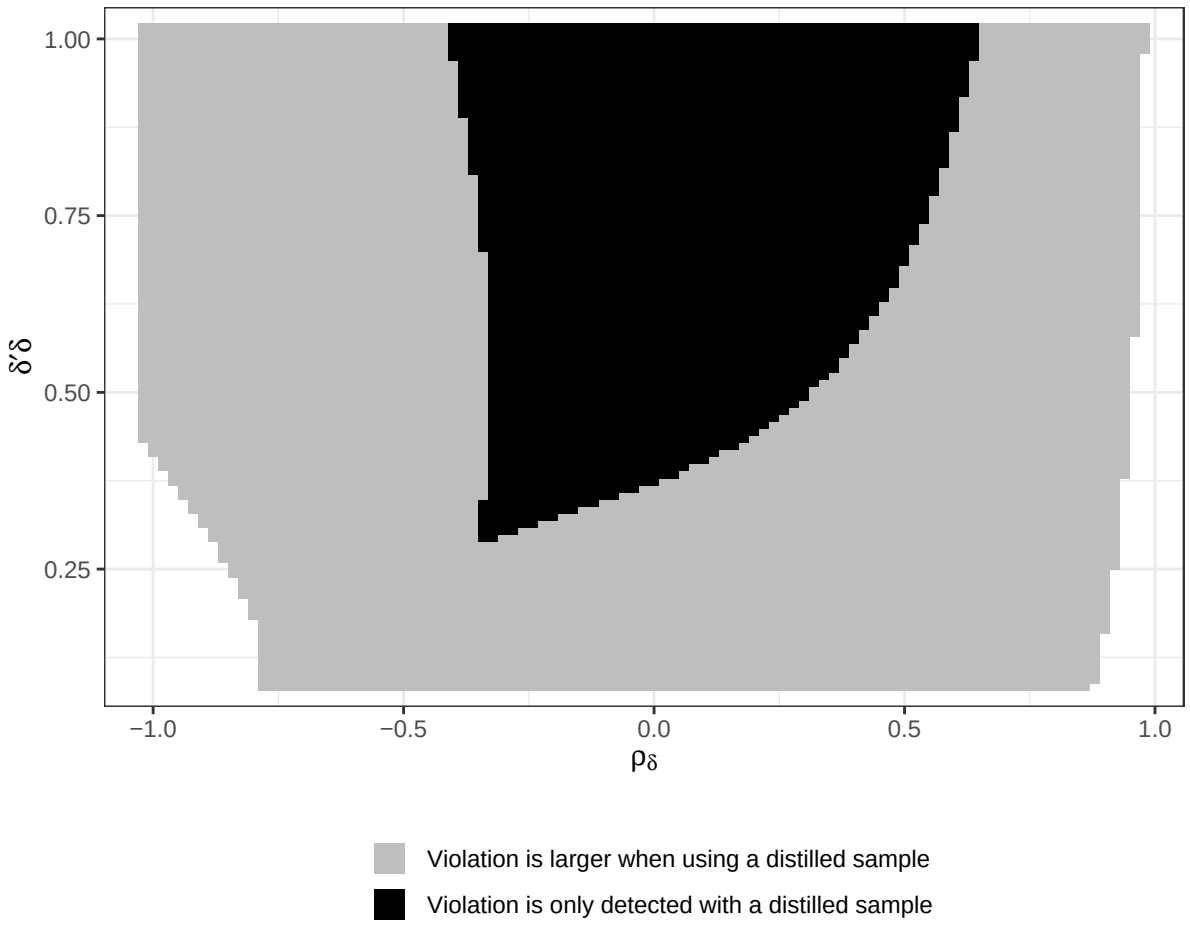
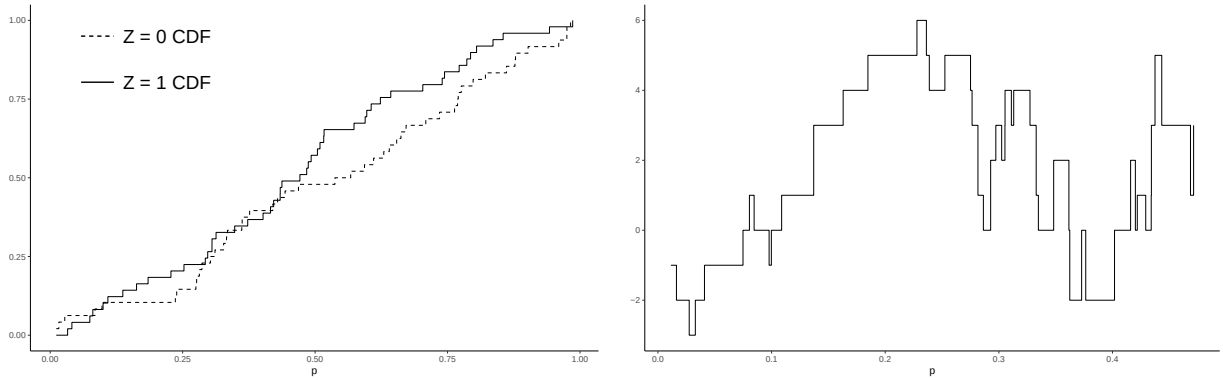
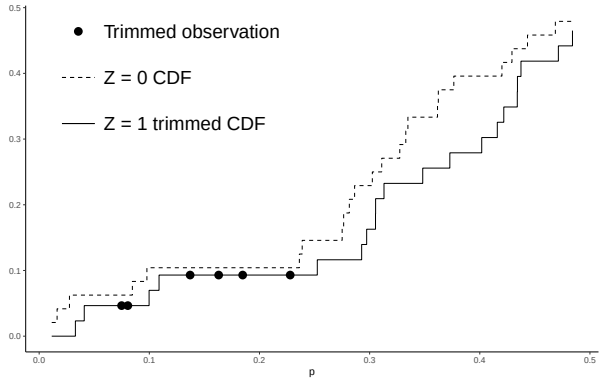


Figure 2: The region of parameter values where the violation of the nesting equality with a distilled sample exceeds the violation of the nesting equality with a coarsened propensity score, and the region where only the nesting equality with a distilled sample is violated. Each point corresponds to a combination of ρ_δ and $\delta'\delta$. Let $\mathcal{N}(\rho_\delta, \delta'\delta)$ denote the violation of the nesting equality with a coarsened propensity score defined in (27), and let $\mathcal{D}(\rho_\delta, \delta'\delta)$ denote the violation of the nesting inequality with a distilled sample defined in (28). A point is shaded gray if $\mathcal{D}(\rho_\delta, \delta'\delta) \geq \mathcal{N}(\rho_\delta, \delta'\delta) \geq 0$. A point is shaded black if $\mathcal{D}(\rho_\delta, \delta'\delta) > 0$ and $\mathcal{N}(\rho_\delta, \delta'\delta) < 0$.

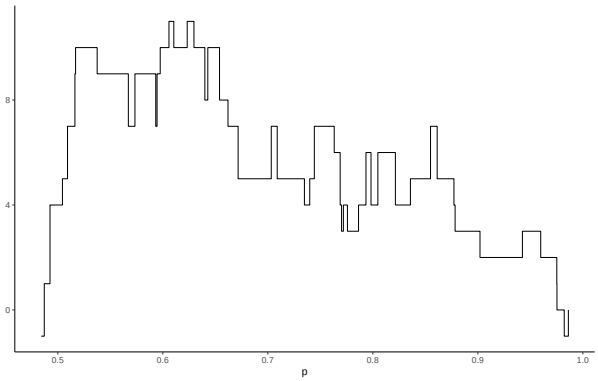


(a) Empirical distributions

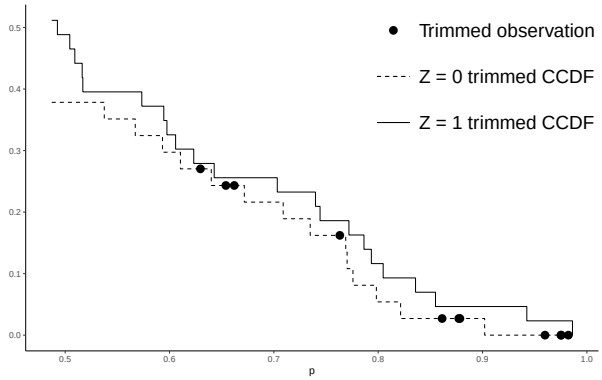
(b) Step 3a



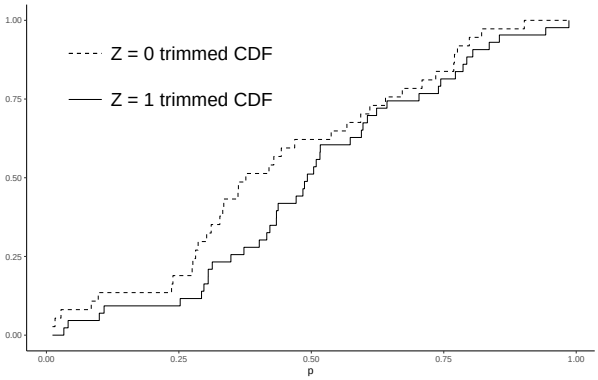
(c) Step 3b



(d) Step 4a



(e) Step 4b



(f) Trimmed distributions

Figure 3: Example of the distillation algorithm. The sample consists of 100 observations. The data generating process sets $\Pr(Z = 1) = 1/2$ with propensity score uniformly distributed between 0 and 1 for $Z = 0, 1$. The interval P^- includes all observations with a propensity weakly less than the median, while P^+ includes all observations with a propensity score above the median. Panel b plots d_{1j} for all $j \in J^-$. Panel d plots d_{0j} for all $j \in J^+$.

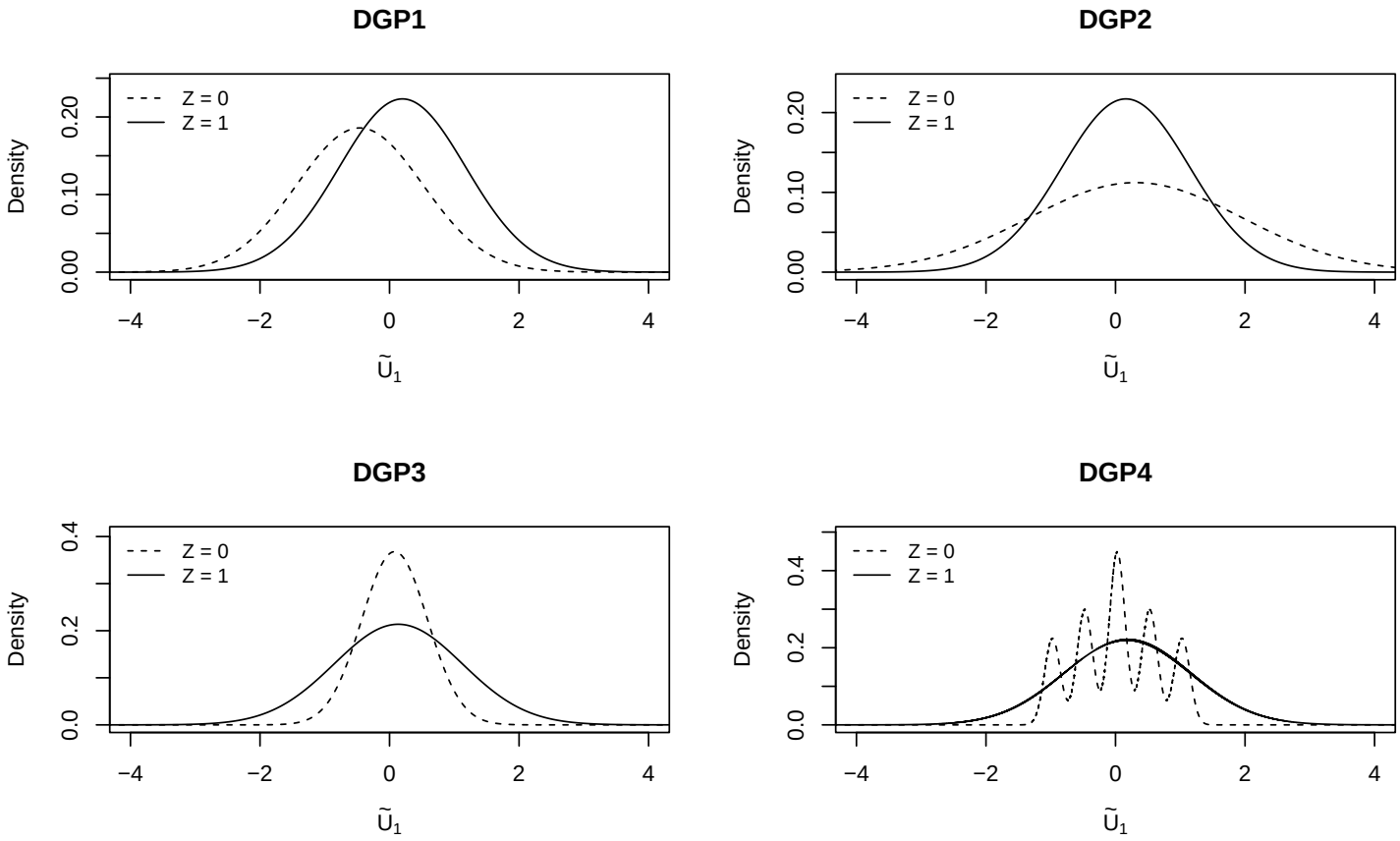


Figure 4: Conditional subdensities of $(\tilde{U}_1, D = 1)$ for DGPs 1 - 4 given values of θ and δ drawn from the $U(-1, 1)$ distribution. The elements of γ are set to 0.

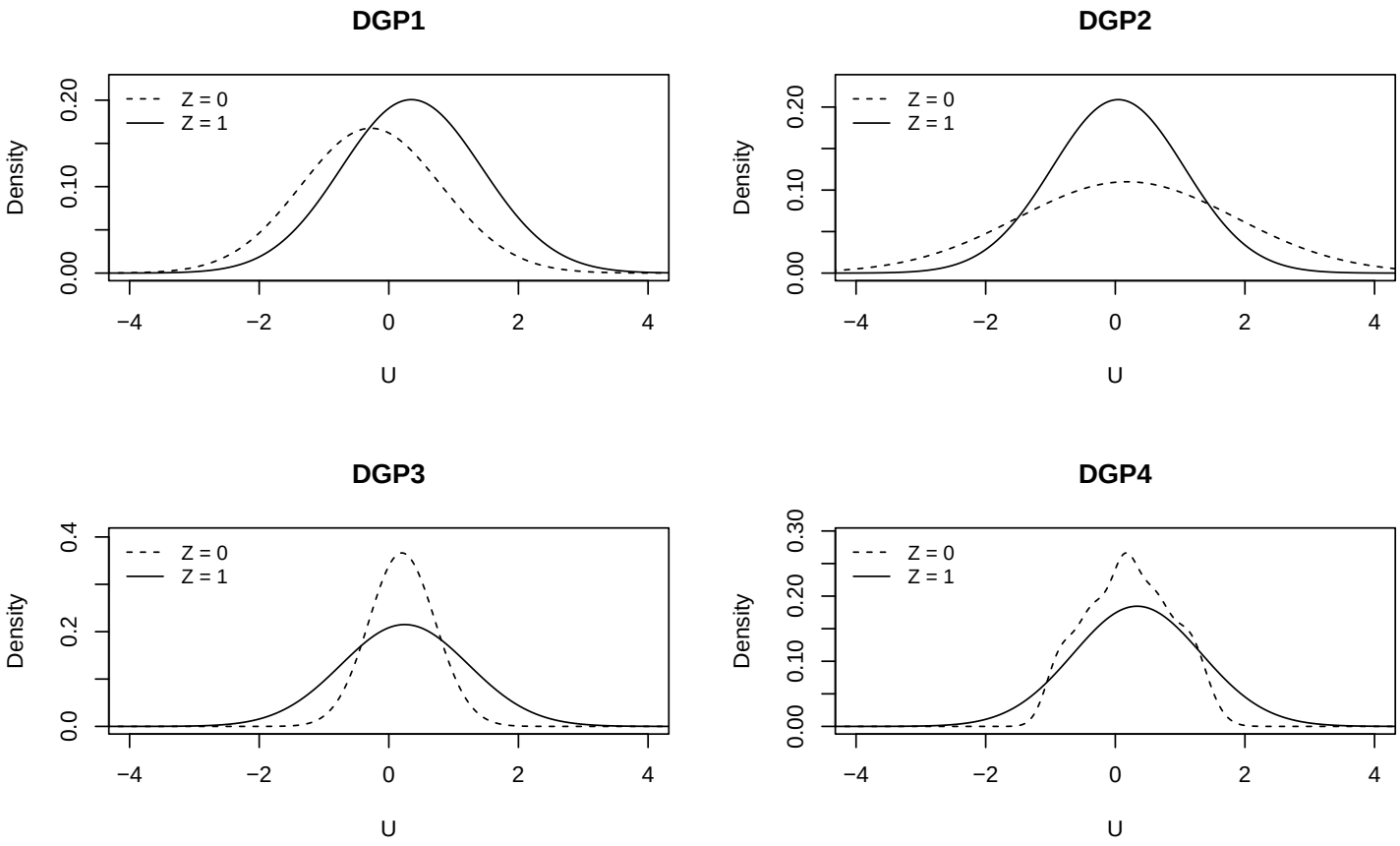


Figure 5: Subdensities of $(U, D = 1)$ for DGPs 1 - 4 conditional on the instrument Z, where $U = Y - X'\theta^*$ with $\theta^* = \text{plim}\,\hat{\theta}$. The elements of θ are drawn from the $U(0.5, 1)$ distribution with random flips of their signs and the elements of δ are drawn from the $U(-1, 1)$ distribution. All elements of γ are set to 0.

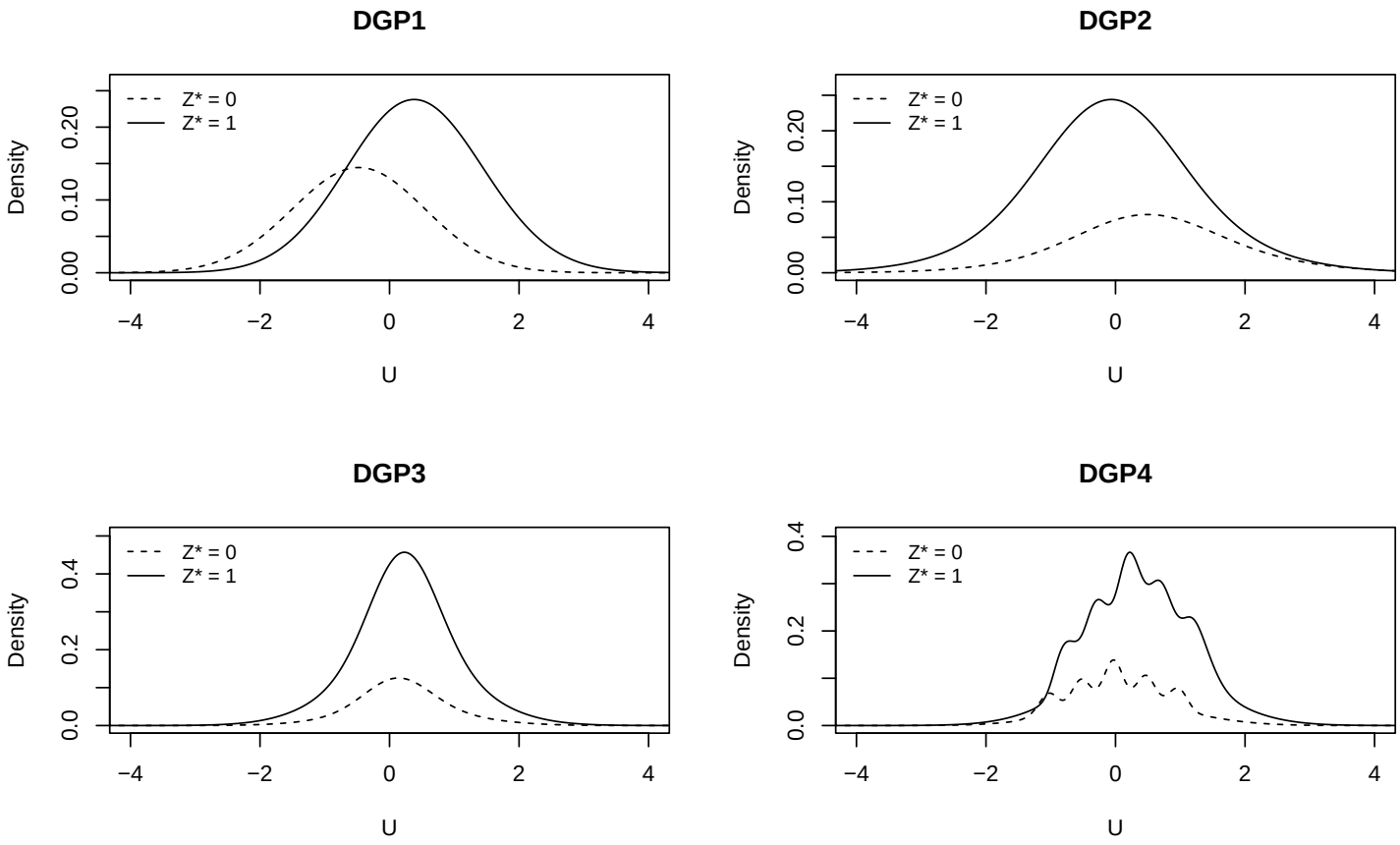


Figure 6: Subdensities of $(U, D = 1)$ for DGPs 1 - 4 conditional on the binarised propensity score, where $U = Y - X'\theta^*$ with $\theta^* = \text{plim}\,\hat{\theta}$. The elements of θ are drawn from the $U(0.5, 1)$ distribution with random flips of their signs and the elements of δ are drawn from the $U(-1, 1)$ distribution. All elements of γ are set to 0.

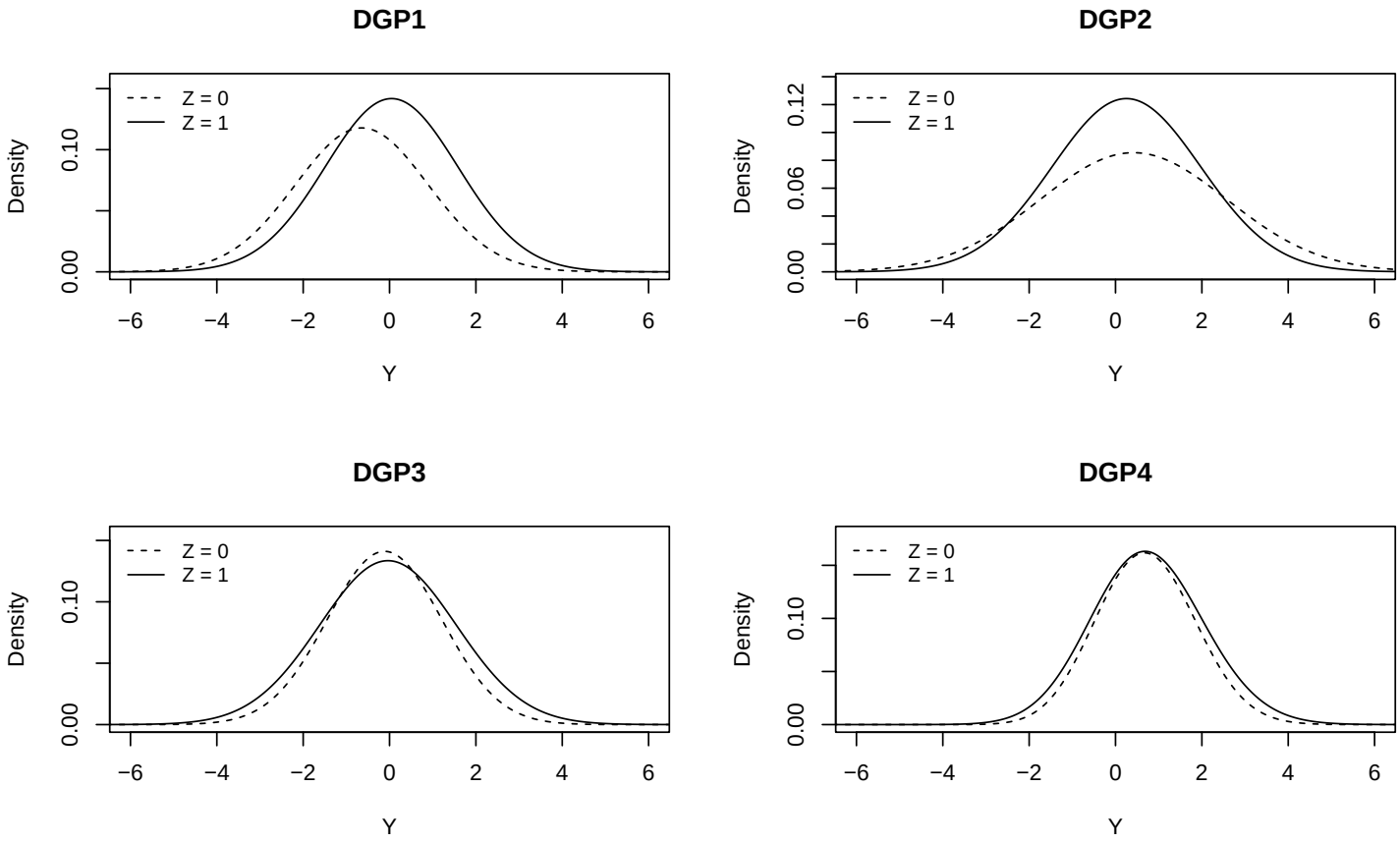


Figure 7: Subdensities of $(Y, D = 1)$ for DGPs 1 - 4 conditional on the instrument Z given θ and δ.he elements of θ are drawn from the $U(0.5, 1)$ distribution with random flips of their signs and the elements of δ are drawn from the $U(-1, 1)$ distribution.

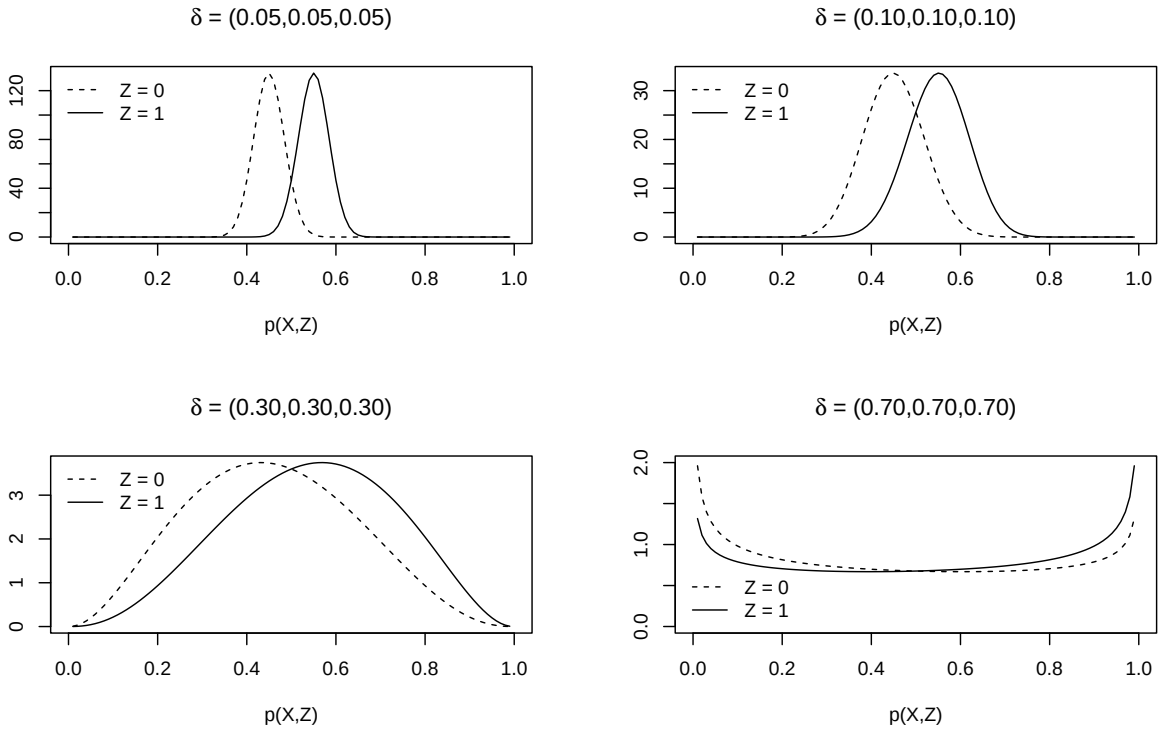


Figure 8: Density of $p(X, Z)$ conditional on $Z = 0$ and $Z = 1$ for four values of δ with γ set to 0.

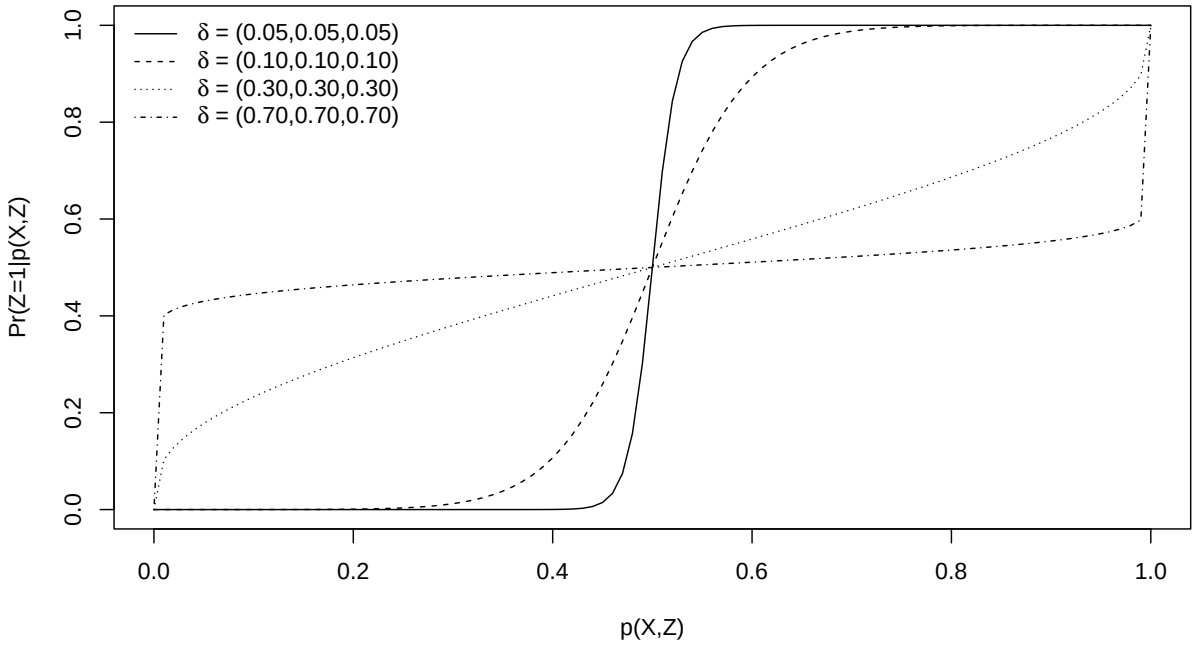
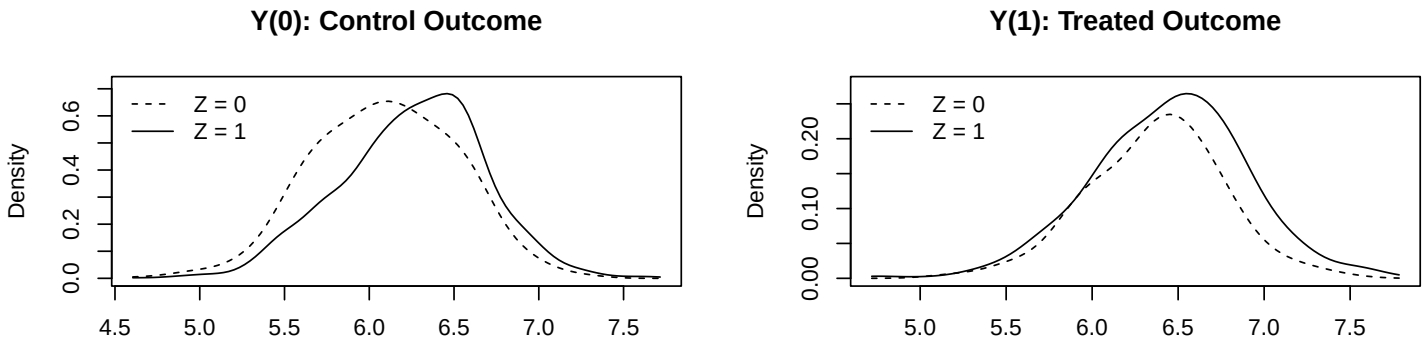
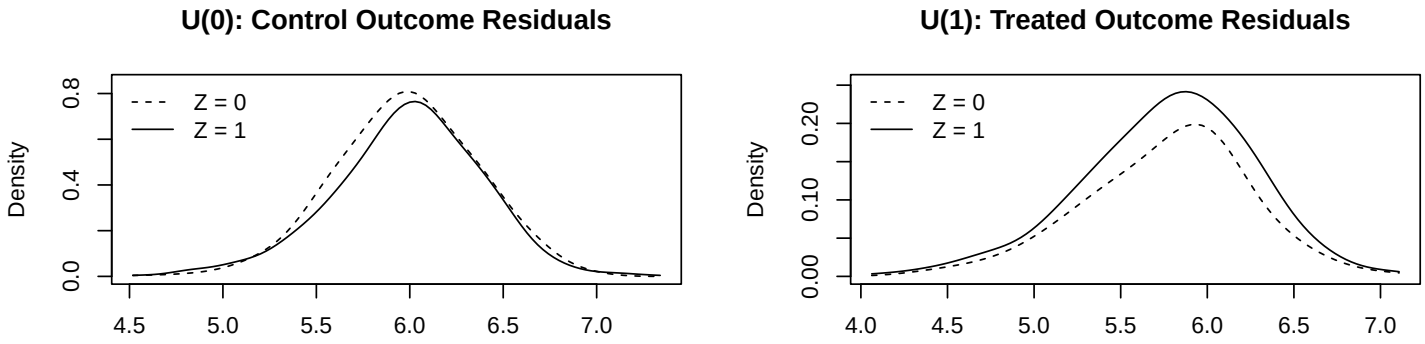


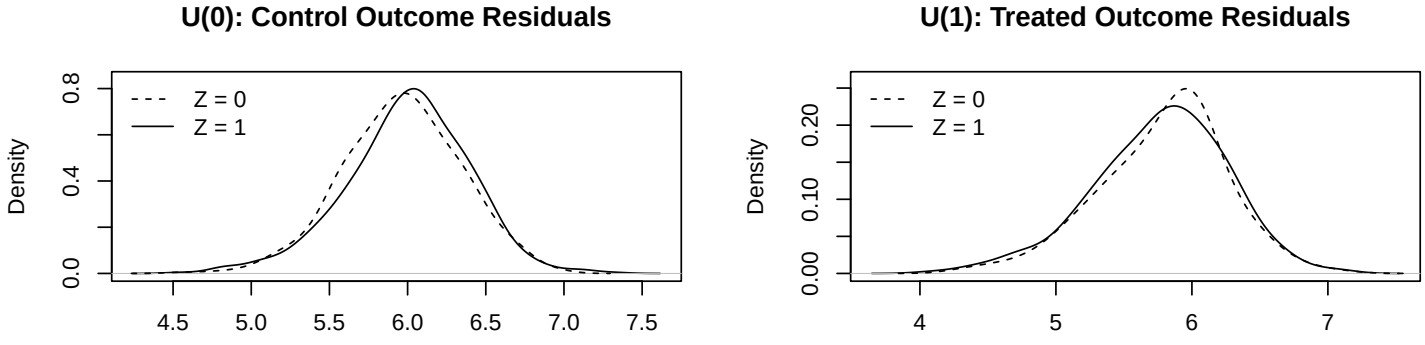
Figure 9: $Pr(Z = 1|p(X, Z))$ for four values of δ with γ set to 0.



(a) Outcome



(b) Nesting Inequality Densities



(c) Rewighted

Figure 10: Card (1993) subdensities. 10a plots conditional (Y, D) subdensities for $Z = 0$ and $Z = 1$ where the outcome Y is log weekly earnings. 10b and 10c plot condidtional subdensities for (U, D), where U are the partial residuals obtained by controlling for residence in the South, residence in a standard metropolitan statistical area in 1966, residence in a standard metropolitan statistical area in 1976, race, potential experience and its square, living in a single mother household at age 14, living with both parents at age 14, father's and mother's years of education, indicators for when father's and mother's years of indication are missing, a nine value categorical variable constructed by interacting father's and mother's education class, and a category for region of residence in 1966. 10b are nesting inequality subdensites which are trimmed to ensure first-order stochastic dominance. 10c are index sufficiency subdensities which are reweighted by $P(Z = 1|p(X, Z))$. Densities use the Gaussian kernel with the rule-of-thumb bandwidth.

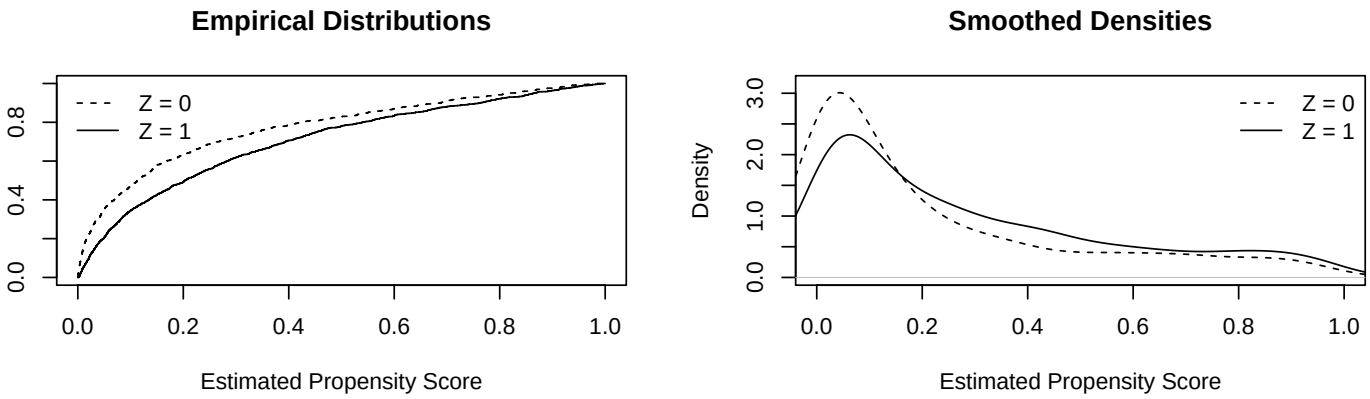
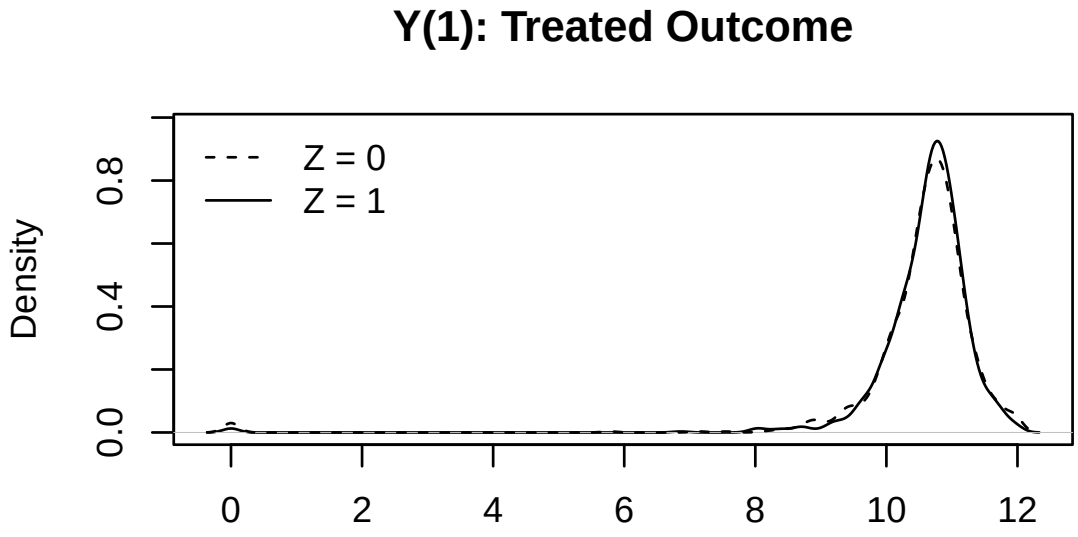
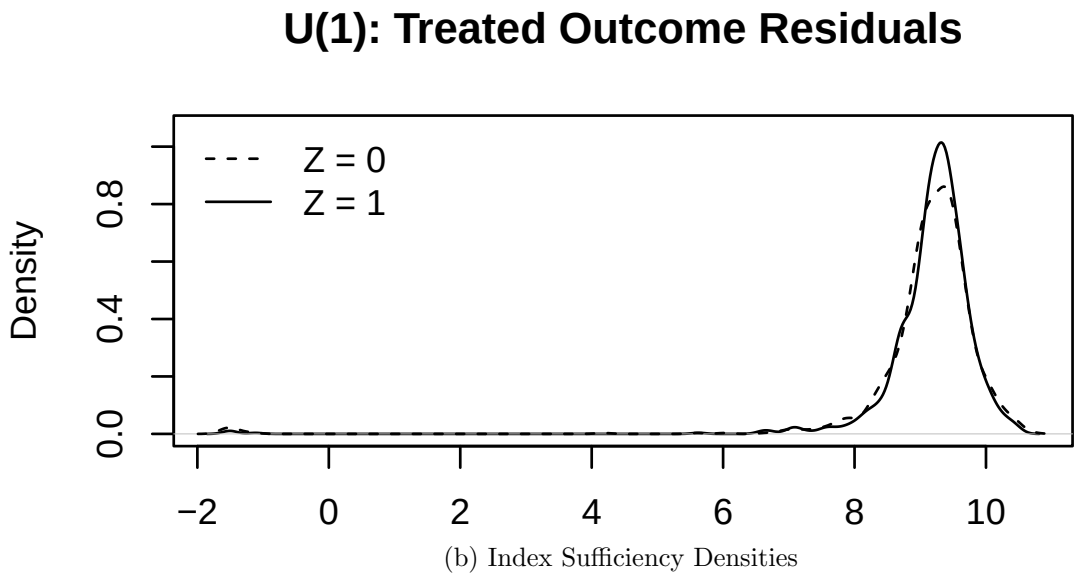


Figure 11: Card (1993) distribution and density of the estimated propensity score. The propensity score is estimated by probit with controls consisting of the instrument, the conditioning covariates, and interactions between the two. Conditioning covariates consist of residence in the South, residence in a standard metropolitan statistical area in 1966, residence in a standard metropolitan statistical area in 1976, race, potential experience and its square, living in a single mother household at age 14, living with both parents at age 14, father's and mother's years of education, indicators for when father's and mother's years of indication are missing, a nine value categorical variable constructed by interacting father's and mother's education class, and a category for region of residence in 1966. The distribution is the simple empirical distribution of estimated propensity scores. The density is an estimated kernel density using the Gaussian kernel and the rule-of-thumb bandwidth.

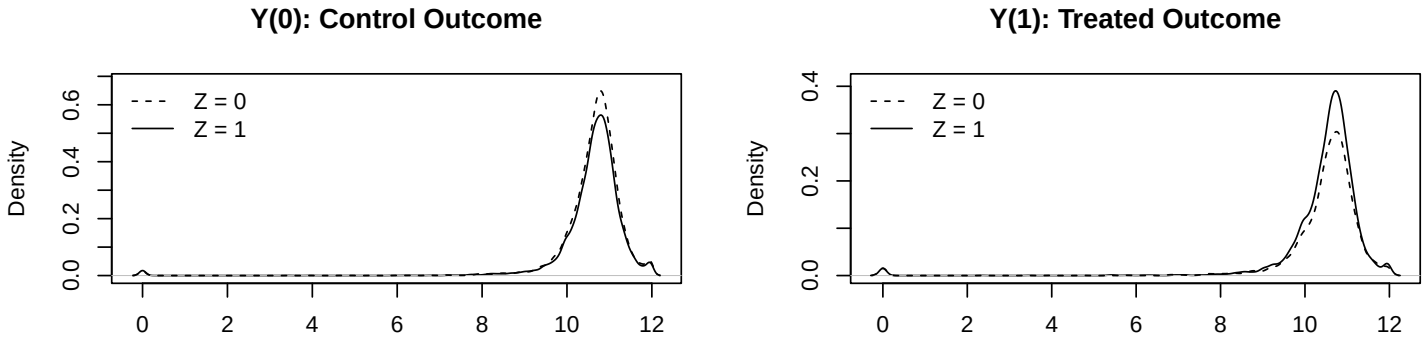


(a) Outcome Densities

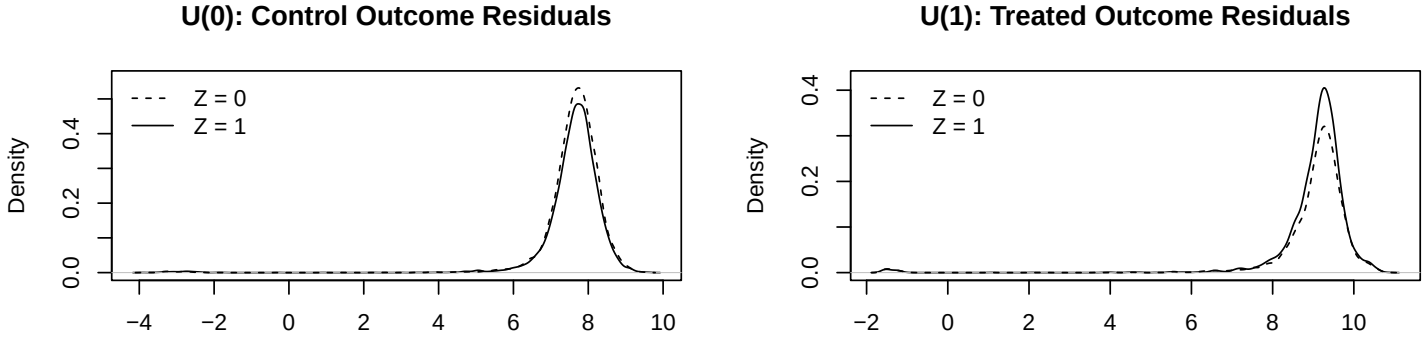


(b) Index Sufficiency Densities

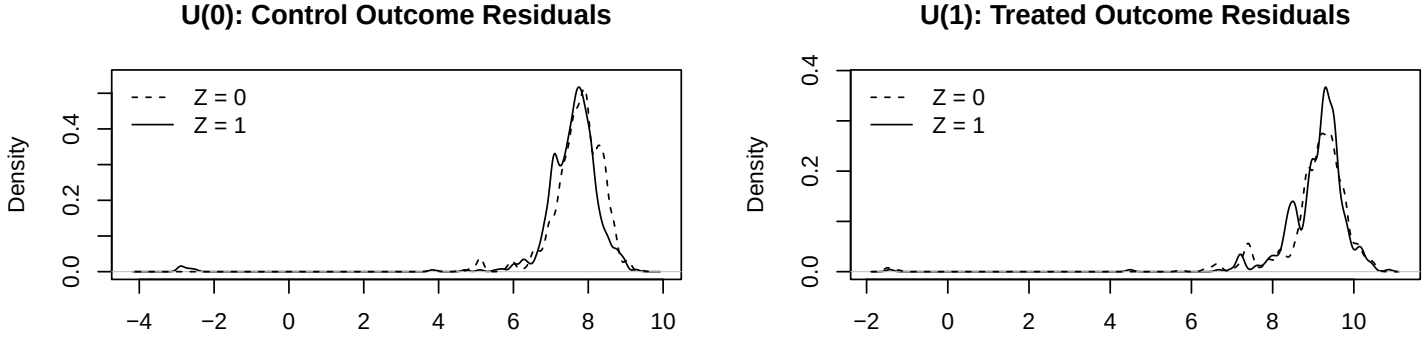
Figure 12: Angrist and Evans (1998) subdensities for the $Z_M = 1$ subsample. 12a plots the conditional $(Y, D = 1)$ subdensities for $Z_S = 0$ and $Z_S = 1$ where the outcome Y is log annual family earnings. Panel 12b plots condidtional subdensities for $(U, D = 1)$, where U are the partial residuals obtained by controlling for mother's age, mother's age at the time of the first birth, two dummies for mother's race, and a dummy for the first child being a boy. Densities use the Gaussian kernel with the rule-of-thumb bandwidth.



(a) Outcome Densities

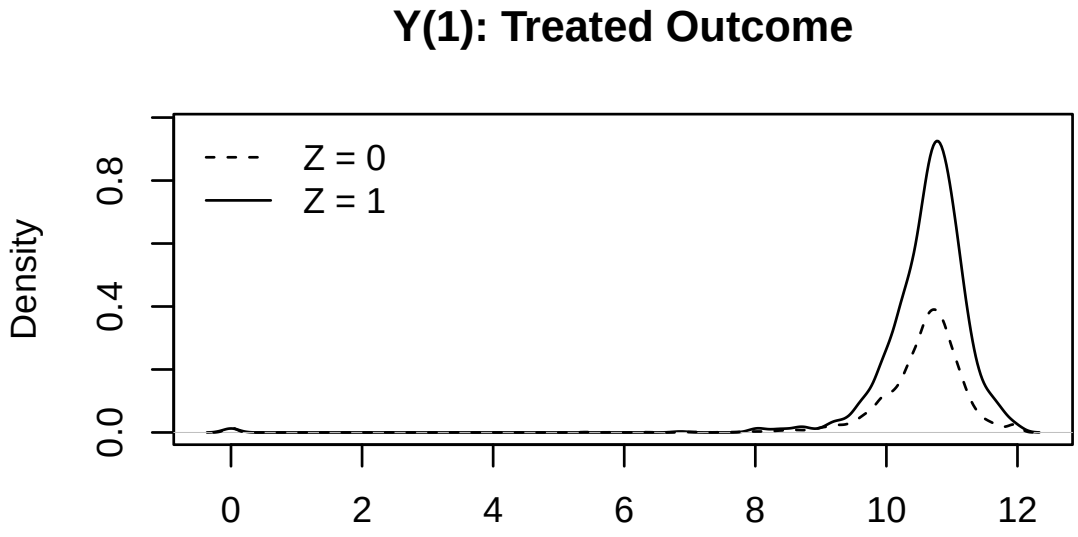


(b) Nesting Inequality Densities

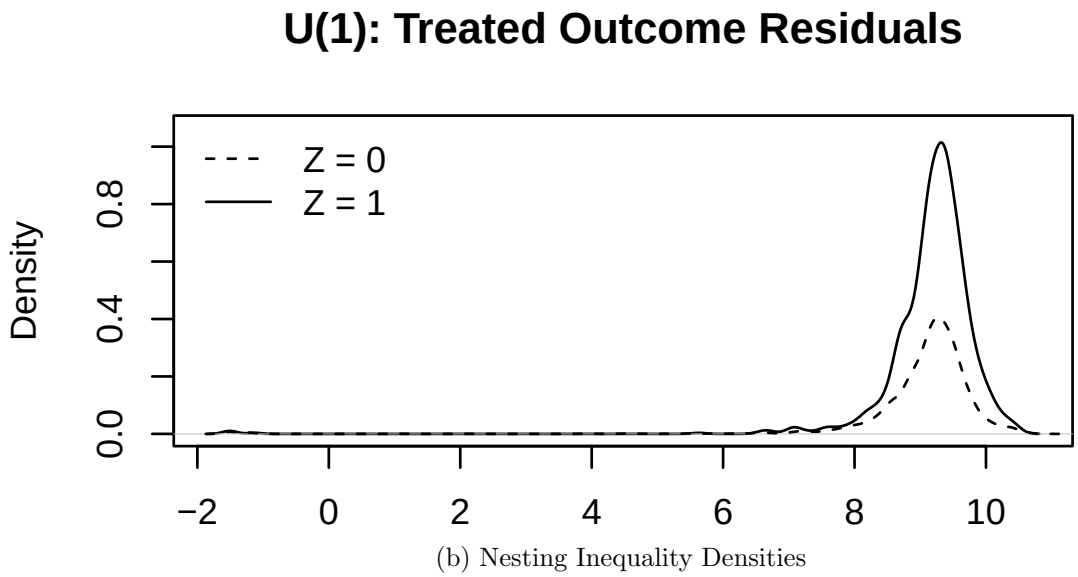


(c) Index Sufficiency Densities

Figure 13: Angrist and Evans (1998) subdensities for $Z_M = 0$. 13a plots conditional (Y, D) subdensities for $Z_S = 0$ and $Z_S = 1$ where the outcome Y is log annual family earnings. 13b and 13c plot condidtional subdensities for (U, D), where U are the partial residuals obtained by controlling for mother's age, mother's age at the time of the first birth, two dummies for mother's race, and dummies for the first and second child being a boy. 13b are nesting inequality subdensites which are trimmed to ensure first-order stochastic dominance. 13c are index sufficiency subdensities which are reweighted by $P(Z = 1|p(X, Z))$. Densities use the Gaussian kernel with the rule-of-thumb bandwidth.

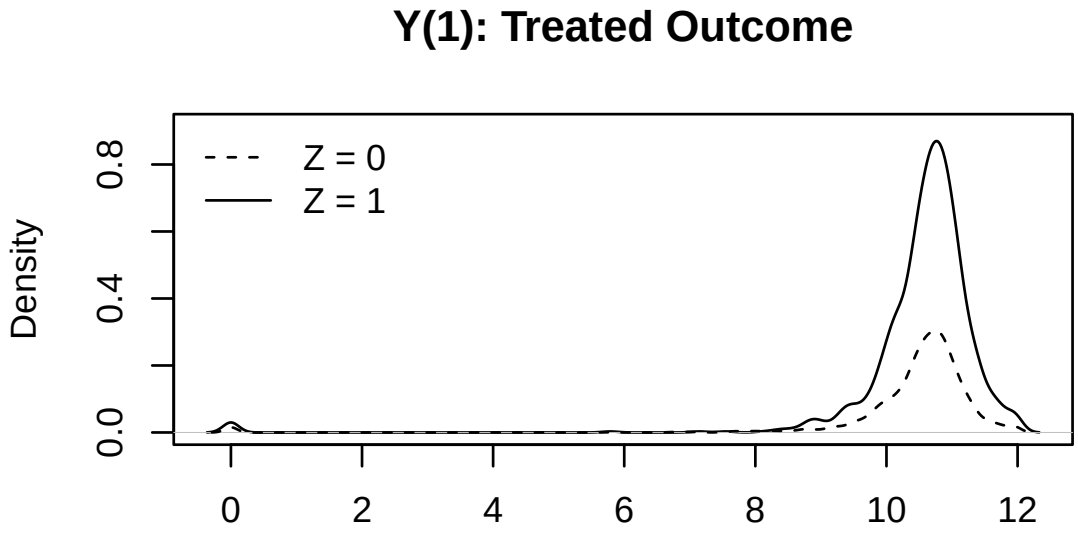


(a) Outcome Densities

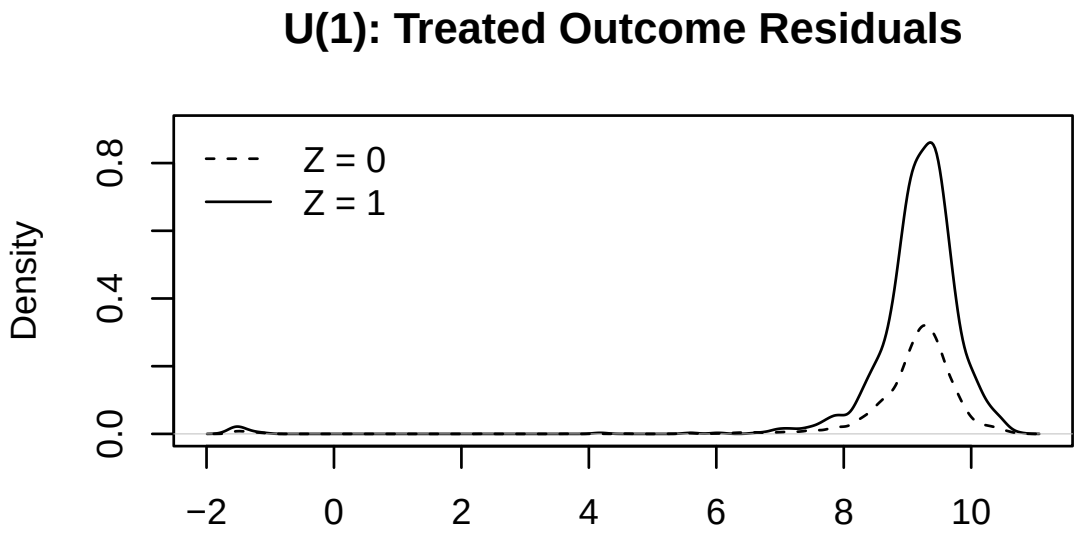


(b) Nesting Inequality Densities

Figure 14: Angrist and Evans (1998) subdensities for $Z_M = 1$ and $Z_M = 0$ with $Z_S = 1$. 14a plots the conditional $(Y, D = 1)$ subdensities where the outcome Y is log annual family earnings. 14b plots condidtional subdensities for $(U, D = 1)$, where U are the partial residuals obtained by controlling for mother's age, mother's age at the time of the first birth, two dummies for mother's race, and a dummy for the first child being a boy. Densities use the Gaussian kernel with the rule-of-thumb bandwidth.



(a) Outcome Densities



(b) Nesting Inequality Densities

Figure 15: Angrist and Evans (1998) subdensities for $Z_M = 1$ and $Z_M = 0$ with $Z_S = 0$. 15a plots the conditional $(Y, D = 1)$ subdensities where the outcome Y is log annual family earnings. 15b plots condidtional subdensities for $(U, D = 1)$, where U are the partial residuals obtained by controlling for mother's age, mother's age at the time of the first birth, two dummies for mother's race, and a dummy for the first child being a boy. Densities use the Gaussian kernel with the rule-of-thumb bandwidth.

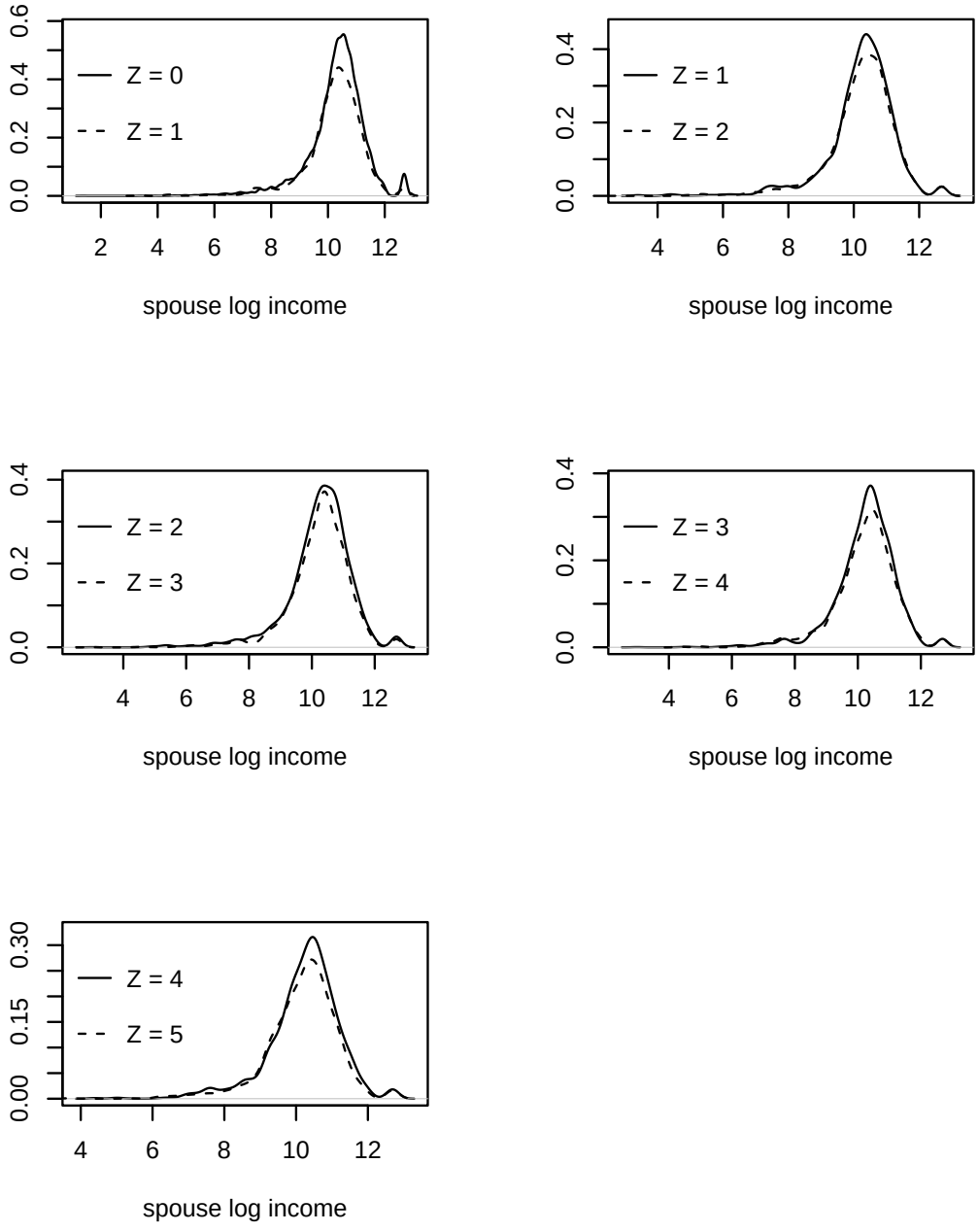


Figure 16: Bleakley and Chin (2010) outcome subdensities. Each panel plots conditional $(Y, D =$ speaks English very well) for a neighbouring pair of Z values, where Y is 'spouse's log wage' and Z is the instrument as defined in equation (57). Subdensities are smoothed using the rule-of-thumb bandwidth.

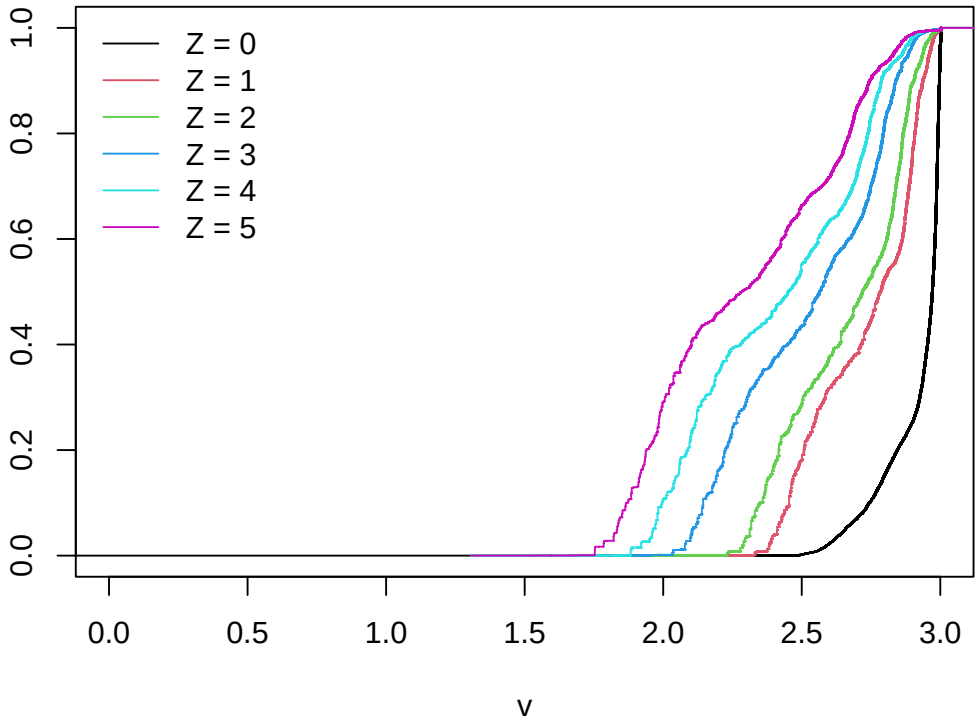


Figure 17: Conditional distributions of the index function $v(X, Z)$ for the data of Bleakley and Chin (2010). Each line is the empirical distribution of estimated values of the index function for a value of Z.

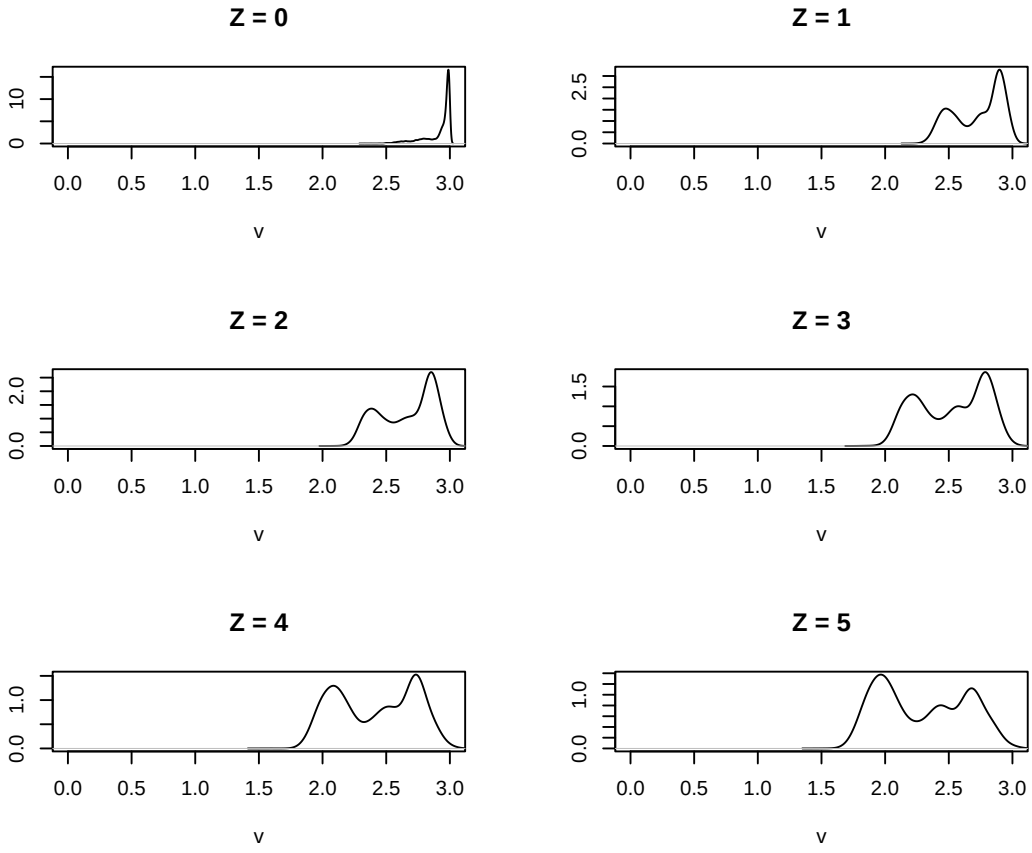


Figure 18: Conditional densities of $v(X, Z)$ for the data of Bleakley and Chin (2010). Each panel shows the smoothed density of estimated values of $v(X, Z)$ conditional on a value of Z.

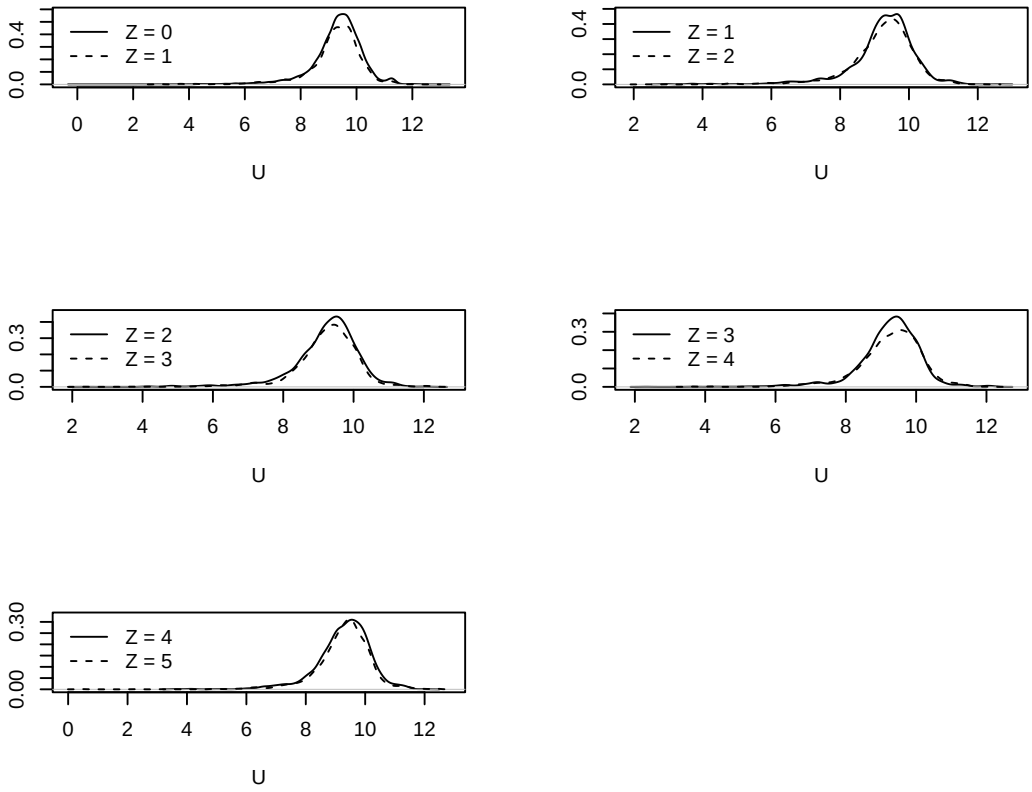
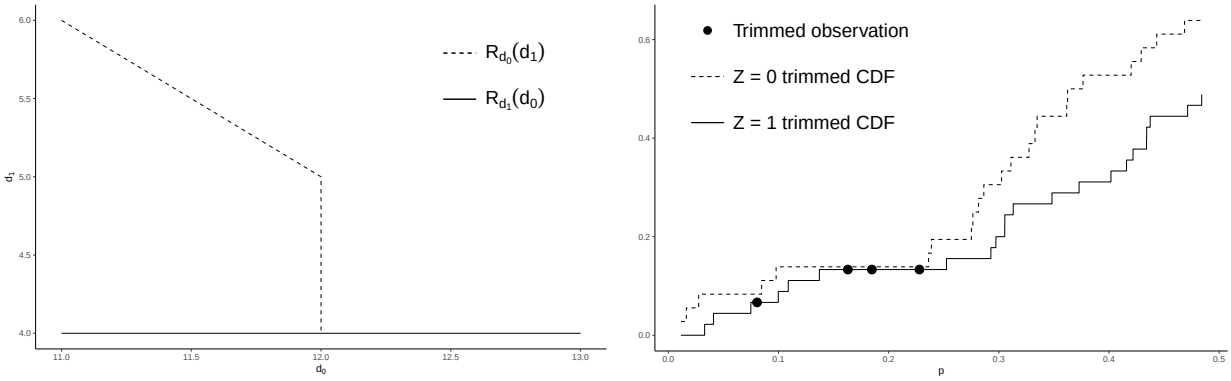
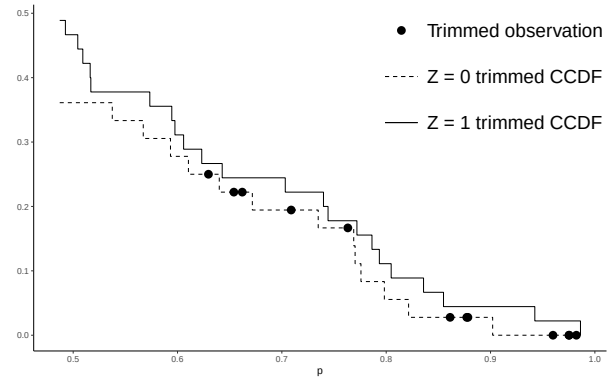


Figure 19: Residual subdensities for the data of Bleakley and Chin (2010). Each panel plots conditional $(U, D = \text{speaks English very well})$ for a neighbouring pair of Z values, where Z is the instrument as defined in equation (57). Subdensities are smoothed using the rule-of-thumb bandwidth.

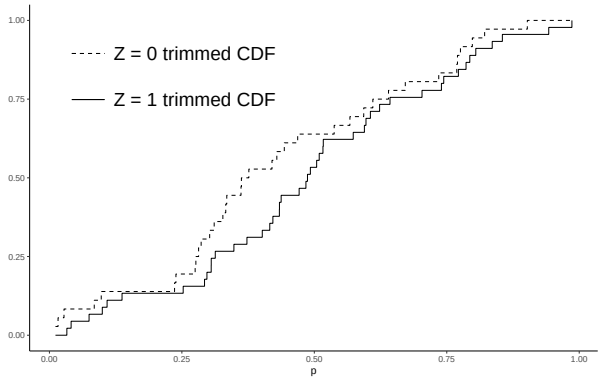


(a) $R_{d_0}(d_1)$ and $R_{d_1}(d_0)$

(b) P^- region



(c) P^+ region



(d) Trimmed distributions

Figure 20: Example of the distillation algorithm. The sample consists of 100 observations. The data generating process sets $\Pr(Z = 1) = 1/2$ with propensity score uniformly distributed between 0 and 1 for $Z = 0, 1$. The interval P^- includes all observations with a propensity weakly less than the median, while P^+ includes all observations with a propensity score above the median.

	$\gamma_k = 0$			$\gamma_k \sim U(-1,1)$		
Nominal Size:	0.1	0.05	0.01	0.1	0.05	0.01
Proposed test procedure						
200	0.00	0.00	0.00	0.00	0.00	0.00
500	0.00	0.00	0.00	0.00	0.00	0.00
1000	0.00	0.00	0.00	0.02	0.00	0.00
Binarised $p(X, Z)$						
200	0.00	0.00	0.00	0.00	0.00	0.00
500	0.00	0.00	0.00	0.00	0.00	0.00
1000	0.00	0.00	0.00	0.00	0.00	0.00
Kitagawa (2015)						
200	0.09	0.06	0.03	0.41	0.39	0.30
500	0.05	0.01	0.00	0.63	0.59	0.53
1000	0.08	0.04	0.02	0.56	0.51	0.46

Table 1: Size

	$\gamma_k = 0$			$\gamma_k \sim U(-1,1)$		
Nominal Size:	0.1	0.05	0.01	0.1	0.05	0.01
Power DGP1						
Proposed test procedure						
200	0.01	0.01	0.00	0.01	0.00	0.00
500	0.45	0.32	0.12	0.19	0.13	0.03
1000	0.82	0.78	0.68	0.76	0.65	0.32
Binarised Propensity Score						
200	0.00	0.00	0.00	0.00	0.00	0.00
500	0.00	0.00	0.00	0.00	0.00	0.00
1000	0.00	0.00	0.00	0.00	0.00	0.00
Kitagawa (2015)						
200	0.12	0.08	0.03	0.47	0.43	0.27
500	0.23	0.16	0.03	0.47	0.44	0.37
1000	0.58	0.47	0.27	0.62	0.59	0.50
Power DGP2						
Proposed test procedure						
200	0.04	0.00	0.00	0.07	0.01	0.00
500	0.61	0.49	0.22	0.49	0.33	0.13
1000	0.97	0.96	0.89	0.88	0.82	0.64
Binarised Propensity Score						
200	0.00	0.00	0.00	0.00	0.00	0.00
500	0.00	0.00	0.00	0.00	0.00	0.00
1000	0.00	0.00	0.00	0.00	0.00	0.00
Kitagawa (2015)						
200	0.02	0.00	0.00	0.32	0.26	0.12
500	0.01	0.00	0.00	0.37	0.34	0.27
1000	0.05	0.01	0.01	0.53	0.48	0.33
Power DGP3						
Proposed test procedure						
200	0.10	0.03	0.00	0.04	0.02	0.01
500	0.89	0.84	0.32	0.51	0.33	0.14
1000	1.00	1.00	0.96	0.93	0.91	0.83
Binarised Propensity Score						
200	0.00	0.00	0.00	0.00	0.00	0.00
500	0.00	0.00	0.00	0.00	0.00	0.00
1000	0.00	0.00	0.00	0.00	0.00	0.00
Kitagawa (2015)						
200	0.11	0.06	0.01	0.45	0.39	0.29
500	0.14	0.06	0.03	0.40	0.40	0.36
1000	0.20	0.18	0.11	0.58	0.53	0.50
Power DGP4						
Proposed test procedure						
200	0.02	0.00	0.00	0.01	0.00	0.00
500	0.22	0.10	0.04	0.14	0.09	0.01
1000	0.73	0.63	0.32	0.55	0.45	0.18
Binarised Propensity Score						
200	0.00	0.00	0.00	0.00	0.00	0.00
500	0.00	0.00	0.00	0.00	0.00	0.00
1000	0.00	0.00	0.00	0.00	0.00	0.00
Kitagawa (2015)						
200	0.06	0.02	0.01	0.31	0.25	0.13
500	0.08	0.03	0.01	0.54	0.47	0.35
1000	0.09	0.06	0.03	0.49	0.46	0.42

Table 2: Power

	$\delta_k \sim U(-1,1)$			$\delta_k \sim U(-0.1,0.1)$		
Nominal Size:	0.1	0.05	0.01	0.1	0.05	0.01
Power DGP 1						
Nesting:						
200	0.02	0.00	0.00	0.00	0.00	0.00
500	0.34	0.23	0.05	0.01	0.00	0.00
1000	0.78	0.68	0.48	0.42	0.18	0.01
Index Sufficiency:						
200	0.01	0.01	0.01	0.00	0.00	0.00
500	0.51	0.37	0.15	0.00	0.00	0.00
1000	0.83	0.79	0.72	0.00	0.00	0.00
Overall:						
200	0.01	0.01	0.00	0.00	0.00	0.00
500	0.45	0.32	0.12	0.00	0.00	0.00
1000	0.82	0.78	0.68	0.00	0.00	0.00
Power DGP 2						
Nesting:						
200	0.00	0.00	0.00	0.00	0.00	0.00
500	0.09	0.04	0.02	0.01	0.00	0.00
1000	0.15	0.08	0.00	0.04	0.02	0.00
Index Sufficiency:						
200	0.04	0.02	0.00	0.01	0.00	0.00
500	0.66	0.54	0.25	0.00	0.00	0.00
1000	0.97	0.96	0.90	0.22	0.10	0.00
Overall:						
200	0.04	0.00	0.00	0.01	0.00	0.00
500	0.61	0.49	0.22	0.00	0.00	0.00
1000	0.97	0.96	0.89	0.20	0.06	0.00
Power DGP 3						
Nesting:						
200	0.05	0.00	0.00	0.03	0.02	0.00
500	0.53	0.33	0.14	0.36	0.22	0.02
1000	0.89	0.88	0.66	0.80	0.68	0.43
Index Sufficiency:						
200	0.12	0.05	0.01	0.00	0.00	0.00
500	0.89	0.87	0.38	0.04	0.01	0.00
1000	1.00	1.00	0.96	0.33	0.18	0.02
Overall:						
200	0.10	0.03	0.00	0.00	0.00	0.00
500	0.89	0.84	0.32	0.04	0.01	0.00
1000	1.00	1.00	0.96	0.33	0.18	0.02
Power DGP 4						
Nesting:						
200	0.00	0.00	0.00	0.02	0.01	0.00
500	0.04	0.03	0.01	0.02	0.01	0.00
1000	0.20	0.17	0.07	0.14	0.07	0.00
Index Sufficiency:						
200	0.02	0.00	0.00	0.01	0.00	0.00
500	0.23	0.14	0.05	0.00	0.00	0.00
1000	0.75	0.68	0.36	0.07	0.01	0.00
Overall:						
200	0.02	0.00	0.00	0.01	0.00	0.00
500	0.22	0.10	0.04	0.00	0.00	0.00
1000	0.73	0.63	0.32	0.07	0.01	0.00

Table 3: Comparison of index sufficiency and nesting inequalities

	$\xi \approx 0.07$	$\xi \approx 0.21$	$\xi = 0.30$	$\xi = 1$
Card (1993)				
Nesting inequalities	0.406	0.998	1.000	1.000
Index sufficiency	0.224	0.346	0.272	0.198
Overall	0.258	0.388	0.304	0.210

Table 4: P-values for the test of nesting inequalities, the test of index sufficiency, and the combined test of both. A P-value below 0.05 indicates that the null hypothesis would be rejected at the 5% level of significance. ξ is a user specified trimming parameter as described in Kitagawa (2015). Conditioning covariates are residence in the South, residence in a standard metropolitan statistical area in 1966, residence in a standard metropolitan statistical area in 1976, race, a quadratic in potential experience, living in a single mother household at age 14, living with both parents at age 14, mother's and father's years of education, dummies for mother's and father's years of education being missing, a nine value categorical variable constructed by interacting father's and mother's education class, and a categorical variable for region of residence in 1966.

	Observations	Nesting inequalities	Index sufficiency
Card (1993)	3010	3010	3010

Table 5: The number of observations in the original sample, the number of observations used in the test of nesting inequalities, and the number of observations used in the test of index sufficiency. Observations are removed for the test of nesting inequalities in order to attain stochastic dominance in the trimmed sample. Observations are removed for the test of index sufficiency if $\Pr(Z = 1|p(X,Z)) \leq 0.05$ or $\Pr(Z = 1|p(X,Z)) \geq 0.95$

	$\xi \approx 0.07$	$\xi \approx 0.21$	$\xi = 0.30$	$\xi = 1$	Unscaled
Joint Test	1.00	1.00	1.00	1.00	1.00
Testing validity of Z_S					
$Z_M = 1$ subsample					
Index sufficiency	1.00	1.00	1.00	1.00	1.00
$Z_M = 0$ subsample					
Nesting inequalities	1.00	1.00	1.00	1.00	1.00
Testing validity of Z_M					
$Z_S = 1$ subsample					
Nesting inequalities	1.00	0.96	0.91	0.68	0.68
$Z_S = 0$ subsample					
Nesting inequalities	0.98	0.97	0.96	0.76	0.76
$Z_M = 1$ Nesting $Z_M = 0, Z_S = 1$	0.43	0.15	0.09	0.00	0.00

Table 6: P-values for the joint test and each individual test. The first row contains P-values for the joint test of all available testable implications. The next rows contain P-values for the individual tests of the validity of the same-sex instrument Z_S with the multiple birth instrument Z_M held constant at 1 and 0. The final rows contain P-values for the individual tests of the validity of Z_M with Z_S held constant at 1 and 0. A P-value below 0.05 indicates that the null hypothesis would be rejected at the 5% level of significance. ξ is a user specified trimming parameter as described in Kitagawa (2015).

	Observations	Nesting inequalities	Index sufficiency
Testing validity of Z_S			
$Z_M = 1$	2121		2121
$Z_M = 0$	25253	25253	1622
Testing validity of Z_M			
$Z_S = 1$	13905	13905	
$Z_S = 0$	13469	13469	

Table 7: The number of observations in the original sample for each value of the same-sex instrument Z_S and multiple birth instrument Z_M, and the number of observations used in each test of nesting inequalities and index sufficiency. The first two rows are tests of the validity of Z_S with Z_M held constant, and the remaining rows are tests of the validity of Z_M with Z_S held constant. As all observations with $Z_M = 1$ have a propensity score of 1, not all testable implications are available. When a testable implication is not available for one instrument with the other held constant, the corresponding cell is left blank. Otherwise, observations are removed when testing the nesting inequalities in order to attain stochastic dominance in the trimmed sample, and removed when testing index sufficiency if $\Pr(Z = 1|p(X, Z)) \leq 0.05$ or $\Pr(Z = 1|p(X, Z)) \geq 0.95$.

	$Z = (0, 1)$	$Z = (1, 2)$	$Z = (2, 3)$	$Z = (3, 4)$	$Z = (4, 5)$	overall
P-value	1.00	0.00	0.26	0.02	0.04	0.00

Table 8: Bleakley and Chin (2010) P-values

	$\gamma_k = 0$						$\gamma_k \sim U(-1,1)$					
Trimming Constant:	$\xi \approx 0.21$			$\xi = 0.30$			$\xi \approx 0.21$			$\xi = 0.30$		
Nominal Size:	0.1	0.05	0.01	0.1	0.05	0.01	0.1	0.05	0.01	0.1	0.05	0.01
Proposed test procedure												
200	0.033	0.014	0.000	0.023	0.008	0.000	0.078	0.040	0.007	0.055	0.025	0.003
500	0.023	0.011	0.001	0.015	0.004	0.001	0.062	0.029	0.002	0.036	0.011	0.000
1000	0.026	0.007	0.001	0.018	0.005	0.001	0.060	0.026	0.007	0.033	0.015	0.002
Binarised $p(X, Z)$												
200	0.023	0.013	0.001	0.018	0.013	0.001	0.026	0.016	0.007	0.022	0.016	0.006
500	0.023	0.020	0.015	0.019	0.016	0.012	0.019	0.016	0.007	0.016	0.011	0.008
1000	0.009	0.007	0.005	0.007	0.005	0.002	0.017	0.012	0.005	0.012	0.009	0.004
Kitagawa (2015)												
200	0.057	0.027	0.005	0.040	0.016	0.004	0.339	0.265	0.159	0.335	0.274	0.165
500	0.028	0.016	0.002	0.022	0.011	0.002	0.406	0.367	0.288	0.405	0.362	0.280
1000	0.024	0.010	0.001	0.011	0.003	0.000	0.527	0.481	0.415	0.522	0.486	0.418

Table 9: Additional Size Results